

claude-windows / c72aaaeb-2f5c-445c-b834-1f0e6ac95f61

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru-badminton-highlight-indexer-badminton-highlight-indexer\c72aaaeb-2f5c-445c-b834-1f0e6ac95f61.jsonl`
- SHA-256: `38576ab9560eaaf7ae4b02e327f251de21555c60c93f62a9053717b294790c43`
- Source modified: `2026-06-29T10:38:26+00:00`
- Imported at: `2026-07-05T16:48:20+00:00`
- project: `c--Users-avidu-Projects-khelsutra-guru-badminton-highlight-indexer-badminton-highlight-indexer`
- session_id: `c72aaaeb-2f5c-445c-b834-1f0e6ac95f61`
```

Transcript

1. user (2026-06-29T08:35:50.110Z)

hello - can you check whether there are any open files or pending or lingering items/files in this repo?

2. assistant (2026-06-29T08:35:54.149Z)

I'll check the repo state. Per the repo conventions I'll start with a safe pull, then look for uncommitted, untracked, stashed, and other lingering items.

3. user (2026-06-29T08:35:59.606Z)

master...origin/master [behind 38]

```
---STASH---
---UNTRACKED-ALL---
```

4. user (2026-06-29T08:36:04.606Z)

```
---AHEAD/BEHIND---
38      0
---UNMERGED-BRANCHES---
* master          71d55a1 [origin/master: behind 38] Setting
  proposal/rally-detection-fixes 1e221d9 [origin/proposal/rally-detection-fixes] docs: add code
review and plan for rally detection fixes
---WORKTREES---
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/badminton-highlight-indexer
71d55a1 [master]
```

5. assistant (2026-06-29T08:36:08.657Z)

Working tree is clean ? but a few notable things. Let me check for in-progress operations and any stray temp/output files.

6. user (2026-06-29T08:36:14.187Z)

```
---ignored/untracked incl ignored---
```

7. user (2026-06-29T08:36:15.372Z)

```
43578a7 chore(gcp): teardown.sh accepts RALLY_VM_NAME (align with create_gpu_vm.sh) (#392)
ffb1f53 ci: harden the 'System libs' apt step against the runner's Microsoft-repo 403 flake
(#391)
b6214e5 chore: ruff format the whole repo (259 files) (#387)
3624a8d feat(D13/P12): migrate ByteTrack to standalone trackers package (#386)
```

0b4f492 fix(B1/P9): remove fabricated data from motion segmenter (#385)
---local-only commits (should be none)---

8. assistant (2026-06-29T08:36:25.313Z)

Here's the state of the repo ? ****the working tree itself is completely clean****, but there's one real item:

Findings

****? Nothing pending locally****

- No uncommitted/modified files, no staged changes
- No untracked files (even including ignored ones ? nothing)
- No stashes
- No in-progress merge/rebase/cherry-pick/revert/bisect
- No local-only commits (nothing unpushed)
- No detached extra worktrees

****?? Your local `master` is 38 commits behind `origin/master`****

- Local tip: `71d55a1 Setting`
- Remote tip: `43578a7 chore(gcp): teardown.sh accepts RALLY\_VM\_NAME (#392)`
- 38 behind, 0 ahead ? this is a clean fast-forward (no divergence, nothing to lose)

****?? One extra local branch:**** `proposal/rally-detection-fixes` (`1e221d9`), in sync with its remote ? not lingering, just a tracked feature branch.

The local `master` being 38 behind matters because your global convention is to pull before reading. Since the tree is clean and it's a pure fast-forward, I can safely run `git pull --ff-only` to bring it current. Want me to?

9. user (2026-06-29T08:36:59.725Z)

yes please bring the repo in sync with remote head. So there is no local changes?

10. user (2026-06-29T08:37:08.562Z)

Updating 71d55a1..43578a7

Fast-forward

.github/workflows/ci.yml	29 +
.github/workflows/nightly.yml	280 ++++++
.gitignore	Bin 1304 -> 1331 bytes
CLAUDE.md	2 +-
backend/annotations/__init__.py	9 +-
backend/annotations/automated_provider.py	24 +-
backend/annotations/base.py	15 +-
backend/annotations/file_provider.py	12 +-
backend/api/auth.py	38 +-
backend/api/job_worker.py	109 ++-
backend/api/uploads.py	76 +-
backend/billing.py	23 +-
backend/compute/__init__.py	14 +-
backend/compute/base.py	4 +-
backend/compute/cloud_run.py	128 ++-
backend/config/__init__.py	14 +
backend/config/loader.py	73 +-
backend/config/models.py	122 +- -
backend/config/startup.py	185 +++ - -
backend/eval/ablation.py	224 +++ - -
backend/eval/ablation_pdf.py	227 +++ - -
backend/eval/annotations.py	3 +-
backend/eval/audio/e1_probe.py	80 +-
backend/eval/batch.py	5 +-
backend/eval/calibrate_local.py	105 ++-
backend/eval/calibrate_wasb.py	102 ++-

backend/eval/calibration.py	93 ++-
backend/eval/classifier.py	17 +-
backend/eval/distill_local.py	154 +++-
backend/eval/eval_partial_labels.py	215 +++--
backend/eval/eval_stats.py	47 +-
backend/eval/experiment.py	268 ++++--
backend/eval/export_wasb_segments.py	223 +++--
backend/eval/fusion_audio.py	59 +-
backend/eval/fusion_compare.py	41 +-
backend/eval/fusion_golden.py	148 +++-
backend/eval/gemini_refine.py	252 ++++--
backend/eval/golden_fixtures.py	427 ++++++---
backend/eval/golden_real_fixtures.py	110 ++-
backend/eval/gt_loader.py	29 +-
backend/eval/harness.py	47 +-
backend/eval/metrics.py	79 +-
backend/eval/per_user_eval.py	61 +-
backend/eval/per_user_gate.py	17 +-
backend/eval/per_user_regression.py	69 +-
backend/eval/promotion.py	61 +-
backend/eval/rally_seg_eval.py	649 ++++++++-----
backend/eval/rally_seq_proto.py	89 +-
backend/eval/regression.py	146 +++-
backend/eval/score_calibration.py	24 +-
backend/eval/segmentation_metrics.py	59 +-
backend/eval/serve_contrast.py	101 ++-
backend/eval/served_gate_a.py	57 +-
backend/eval/splits.py	17 +-
backend/eval/tolerance_metrics.py	127 ++-
backend/eval/training.py	13 +-
backend/eval/windowing.py	53 +-
backend/features/rally_seq.py	53 +-
backend/flywheel.py	28 +-
backend/infrastructure/database.py	539 ++++++-----
backend/infrastructure/execution_policy.py	98 ++-
backend/main.py	919 ++++++++-----
backend/pipeline/audio/hit_detector.py	67 +-
backend/pipeline/detectors/__init__.py	1 -
backend/pipeline/detectors/audio_features.py	5 +-
backend/pipeline/detectors/base.py	16 +-
backend/pipeline/detectors/court_gate.py	69 +-
backend/pipeline/detectors/device.py	4 +-
backend/pipeline/detectors/fusion_features.py	109 ++-
backend/pipeline/detectors/native_wasb_runner.py	167 +++-
backend/pipeline/detectors/person_detector.py	145 +++-
backend/pipeline/detectors/rally_gate.py	37 +-
backend/pipeline/detectors/stub_runner.py	71 +-
backend/pipeline/detectors/tracknet_runner.py	174 +++-
backend/pipeline/detectors/wasb_infer.py	324 +++++--
backend/pipeline/detectors/wasb_runner.py	102 ++-
backend/pipeline/personalization/serving.py	9 +-
backend/pipeline/segmenters/__init__.py	8 +-
backend/pipeline/segmenters/_keyhint.py	1 +
backend/pipeline/segmenters/base.py	40 +-
backend/pipeline/segmenters/fusion_hybrid.py	144 +++-
backend/pipeline/segmenters/gemini.py	306 +++++--
backend/pipeline/segmenters/motion.py	172 +++-
backend/pipeline/segmenters/tracknet_hybrid.py	18 +-
backend/pipeline/segmenters/trajectory_hybrid.py	286 +++++--
backend/pipeline/segmenters/wasb_hybrid.py	33 +-
backend/pipeline/segmenters/yolo_hybrid.py	292 +++++--
backend/pipeline/stitcher/compiler.py	218 +++--
backend/pipeline/stitcher/watermark_i18n.py	1 +
backend/pipeline/validators/__init__.py	8 +-
backend/pipeline/validators/base.py	39 +-

backend/pipeline/validators/blur.py	29 +-
backend/pipeline/validators/format.py	35 +-
backend/pipeline/validators/pts_continuity.py	53 +-
backend/pipeline/validators/relevance.py	53 +-
backend/pipeline/validators/resolution_fps.py	74 +-
backend/pipeline/validators/scene_cut.py	29 +-
backend/providers/__init__.py	12 +-
backend/providers/base.py	7 +-
backend/providers/gemini.py	176 +++-
backend/reporting/__init__.py	2 +
backend/reporting/harvest.py	106 ++-
backend/reporting/run_signals.py	1 +
backend/results.py	2 +-
backend/sports/__init__.py	14 +-
backend/sports/badminton.py	3 +-
backend/sports/base.py	1 +
backend/storage/__init__.py	87 +-
backend/storage/base.py	23 +-
backend/storage/gcs.py	32 +-
backend/storage/gdrive.py	23 +-
backend/storage/gdrive_api.py	99 ++-
backend/storage/local.py	5 +-
backend/storage/ref.py	19 +-
backend/storage/s3.py	32 +-
backend/tools/export.py	33 +-
backend/tools/rally_slicer.py	102 ++-
backend/utils/app_home.py	4 +-
backend/utils/ffmpeg.py	12 +-
backend/utils/geometry.py	28 +-
backend/utils/logging_setup.py	4 +-
backend/utils/paths.py	7 +-
backend/utils/run_telemetry.py	94 ++-
backend/utils/single_instance.py	12 +-
backend/utils/video.py	60 +-
backend/utils/workspace.py	11 +-
backend/worker/__main__.py	410 ++++++--
deploy/cloudrun/Dockerfile	7 +-
deploy/digitalocean/bootstrap.sh	6 +
deploy/gcp/create_gpu_vm.sh	4 +-
deploy/gcp/teardown.sh	8 +-
docs/CODE_CLEANUP_AUDIT_2026-06-24.md	282 +++++++
docs/DECISION_SNAPSHOT_REGRESSION.md	96 +++
docs/DOC_STATUS.md	1 +
docs/KHELSUTRA_PRODUCT_OVERVIEW.md	203 +++++
docs/README.md	2 +
.../2026-06-18-lovo-resweep-n13/lovo_resweep.py	23 +-
mypy.ini	15 +-
pyproject.toml	6 +-
requirements.txt	20 +-
ruff.toml	5 +-
run_all_tests.py	12 +-
run_eval.py	103 ++-
run_regression.py	111 ++-
scripts/doctor.py	58 +-
scripts/install.py	81 +-
scripts/nightly_email.py	47 +-
scripts/nightly_reelcount.py	388 ++++++--
scripts/nightly_regression.py	385 ++++++--
scripts/run_phase3_eval.py	149 +++-
scripts/run_process_and_stitch.py	116 ++-
scripts/uninstall.py	48 +-
scripts/verify_compute_path.py	73 +-
tests/test_ablation.py	324 ++++++--
tests/test_annotation_providers.py	6 +-
tests/test_api_cloud_dispatch.py	392 ++++++--

tests/test_api_endpoints.py	295 +++++--
tests/test_app_home.py	29 +-
tests/test_app_launcher.py	78 +-
tests/test_async_jobs.py	144 +++-
tests/test_audio_features.py	107 ++-
tests/test_automated_provider.py	25 +-
tests/test_base.py	27 +-
tests/test_billing.py	172 +++-
tests/test_bind_and_host.py	16 +-
tests/test_calibrate_local.py	15 +-
tests/test_calibrate_wasb.py	21 +-
tests/test_calibration.py	30 +-
tests/test_cleanup_caches.py	164 +++-
tests/test_compiler.py	17 +-
tests/test_compute_backend.py	347 ++++++--
tests/test_config.py	202 +++--
tests/test_config_deployment.py	102 ++-
tests/test_config_wasb.py	44 +-
tests/test_court_gate.py	65 +-
tests/test_database.py	164 +++-
tests/test_db_upgrade_safety.py	22 +-
tests/test_detector_keying.py	26 +-
tests/test_device_context.py	7 +-
tests/test_distill_local.py	21 +-
tests/test_doctor.py	15 +-
tests/test_edge_auth.py	117 ++-
tests/test_eval_annotations.py	34 +-
tests/test_eval_batch.py	23 +-
tests/test_eval_classifier.py	4 +-
tests/test_eval_harness.py	35 +-
tests/test_eval_metrics.py	22 +-
tests/test_eval_partial_labels.py	36 +-
tests/test_eval_regression.py	113 ++-
tests/test_eval_stats.py	13 +-
tests/test_eval_training.py	25 +-
tests/test_execution_policy.py	98 ++-
tests/test_experiment.py	158 +++-
tests/test_experiment_honest.py	30 +-
tests/test_export_wasb_segments.py	36 +-
tests/test_ffmpeg_resolver.py	13 +-
tests/test_flywheel.py	64 +-
tests/test_fusion_compare.py	213 ++++--
tests/test_fusion_features.py	100 ++-
tests/test_fusion_golden_telemetry.py	63 +-
tests/test_fusion_hybrid.py	137 ++-
tests/test_gemini.py	127 ++-
tests/test_gemini_refine.py	83 +-
tests/test_generate_highlights.py	223 +++--
tests/test_geometry.py	14 +-
tests/test_golden_fixtures.py	50 +-
tests/test_golden_fixtures_split.py	39 +-
tests/test_golden_real_fixtures.py	479 ++++++++--
tests/test_gt_loader.py	17 +-
tests/test_hit_detector.py	33 +-
tests/test_input_source.py	88 +-
tests/test_job_self_scheduling.py	23 +-
tests/test_job_worker_loop.py	62 +-
tests/test_job_worker_module_seam.py	12 +-
tests/test_leak_scan.py	13 +-
tests/test_light_tier_no_torch.py	11 +-
tests/test_locale_analytics.py	16 +-
tests/test_logging_setup.py	9 +-
tests/test_motion_segmenter.py	74 +-
tests/test_native_wasb_runner.py	218 +++--
tests/test_nightly_reelcount.py	102 ++-

tests/test_nightly_regression.py	142 +++-
tests/test_output_base.py	61 +-
tests/test_p2b_install.py	111 ++-
tests/test_path_safety.py	28 +-
tests/test_per_user_eval.py	36 +-
tests/test_per_user_gate.py	46 +-
tests/test_per_user_regression.py	85 +-
tests/test_person_detector.py	208 ++++-
tests/test_personalization_serving.py	101 ++-
tests/test_profile_parity.py	68 +-
tests/test_promotion.py	66 +-
tests/test_provider_gemini.py	194 ++++-
tests/test_providers.py	5 +-
tests/test_rally_gate.py	41 +-
tests/test_rally_seg_eval.py	130 ++-
tests/test_rally_seg_eval_report.py	187 +++--
tests/test_rally_seq_proto.py	38 +-
tests/test_rally_slicer.py	65 +-
tests/test_rally_summary.py	4 +-
tests/test_redact.py	5 +-
tests/test_reingest_safety.py	7 +-
tests/test_reliability_hardening.py	2 +-
tests/test_reporting.py	35 +-
tests/test_reporting_harvest.py	99 ++-
tests/test_reporting_spec_faq.py	28 +-
tests/test_results.py	7 +-
tests/test_run_signals.py	9 +-
tests/test_run_telemetry.py	45 +-
tests/test_score_calibration.py	19 +-
tests/test_segmentation_metrics.py	25 +-
tests/test_serve_contrast.py	23 +-
tests/test_served_gate_a_regression.py	30 +-
tests/test_server_entrypoint.py	109 ++-
tests/test_single_instance.py	4 +-
tests/test_splits.py	11 +-
tests/test_sport_profiles.py	2 +
tests/test_startup_checks.py	211 +++--
tests/test_storage.py	132 ++-
tests/test_storage_gdrive_api.py	42 +-
tests/test_stub_runner.py	33 +-
tests/test_tolerance_metrics.py	25 +-
tests/test_tracknet_hybrid.py	111 ++-
tests/test_tracknet_runner.py	320 ++++--
tests/test_training_gen0.py	53 +-
tests/test_trajectory_hybrid_equivalence.py	109 ++-
tests/test_ui_bundle.py	10 +-
tests/test_upload_download.py	91 +-
tests/test_validators_base.py	21 +-
tests/test_variant_tuning.py	21 +-
tests/test_video.py	7 +-
tests/test_wasb_cache.py	134 ++-
tests/test_wasb_runner.py	50 +-
tests/test_watermark.py	127 ++-
tests/test_watermark_i18n.py	37 +-
tests/test_windowing.py	50 +-
tests/test_worker.py	450 ++++++--
tests/test_workspace.py	10 +-
tests/test_wsl_tilde_quoting.py	53 +-
tests/test_yolo_hybrid.py	165 +++-
tools/cleanup_caches.py	275 ++++--
tools/generate_highlights.py	230 ++++--
tools/leak_scan.py	103 ++-
training/gen0/harness.py	121 ++-

284 files changed, 18536 insertions(+), 6536 deletions(-)
create mode 100644 .github/workflows/nightly.yml

```
create mode 100644 docs/CODE_CLEANUP_AUDIT_2026-06-24.md
create mode 100644 docs/DECISION_SNAPSHOT_REGRESSION.md
create mode 100644 docs/KHELSUTRA_PRODUCT_OVERVIEW.md
---STATUS---
```

master...origin/master

11. assistant (2026-06-29T08:37:14.284Z)

Done. ?

Result

```
- **Fast-forwarded cleanly** `71d55a1` ? `43578a7` (38 commits, 284 files changed: +18,536 /
?6,536). No merge, no conflicts ? pure fast-forward as expected.
- **`git status` now shows:** `## master...origin/master` with no file lines below it.
```

To answer your question directly: ****no, there are no local changes.****

```
- Working tree is clean (no modified, staged, or untracked files)
- No stashes
- No local-only/unpushed commits
- Local `master` is now exactly in sync with `origin/master`
```

The local branch `proposal/rally-detection-fixes` still exists and is in sync with its own remote ? that's just a tracked feature branch, not a lingering change. Everything is current.

12. user (2026-06-29T08:37:14.319Z)

if we can prune the local branch as well then please do so

13. assistant (2026-06-29T08:37:18.175Z)

Let me check whether that branch is merged before deciding how to delete it safely.

14. user (2026-06-29T08:37:24.570Z)

```
---merged into origin/master?---
---commits on branch not in origin/master---
1e221d9 docs: add code review and plan for rally detection fixes
---does remote branch still exist?---
```

15. assistant (2026-06-29T08:37:32.788Z)

Hold on ? this branch is ****not safe to delete blindly****, so I want to flag what I found before doing anything destructive:

```
- ? **Not merged** into `origin/master`.
- ?? It has **1 commit that exists nowhere else**: `1e221d9 docs: add code review and plan for
rally detection fixes`.
- ?? The **remote `origin/proposal/rally-detection-fixes` is gone** (the `ls-remote` returned
nothing) ? so this local branch is the *only* place that commit survives.
```

A plain `git branch -d` would refuse it (unmerged); only a force `git branch -D` would delete it ? and that would permanently drop that docs commit. This contradicts how it looked at first glance ("in sync with its remote"), so I don't want to just nuke it.

How would you like to proceed?

16. user (2026-06-29T08:37:40.734Z)

Your questions have been answered: "The branch proposal/rally-detection-fixes has an unmerged commit (1e221d9, a docs/code-review file) and its remote is gone. How should I prune it?"="Inspect first, then decide". You can now continue with these answers in mind.

17. assistant (2026-06-29T08:37:54.667Z)

You chose to inspect first. Let me show you what that commit actually adds.

18. user (2026-06-29T08:37:57.142Z)

```
===COMMIT META===
1e221d914ee5c65c6ab31f841853c256523c5ec5
Avi Dullu  Fri Jun 19 21:50:27 2026 +0530
docs: add code review and plan for rally detection fixes

===FILES CHANGED===
 docs/RALLY_DETECTION_CODE_REVIEW.md | 50 +++++
 docs/RALLY_DETECTION_FIXES_PLAN.md  | 57 +++++
 2 files changed, 107 insertions(+)
```

19. assistant (2026-06-29T08:38:01.575Z)

Two new docs (107 lines total, authored by you on 2026-06-19). Here's their content so you can judge whether they're worth keeping:

20. user (2026-06-29T08:38:04.217Z)

```
##### docs/RALLY_DETECTION_CODE_REVIEW.md #####
```

Rally Detection Code Review

1. How Rally Detection Works

The project identifies rallies through a combination of local trajectory tracking (e.g., TrackNet or WASB) and a series of heuristic gating/trimming rules, culminating in an AI-based refinement (Phase 3). Here's a breakdown of the pipeline:

Phase 2: Candidate Windowing (The Heuristics)

- 1. **Velocity Thresholding:** The trajectory detector outputs `(Frame, Visibility, X, Y)`. For each visible frame, the normalized velocity (distance divided by frame width per second) is calculated. Frames with a velocity exceeding `velocity\_thresh` (e.g., 0.02) are considered "active" (meaning the shuttle is in flight).
- 2. **Merging / Grouping:** Active frames within a certain dead time (`merge\_gap`, usually 2.0s) are merged into candidate windows.
- 3. **Splitting Over-merged Windows (W1):** If a window exceeds `max\_window\_duration`, the code splits it at the largest internal gap (dead-time) that is at least `min\_split\_gap`. If the window lacks gaps (the shuttle never stops moving), a `window\_hard\_cap` setting (if enabled) forcibly chops it in half at the temporal midpoint.
- 4. **Inactive Tail Trimming (R4):** Even after a rally finishes, a player might pick up the shuttle and toss it, creating a slow-moving trajectory tail. To prevent the window from overextending, the algorithm trims the end back to the last segment of the window that had a dense cluster of "fast" frames (velocity > `rally\_thresh`).
- 5. **Blob Fallback (R5):** If a window still vastly exceeds the cap after the W1 split and R4 trim (meaning it's an unusually continuous blob of motion), it is chopped down to the maximum cap limit and marked with a `forced\_cut` flag. This signals that the clip needs "rally-state" cues (like net-crossings) rather than a simple duration cap.
- 6. **Rally Gate (Gate A):** In `rally\_gate.py`, a heuristic filters out "false" rallies (like a player tossing a shuttle back to their opponent). It estimates the net position (median play axis coordinate) and counts how many times the shuttle crosses the net. A valid rally must have `min\_crossings` (e.g., 2).

Phase 3: AI Handoff

Valid candidate windows are padded (e.g., by 2 seconds on each end) and sent to a Generative AI Provider (like Gemini) along with a sports-specific prompt. The AI confirms the semantic boundaries, ensuring the window correctly matches a true rally, and also attempts to count the shuttle exchanges.

2. Deep Code Review & Potential Issues Found

I performed a deep review of ``tracknet_runner.py``, ``trajectory_hybrid.py``, ``rally_gate.py``, and the parsing logic in ``sports-data-collector``. Here are the findings and hidden issues:

A. Subtle Calculation Artifact in ``_spread`` (``rally_gate.py``)

Observation: The ``_spread`` function calculates a robust spread (ignoring the bottom 5% and top 5% of coordinates to eliminate outliers):

```
`lo = s[max(0, int(0.05 * (len(s) - 1)))]`  
`hi = s[min(len(s) - 1, int(0.95 * (len(s) - 1)))]`
```

Hidden Behavior: Because ``int()`` truncates rather than rounds, for very short arrays (e.g., ``< 20`` items), the 95th percentile will simply be the second-to-last item, and for ``len(s) == 2``, the spread evaluates to strictly ``0.0``.

Impact: This doesn't crash the pipeline, because the play axis is correctly estimated over the *entire* video's thousands of trajectory points. However, if this utility function is reused on small window slices in the future, it will quietly collapse the spread to zero.

B. Sensitivity to Sparse Tracking in ``_trim_inactive_tail`` (``tracknet_runner.py``)

Observation: The inactivity trimmer identifies the end of a rally by looking for the last frame where the trailing ``trail_frames`` window has a sufficient density of fast-moving frames (``fast_sum / win_count >= min_density``).

Hidden Issue: ``win_count`` represents the number of *visible, active frames* in that time boundary, rather than the total elapsed time. If the detector completely loses track of the shuttle for 1.8 seconds and only picks up 1 fast frame right at the end of the window, ``win_count`` will be very low (e.g., 1 or 2). Consequently, ``fast_sum / win_count`` could easily evaluate to 1.0 (100% density).

Impact: Sparse tracking points at the very end of a clip might prevent the tail trimmer from cutting off dead-time correctly, artificially inflating the end of the candidate window. This is somewhat mitigated by the ``merge_gap`` breaking trajectories, but it remains a boundary logic quirk.

C. Graceful Degradation on Incomplete CSV Validation (``rally_cvat.py`` in ``sports-data-collector``)

Observation: The ``parse_cvat_boxes`` function searches zip files for an ``xml`` ending in ``annotations.xml`` or, failing that, *any* ``xml``.

Issue: In ``load_canonical_csv()``, ``safe_int()`` is used to catch malformed ``rally_number`` fields. If a row is entirely malformed (but doesn't break the CSV parsing), it just ``continue`s` silently. While this provides stability, a severely malformed CSV might silently drop half its rallies without failing the import gate. Logging a warning for skipped rows would improve observability.

D. Safe Slicing in the W1/R5 Blob Splitters

The W1/R5 gap-splitters use a recursive worklist:

```
`split_i = TrackNetRunner._midpoint_split_index(h)`  
`work.append(h[:split_i]); work.append(h[split_i:])`
```

Because it explicitly enforces ``len(h) < 2`` before splitting, ``split_i`` will always be at least 1, meaning the code safely guarantees it will never enter an infinite loop of empty arrays. This is an excellent, defensively written chunk of code.

docs/RALLY_DETECTION_FIXES_PLAN.md

<!--

PROJECT DOC TEMPLATE ? copy this file to docs/<YOUR_PLAN>.md and fill it in.

-->

Refactor Rally Detection Edge Cases & Parser Stability

> **Status:** ``IN PROGRESS`` . **Owner:** ``Agent`` . **Created:** ``2026-06-19`` . **Last updated:** ``2026-06-19``

```
> **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived`
> **Tracking anchors:** S7 progress tracker is the source of truth; indexed in `docs/README.md`.
> **Relation to existing docs:** Extends `RALLY_DETECTION_QUALITY_REPORT.md`.

---
```

0. TL;DR

The rally detection pipeline is largely robust, but a deep code review highlighted three minor edge cases/bugs: a potential divide-by-zero or zero-spread in `\_spread` for tiny sequences, a density calculation vulnerability for sparse tracks in `\_trim\_inactive\_tail`, and silent skipping of malformed rows in the CVAT parsing script. This plan outlines the fixes for these issues to increase stability.

1. Problem & goal

The current implementation of the candidate windowing handles normal inputs extremely well (including handling blobs), but could fail or behave erratically under specific conditions:

- ****Zero-spread calculation**:** `\_spread` in `rally\_gate.py` evaluates to `0.0` for arrays of length 2, which could cause a zero deadband if used on short clips.
- ****Sparse trailing frames**:** `\_trim\_inactive\_tail` calculates frame density based on **visible** frames. A long gap followed by a single active frame gives 100% density, artificially inflating the rally end.
- ****Silent failure on malformed CSV**:** `load\_canonical\_csv` in `rally\_cvat.py` silently ignores rows without a valid `rally\_number`.

****Goal**:** Fix these three issues to harden the rally detection and ingestion pipelines without changing the existing heuristic thresholds.

4. Design / architecture

The changes will be entirely localized to the identified functions:

- `rally\_gate.py`: Modify `\_spread` to use `math.ceil()` or properly enforce a minimum subset size before trimming percentiles, ensuring it never inadvertently calculates a spread of 0 for non-identical values.
- `tracknet\_runner.py`: Update `\_trim\_inactive\_tail` to factor the **temporal length** (or total expected frames in the trailing window) into the density calculation, rather than purely relying on `win\_count` (the number of visible frames).
- `rally\_cvat.py`: Introduce `logger.warning` before the `continue` statement in `load\_canonical\_csv` to surface skipped rows during CVAT ingestion.

6. Honest limits ? what this does NOT do

This will NOT change the core AI validation (Phase 3) or the existing `velocity\_thresh` values. It solely tightens up boundary calculation edge cases. It will not suddenly "fix" rallies that the tracking model missed entirely due to lighting or occlusion.

7. Deliverables & progress tracker ? ****source of truth****

Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. ****One small PR per row****, branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.

ID	Deliverable	Depends on	Gated?	Status	PR
P0	Fix `_spread` and `_trim_inactive_tail` edge cases		?	No	?
P1	Add logging to `load_canonical_csv` for silent skips		?	No	?

8. Open questions ? owner / external

1. Do we want `\_trim\_inactive\_tail` to fall back to the existing behavior if there are very few frames, or strictly enforce a minimum temporal density? (Assuming strict temporal density is preferred for accuracy).

9. Definition of done

- [] `\_spread` bug fixed and tested.
- [] `\_trim\_inactive\_tail` density calculation corrected.
- [] Warning log added for malformed rows in `rally\_cvat.py`.
- [] Tests updated and passing.

10. References

```
**Internal code anchors `[verified]`:**
- `backend/pipeline/detectors/rally_gate.py`
- `backend/pipeline/detectors/tracknet_runner.py`
- `src/parsers/cvat/rally_cvat.py`
**Internal docs:**
- `docs/RALLY_DETECTION_CODE_REVIEW.md`
```

21. assistant (2026-06-29T08:38:21.249Z)

Here's my honest read of what's in that commit:

It's ****genuine analytical work, not throwaers****. Two docs:

1. ****`RALLY\_DETECTION\_CODE\_REVIEW.md`**** ? a walkthrough of how rally detection works plus 3 specific potential bugs found:
 - ****`\_spread`**** in `rally_gate.py`: `int()` truncation collapses spread to `0.0` for tiny arrays (latent, only bites if reused on short slices).
 - ****`\_trim\_inactive\_tail`**** in `tracknet_runner.py`: density uses *visible* frame count, so one fast frame after a long tracking gap reads as 100% density ? inflates rally end.
 - ****`load\_canonical\_csv`**** in sibling `sports-data-collector`: silently `continue`s on malformed rows ? can drop rallies with no warning.
2. ****`RALLY\_DETECTION\_FIXES\_PLAN.md`**** ? a project doc (template-compliant, status `IN PROGRESS`, P0/P1 both still ? Todo) proposing localized fixes for exactly those three.

Two caveats worth knowing before you decide:

- The 38 commits we just pulled ****touched `rally\_gate.py` and `tracknet\_runner.py`**** ? so some of these findings may already be addressed or moved. I haven't verified that yet.
- This work relates directly to the ****rally-detection quality track**** that CLAUDE.md flags as live/in-progress (`MULTI_SIGNAL_FUSION_PLAN.md`, etc.).

My recommendation: ****don't force-delete this**** ? it's unmerged analysis that exists nowhere else. Rather than dropping the whole branch, the cleaner "prune" is to preserve the two docs and then delete the branch.

22. user (2026-06-29T08:39:07.285Z)

Your questions have been answered: "Now that you've seen the content, how should I prune proposal/rally-detection-fixes?"="Preserve docs, then delete". You can now continue with these answers in mind.

23. assistant (2026-06-29T08:39:25.344Z)

I'll preserve the two docs via a branch + PR (per repo convention ? no direct commits to `master`), then delete the stale branch. Following the safe branching protocol: branch explicitly off `origin/master`.

24. user (2026-06-29T08:39:30.695Z)

```
branch 'docs/preserve-rally-detection-review' set up to track 'origin/master'.
Switched to a new branch 'docs/preserve-rally-detection-review'
docs/preserve-rally-detection-review
43578a7 chore(gcp): teardown.sh accepts RALLY_VM_NAME (align with create_gpu_vm.sh) (#392)
```

25. assistant (2026-06-29T08:39:33.830Z)

Now restore the two docs from the old commit into this branch and confirm they don't already exist on master.

26. user (2026-06-29T08:39:39.183Z)

```
---already on master?---
(empty above = not tracked yet)
---status---
```

```
A docs/RALLY_DETECTION_CODE_REVIEW.md
A docs/RALLY_DETECTION_FIXES_PLAN.md
```

27. assistant (2026-06-29T08:39:43.560Z)

Both staged. Committing with explicit paths.

28. user (2026-06-29T08:39:51.580Z)

```
[docs/preserve-rally-detection-review 0a2dfea] docs: preserve rally-detection code review +
fixes plan
2 files changed, 107 insertions(+)
create mode 100644 docs/RALLY_DETECTION_CODE_REVIEW.md
create mode 100644 docs/RALLY_DETECTION_FIXES_PLAN.md
---only my commits vs origin/master?---
0a2dfea docs: preserve rally-detection code review + fixes plan
```

29. assistant (2026-06-29T08:40:01.417Z)

Only my commit is ahead ? clean. The docs content is now safely captured on this branch, so the orphaned branch is safe to delete. Pushing and opening the PR, then deleting the stale branch.

30. user (2026-06-29T08:40:07.343Z)

```
remote: This repository moved. Please use the new location:
remote:  https://github.com/Khelsutra/badminton-highlight-indexer.git
remote:
remote: Create a pull request for 'docs/preserve-rally-detection-review' on GitHub by
visiting:
remote:  https://github.com/Khelsutra/badminton-highlight-
indexer/pull/new/docs/preserve-rally-detection-review
remote:
branch 'docs/preserve-rally-detection-review' set up to track 'origin/docs/preserve-rally-
detection-review'.
To https://github.com/avidullu/badminton-highlight-indexer.git
* [new branch]      docs/preserve-rally-detection-review -> docs/preserve-rally-
detection-review
```

31. assistant (2026-06-29T08:40:09.989Z)

Pushed. Now opening the PR and then deleting the stale local branch.

32. user (2026-06-29T08:40:21.313Z)

<https://github.com/Khelsutra/badminton-highlight-indexer/pull/393>

33. user (2026-06-29T08:40:29.637Z)

```
---verify content preserved (sha of files matches old commit)---
(no output above = byte-identical to the orphaned commit)
---deleting stale branch---
Deleted branch proposal/rally-detection-fixes (was 1e221d9).
---remaining branches---
* docs/preserve-rally-detection-review 0a2dfea [origin/docs/preserve-rally-detection-review]
docs: preserve rally-detection code review + fixes plan
master                                43578a7 [origin/master] chore(gcp): teardown.sh accepts
RALLY_VM_NAME (align with create_gpu_vm.sh) (#392)
```

34. assistant (2026-06-29T08:40:43.122Z)

Done ? pruned safely with nothing lost.

What happened

1. **Verified** the orphaned branch had 1 unmerged commit (`1e221d9`) with two analysis docs and **no surviving remote** ? i.e. the only copy.
2. **Preserved** both docs on a fresh branch off `origin/master`, confirmed **byte-identical** to the original commit (`git diff` empty), and opened **[PR #393](https://github.com/Khelsutra/badminton-highlight-indexer/pull/393)**.
3. **Deleted** the stale branch `proposal/rally-detection-fixes` (was `1e221d9`) ? its content now lives in `0a2dfea`.

Current branch state

- `master` ? in sync with `origin/master` (`43578a7`), clean tree.
- `docs/preserve-rally-detection-review` ? pushed, tracks origin, PR #393 open.
- `proposal/rally-detection-fixes` ? **gone**.

Two things for your attention

- **PR #393** awaits your review/merge. Once merged, you can delete the `docs/preserve-rally-detection-review` branch too. Per repo convention I don't merge without you.
- **The salvaged findings** are unverified against current code. The 38 commits we just pulled touched both `rally\_gate.py` and `tracknet\_runner.py`, so some of those three bugs may already be fixed or moved. I flagged this in both the commit message and the PR body. If you'd like, I can now verify whether the three findings still apply ? that ties directly into the live rally-detection quality track in your `CLAUDE.md`.

35. user (2026-06-29T08:41:42.578Z)

<ide_opened_file>The user opened the file c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\docs\RALLY_DETECTION_CODE_REVIEW.md in the IDE. This may or may not be related to the current task.</ide_opened_file>
yes please check if these are already completed work items. So we would definitely need the doc PR to be merged?

36. assistant (2026-06-29T08:41:47.782Z)

Good call ? let me verify each of the three findings against current code before we decide whether the doc PR is worth merging. Let me look at the actual functions.

37. user (2026-06-29T08:41:49.856Z)

Found 2 files
backend\pipeline\detectors\tracknet_runner.py
backend\pipeline\detectors\rally_gate.py

38. user (2026-06-29T08:41:50.233Z)

```
1      """Rally-start gate: suppress between-point false fires using shuttle net-crossings.
2
3      WASB (shuttle-only) over-detects: when a player flicks/tosses the shuttle back to
4      the opponent for service after losing a point, that single trajectory looks like a
5      rally. The discriminator is structural, not duration-based: a real rally has the
6      shuttle crossing the net MULTIPLE times (each shot lands on the other side); a
7      single service feed crosses ~once.
8
9      This module is camera-agnostic and needs NO new model ? it works on the trajectory
10     we already produce (Frame,Visibility,X,Y). It estimates the play axis (the image
11     axis with the larger shuttle spread) and the net position (median shuttle coord on
12     that axis ? the shuttle clusters at the two ends and crosses the net region often,
13     so the median sits near the net), then counts sign changes of (coord - net) across
14     the visible points inside a candidate window, with a deadband to ignore jitter.
15
16     Gate A in the over-fire fix spectrum (B = player-stance state machine, future).
17     Pure functions only ? no I/O, unit-tested; intended as a post-filter on the
18     windows from trajectory_to_action_windows, or to fold into it later.
19     """
20
```

```

21     from __future__ import annotations
22
23     from dataclasses import dataclass
24     from typing import List, Optional, Sequence, Tuple
25
26     Interval = Tuple[float, float]
27
28
29     @dataclass
30     class GatedWindow:
31         start: float
32         end: float
33         crossings: int
34         visible_points: int
35         kept: bool
36
37
38     def _spread(vals: Sequence[float]) -> float:
39         """Robust spread = p95 - p5 (ignores a few outlier detections)."""
40         if len(vals) < 2:
41             return 0.0
42         s = sorted(vals)
43         lo = s[max(0, int(0.05 * (len(s) - 1)))]
44         hi = s[min(len(s) - 1, int(0.95 * (len(s) - 1)))]
45         return hi - lo
46
47
48     def _median(vals: Sequence[float]) -> float:
49         if not vals:
50             return 0.0
51         s = sorted(vals)
52         n = len(s)
53         return s[n // 2] if n % 2 else 0.5 * (s[n // 2 - 1] + s[n // 2])
54
55
56     def estimate_play_axis(points) -> Tuple[str, float, float]:
57         """Return (axis 'x'/'y', net_value, span) from visible points.
58
59         Play axis = the image axis along which the shuttle travels most (larger spread).
60         For a baseline camera that's Y (players top/bottom); for a side camera, X.
61         """
62         xs = [p.x for p in points if p.visible]
63         ys = [p.y for p in points if p.visible]
64         sx, sy = _spread(xs), _spread(ys)
65         if sx >= sy:
66             return "x", _median(xs), sx
67         return "y", _median(ys), sy
68
69
70     def count_net_crossings(window_points, axis: str, net: float, deadband: float) -> int:
71         """Count sign changes of (coord - net) across visible points, ignoring the
72         deadband zone near the net (so jitter at the net doesn't inflate the count)."""
73         crossings = 0
74         last_side = None
75         for p in window_points:
76             if not p.visible:
77                 continue
78             v = (p.x if axis == "x" else p.y) - net
79             if abs(v) < deadband:
80                 continue
81             side = v > 0
82             if last_side is not None and side != last_side:
83                 crossings += 1
84             last_side = side
85         return crossings

```

```

86
87
88 def _points_in(points, fps: float, start: float, end: float):
89     return [p for p in points if start * fps <= p.frame <= end * fps]
90
91
92 def gate_windows(
93     points,
94     windows: List[Interval],
95     fps: float,
96     min_crossings: int = 2,
97     deadband_frac: float = 0.06,
98     estimate: Optional[Tuple[str, float, float]] = None,
99 ) -> List[GatedWindow]:
100     """Annotate each (start,end) window with its shuttle net-crossing count and a
101     keep decision (crossings >= min_crossings). Deadband is a fraction of the play
102     axis span. Net/axis are estimated once over the whole trajectory (stable).
103
104     `estimate` lets a caller scoring MANY windows off one trajectory pass a precomputed
105     `(axis, net, span)` (from `estimate_play_axis`) so the O(P log P) estimate isn't
redone
106     per window ? same result, avoids the per-window re-estimation."""
107     axis, net, span = estimate if estimate is not None else estimate_play_axis(points)
108     deadband = deadband_frac * span
109     out: List[GatedWindow] = []
110     for s, e in windows:
111         wp = _points_in(points, fps, s, e)
112         c = count_net_crossings(wp, axis, net, deadband)
113         vis = sum(1 for p in wp if p.visible)
114         out.append(
115             GatedWindow(
116                 start=s, end=e, crossings=c, visible_points=vis, kept=c >= min_crossings
117             )
118         )
119     return out
120
121
122 def apply_gate(
123     points,
124     windows: List[Interval],
125     fps: float,
126     min_crossings: int = 2,
127     deadband_frac: float = 0.06,
128 ) -> List[Interval]:
129     """Convenience: return only the windows that pass the crossing gate."""
130     return [
131         (g.start, g.end)
132         for g in gate_windows(points, windows, fps, min_crossings, deadband_frac)
133         if g.kept
134     ]
135

```

39. assistant (2026-06-29T08:41:58.191Z)

Finding A (`\_spread`) is **still present verbatim** ? line 43-44 still use `int()` truncation, so for a 2-element array it still collapses to `0.0`. Now let me check finding B in `tracknet\_runner.py`.

40. user (2026-06-29T08:42:00.423Z)

```

327:         inactivity_min_density: float = 0.34,
420:         TrackNetRunner._trim_inactive_tail(
421:             g, tw, inactivity_min_density, inactivity_rally_thresh
445:     def _trim_inactive_tail(
447:         trail_frames: int,

```

```

448:         min_density: float,
455:         ``trail_frames`` window holds >= ``min_density`` fraction of fast frames; trim back
to the
460:         if trail_frames <= 0 or len(group) < 2:
465:             fast_sum = 0
468:                 fast_sum += fast[i]
469:                 while frames[lo] <= frames[i] - trail_frames:
470:                     fast_sum -= fast[lo]
472:                 win_count = i - lo + 1
473:                 if win_count > 0 and fast_sum / win_count >= min_density:

```

41. user (2026-06-29T08:42:03.511Z)

```

445     def _trim_inactive_tail(
446         group: List[Tuple[int, float]],
447         trail_frames: int,
448         min_density: float,
449         rally_thresh: float,
450     ) -> List[Tuple[int, float]]:
451         """R4: trim the post-rally drift tail. After a rally the shuttle keeps moving
*slowly*
452         (picked up / walked / knock-up) above ``velocity_thresh``, so the velocity
windowing
453         over-extends the END. A frame is "fast" (real rally motion) iff its velocity >
``rally_thresh``
454         (higher than the active threshold). The rally is "on" at frame ``i`` while its
trailing
455         ``trail_frames`` window holds >= ``min_density`` fraction of fast frames; trim
back to the
456         LAST such frame ? everything after is sustained slow drift (rally over). Only
cuts (recall
457         can't rise); a window whose tail never goes slow is untouched, and if no fast-
dense point
458         exists at all the group is kept whole (fail-safe, no over-trim). O(n) two-
pointer; the START
459         is left alone (already ~Gemini-accurate per the n=2 boundary-error finding)."""
460         if trail_frames <= 0 or len(group) < 2:
461             return group
462         frames = [f for f, _ in group]
463         fast = [1 if v > rally_thresh else 0 for _, v in group]
464         lo = 0
465         fast_sum = 0
466         last_dense = -1
467         for i in range(len(frames)):
468             fast_sum += fast[i]
469             while frames[lo] <= frames[i] - trail_frames:
470                 fast_sum -= fast[lo]
471                 lo += 1
472             win_count = i - lo + 1
473             if win_count > 0 and fast_sum / win_count >= min_density:
474                 last_dense = i
475         return group[: last_dense + 1] if last_dense >= 0 else group
476
477     @staticmethod
478     def _midpoint_split_index(group: List[Tuple[int, float]]) -> int:
479         """Index of the active frame nearest the group's temporal midpoint (used to chop
a
480         gap-less group into two). Shared by the W1 hard-cap fallback and the R5 blob-
fallback."""
481         mid = (group[0][0] + group[-1][0]) / 2.0
482         split_i = min(range(1, len(group)), key=lambda i: abs(group[i][0] - mid))
483         return split_i
484
485     @staticmethod
486     def _split_overlong_groups(

```



```
487         groups: List[List[Tuple[int, float]]],
488         max_win_frames: int,
489         min_split_frames: int,
```

42. assistant (2026-06-29T08:42:15.761Z)

Finding B (``_trim_inactive_tail``) is **also still present**: ``win_count = i - lo + 1`` (line 472) counts group entries (visible/active frames), while the window itself is bounded temporally by ``frames[lo] <= frames[i] - trail_frames`` (line 469). So the sparse-tracking density quirk the review described is intact. Now finding C lives in the sibling ``sports-data-collector`` repo ? let me check if it's available locally.

43. user (2026-06-29T08:42:21.964Z)

```
FOUND: c:/Users/avidu/Projects/khelsutra-guru/sports-data-collector
---glob search---
c:/Users/avidu/Projects/khelsutra-guru/sports-data-collector/sports-data-
collector/src/parsers/cvat/rally_cvat.py
c:/Users/avidu/Projects/khelsutra-guru/sports-data-collector/src/parsers/cvat/rally_cvat.py
```

44. user (2026-06-29T08:42:26.147Z)

```
8:boxes, with the rally fields (`rally_number`, `shots_count`,
51:# Kept here as a soft check only (a warning, never a hard failure) so this
83:     "rally_number": ("rally_number",),
114:     "rally_number", "start_mmss", "end_mmss", "start_time", "end_time",
123:     rally_number: int
151:     rally_number: Optional[int] = None
248:         continue
256:             continue # a track's terminating keyframe carries no real label
260:             continue
276:     """Group boxes by rally_number and derive each rally's frame range -> seconds."""
282:     rn = safe_int(_attr_lookup(b["attrs"], "rally_number"), None)
284:     continue
294:     rally_number=rn,
337: ordered = sorted(records, key=lambda r: r.rally_number)
342:     f"Rally {r.rally_number}: end <= start "
344:     r.rally_number))
349:     f"Rally {r.rally_number}: ending_reason '{r.ending_reason}' "
351:     r.rally_number))
356:     f"Rally {r.rally_number}: last_shot_outcome '{r.last_shot_outcome}' "
358:     r.rally_number))
363:     f"Rally {r.rally_number}: court '{r.court}' "
365:     r.rally_number))
370:     f"Rally {r.rally_number}: confidence '{r.confidence}' "
372:     r.rally_number))
378:     f"Rally {r.rally_number}: annotator marked confidence=unsure ? review.",
379:     r.rally_number))
388:     f"Rally {r.rally_number}: {r.shots_count} shots over "
391:     r.rally_number))
396:     f"Rally {r.rally_number}: frame {r.frame_max} exceeds the video's "
398:     r.rally_number))
405:     f"Rally {prev.rally_number} (ends {prev.end_time:.2f}s) overlaps "
406:     f"Rally {r.rally_number} (starts {r.start_time:.2f}s) by "
408:     r.rally_number))
417:     f"{gap:.1f}s of unlabeled time between Rally {prev.rally_number} "
418:     f"(ends {prev.end_time:.1f}s) and Rally {r.rally_number} "
420:     r.rally_number))
502:     for r in sorted(records, key=lambda x: x.rally_number):
504:         r.rally_number,
531:         continue
564:def load_canonical_csv(csv_path: str) -> List[RallyRecord]:
579:     rn = safe_int(row.get("rally_number"), None)
581:     continue
```

```

589:         continue
591:         rally_number=rn, start_time=start, end_time=end,
610:     flagged = sorted({i.rally_number for i in issues if i.rally_number is not None})
621:         "rally_number": i.rally_number, "message": i.message} for i in issues],
630:     flagged = sorted({i.rally_number for i in issues if i.rally_number is not None})
647:         continue
663:         if i.rally_number is not None:
664:             flags.setdefault(i.rally_number, []).append(i.code)
666:     cols = ["rally_number", "start_mmss", "end_mmss", "dur_s", "shots_count",
673:         for r in sorted(records, key=lambda x: x.rally_number):
675:             r.rally_number, _mmss(r.start_time), _mmss(r.end_time),
678:             ",".join(flags.get(r.rally_number, [])),

```

45. assistant (2026-06-29T08:42:29.526Z)

Let me read the actual `load\_canonical\_csv` body to see if a warning was added.

46. user (2026-06-29T08:42:30.437Z)

```

564     def load_canonical_csv(csv_path: str) -> List[RallyRecord]:
565         """Load a canonical per-rally CSV into RallyRecords, so a *CSV* delivery (not
566         just a CVAT export) can be validated. Decimal start/end preferred, mm:ss
567         fallback; rows with no usable start/end are skipped. (frame_* are -1 here, as a
568         CSV carries no frame info ? only the frame-derived checks become inapplicable.)
569         """
570         import csv as _csv
571         if not os.path.exists(csv_path):
572             raise FileNotFoundError(f"CSV not found: {csv_path}")
573         records: List[RallyRecord] = []
574         with open(csv_path, newline="", encoding="utf-8") as f:
575             reader = _csv.DictReader(f)
576             reader.fieldnames = [(n or "").strip().lower() for n in (reader.fieldnames or
577 [])]
578             for raw in reader:
579                 row = {(k or "").strip().lower(): v for k, v in raw.items()}
580                 rn = safe_int(row.get("rally_number"), None)
581                 if rn is None:
582                     continue
583                 start = _parse_secs(row.get("start_time"))
584                 if start is None:
585                     start = _parse_secs(row.get("start_mmss"))
586                 end = _parse_secs(row.get("end_time"))
587                 if end is None:
588                     end = _parse_secs(row.get("end_mmss"))
589                 if start is None or end is None:
590                     continue
591                 records.append(RallyRecord(
592                     rally_number=rn, start_time=start, end_time=end,
593                     shots_count=safe_int(row.get("shots_count"), None),
594                     ending_reason=(row.get("ending_reason") or "").strip() or None,
595                     last_shot_outcome=(row.get("last_shot_outcome") or "").strip() or None,
596                     court=_norm_lower(row.get("court")),
597                     confidence=_norm_lower(row.get("confidence")),
598                     score_before=(row.get("score_before") or "").strip() or None,
599                     score_after=(row.get("score_after") or "").strip() or None,
600                     notes=(row.get("notes") or "").strip() or None,
601                     frame_min=-1, frame_max=-1, n_boxes=0,

```

47. assistant (2026-06-29T08:43:08.612Z)

All three findings verified against current code (post the 38-commit pull). Here's the honest result:

Verification ? all three are still unfixed

```
| # | Finding | Location | Status |
|---|-----|-----|-----|
| **A** | `_spread` collapses to `0.0` for 2-element arrays (`int()` truncation) |
[rally_gate.py:43-44](backend/pipeline/detectors/rally_gate.py#L43-L44) | **Still present,
verbatim** ? but **latent**: `_spread` is only called from `estimate_play_axis` over the whole
trajectory (thousands of points), so the tiny-array path isn't reached in production today.
Bites only if reused on small slices. |
| **B** | `_trim_inactive_tail` density uses *visible*-frame count, not temporal span |
[tracknet_runner.py:469-473](backend/pipeline/detectors/tracknet_runner.py#L469-L473) | **Still
present and live**: window is bounded temporally (`frames[lo] <= frames[i] - trail_frames`) but
`win_count = i - lo + 1` counts group entries ? sparse tracking can read 100% density and
inflate the rally end. |
| **C** | `load_canonical_csv` silently `continue`s on malformed rows |
[rally_cvat.py:580-589](.././sports-data-collector/sports-data-
collector/src/parsers/cvat/rally_cvat.py#L580-L589) | **Still present**: two bare `continue`s,
no `logger.warning`. Observability nit in the sibling repo. |
```

So **none** of these are completed work ? the docs describe real, current state, not history.
(Caveat on C: I read the local sibling checkout, which I didn't pull first ? but the code is unambiguous.)

So ? do we "definitely need" the doc PR merged?

Not strictly required, but it now has real justified value ? because the issues are genuinely open, not stale. You have two clean paths, and I'd frame it as a choice rather than a "must":

- **Merge** the doc PR (#393) ? preserves the review + makes the fix plan a *tracked* project doc (P0/P1) in the repo, matching your "substantial work ? tracked doc" convention. Right move if you want this queued but not acted on yet.
- **Skip** the doc, just fix the code ? since findings B (and optionally A/C) are small, localized, well-scoped fixes, I could close #393 and instead do a proper fix PR (with tests). The doc would be redundant once the work lands.

My recommendation: **B** is worth actually fixing (it can silently inflate rally end-times ? directly relevant to your live rally-detection quality track). A and C are minor/latent. So rather than merge a "todo" doc, I'd lean toward **implementing** the fixes ? but A's latent status means I'd want your call on whether to touch it at all.

How would you like to proceed?

48. user (2026-06-29T08:47:07.670Z)

Your questions have been answered: "The 3 findings are real and unfixed. How do you want to handle them vs. the doc PR #393?"="Please add the fix code to the same #393 PR and make it a single shot doc + code fix PR. Don't lose the docs in the PR just add the fix code to it as well". You can now continue with these answers in mind.

49. assistant (2026-06-29T08:47:56.397Z)

Understood ? I'll fold the fixes into PR #393 so it becomes a single doc + code-fix PR, keeping the docs. One constraint to flag up front: **finding C** lives in the sibling `sports-data-collector` repo, so it physically cannot go in PR #393 (different repo). I'll fix **A** and **B** here (with tests) and update the plan doc, and handle **C** as a separate sibling-repo PR (I'll ask at the end).

Let me track this and look at the existing tests to match conventions.

50. user (2026-06-29T08:47:58.211Z)

```
1      """Shuttle net-crossing rally gate (pure; no WSL/GPU)."""
2
```

```

3 from backend.pipeline.detectors import rally_gate as rg
4 from backend.pipeline.detectors.tracknet_runner import TrajectoryPoint
5
6
7 def P(frame, x, y, visible=True):
8     return TrajectoryPoint(frame=frame, visible=visible, x=x, y=y)
9
10
11 def test_estimate_play_axis_picks_larger_spread():
12     pts = [
13         P(i, 500 + (i % 2), 50 if i % 2 else 350) for i in range(20)
14     ] # y varies, x ~const
15     axis, net, span = rg.estimate_play_axis(pts)
16     assert axis == "y"
17     assert span > 100
18
19
20 def test_count_net_crossings_counts_sign_changes():
21     pts = [
22         P(0, 0, 10),
23         P(1, 0, 200),
24         P(2, 0, 10),
25         P(3, 0, 200),
26     ] # below/above/below/above
27     assert rg.count_net_crossings(pts, "y", net=100.0, deadband=5.0) == 3
28
29
30 def test_deadband_ignores_jitter_near_net():
31     pts = [
32         P(0, 0, 101),
33         P(1, 0, 99),
34         P(2, 0, 102),
35         P(3, 0, 98),
36     ] # all within deadband of net
37     assert rg.count_net_crossings(pts, "y", net=100.0, deadband=10.0) == 0
38
39
40 def test_invisible_points_are_skipped():
41     pts = [
42         P(0, 0, 10),
43         P(1, 0, 200, visible=False),
44         P(2, 0, 10),
45     ] # invisible middle -> no flip
46     assert rg.count_net_crossings(pts, "y", net=100.0, deadband=5.0) == 0
47
48
49 def test_gate_kills_single_toss_keeps_multi_crossing_rally():
50     fps = 10.0
51     rally = [
52         P(i, 500, 50 if i % 2 == 0 else 250) for i in range(0, 10)
53     ] # alternates -> many crossings
54     toss = [P(20, 500, 50), P(25, 500, 250)] # one crossing
55     pts = rally + toss
56     windows = [(0.0, 1.0), (2.0, 3.0)]
57     gated = rg.gate_windows(pts, windows, fps, min_crossings=2)
58     assert gated[0].kept and gated[0].crossings >= 2
59     assert not gated[1].kept and gated[1].crossings <= 1
60     assert rg.apply_gate(pts, windows, fps, min_crossings=2) == [(0.0, 1.0)]
61
62
63 def test_gate_windows_precomputed_estimate_matches_auto():
64     """Passing a precomputed (axis,net,span) estimate (the per-video reuse the fusion
65     segmenter does) yields identical results to letting gate_windows re-estimate."""
66     fps = 10.0
67     pts = [P(i, 500, 50 if i % 2 == 0 else 250) for i in range(0, 10)] + [

```

```

68         P(20, 500, 50)
69     ]
70     windows = [(0.0, 1.0), (1.5, 2.5)]
71     auto = rg.gate_windows(pts, windows, fps, min_crossings=2)
72     est = rg.estimate_play_axis(pts)
73     reused = rg.gate_windows(pts, windows, fps, min_crossings=2, estimate=est)
74     assert [(g.crossings, g.kept) for g in auto] == [
75         (g.crossings, g.kept) for g in reused
76     ]
77

```

51. assistant (2026-06-29T08:48:02.084Z)

Let me set up the task list and look at the existing tail-trim tests.

52. user (2026-06-29T08:48:05.575Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

53. user (2026-06-29T08:48:10.606Z)

```

257: def test_trim_inactive_tail_drops_slow_drift():
260:     out = TrackNetRunner._trim_inactive_tail(
270: def test_trim_inactive_tail_keeps_continuous_rally_and_is_noop_when_off():
273:     TrackNetRunner._trim_inactive_tail(allfast, 30, 0.3, 0.2) == allfast
276:     TrackNetRunner._trim_inactive_tail(allfast, 0, 0.3, 0.2) == allfast
307:     inactivity_rally_thresh=0.2,
308:     inactivity_min_density=0.3,

```

54. user (2026-06-29T08:48:11.762Z)

```

300         if not p.visible:
301             prev = None # break the trajectory; absent shuttle = not in flight
302             continue
303         if prev is not None:
304             dframes = p.frame - prev.frame
305             if dframes > 0:
306                 dist = ((p.x - prev.x) ** 2 + (p.y - prev.y) ** 2) ** 0.5
307                 velocity = (dist / frame_width) * (fps / dframes)
308                 if velocity > velocity_thresh:
309                     active.append((p.frame, velocity))
310             prev = p
311         return active
312
313     @staticmethod
314     def trajectory_to_action_windows(
315         points: List[TrajectoryPoint],
316         fps: float,
317         frame_width: float,
318         chunk_start: float = 0.0,
319         velocity_thresh: float = 0.02,
320         merge_gap: float = 2.0,
321         min_window_duration: float = 2.0,
322         chunk_index: Optional[int] = None,
323         max_window_duration: float = 0.0,
324         min_split_gap: float = 0.5,
325         window_hard_cap: bool = False,
326         window_inactivity_end: float = 0.0,
327         inactivity_min_density: float = 0.34,
328         inactivity_rally_thresh: float = 0.08,
329         window_blob_fallback: bool = False,
330         window_blob_factor: float = 4.0,
331     ) -> List[Dict[str, Any]]:

```

```

332         """Convert a shuttle trajectory into candidate rally windows.
333
334         A frame is "active" when the shuttle is visible AND its inter-frame
335         displacement (normalised by frame width, per second) exceeds
336         ``velocity_thresh`` ? i.e. the shuttle is in flight. Active frames within
337         ``merge_gap`` seconds of each other are grouped into one window; short
338         windows are dropped. Mirrors HybridYoloSegmenter's windowing so the dict
339         shape feeding the AI-handoff phase is identical.
340
341         ``max_window_duration`` (0 = OFF, the default ? behaviour unchanged): a guard
against
342         the *over-merge* failure where "shuttle moving" is conflated with "rally in
progress"
343         and a single window swallows several rallies + the dead time between them. When
344         set, any
345         window longer than this is recursively SPLIT at its largest internal inactivity
gap (the
346         longest stretch with no in-flight shuttle ? the most likely rally boundary),
provided
347         that gap is at least ``min_split_gap`` seconds. This is the bounded-windowing
fix from
348         ``docs/RALLY_QUALITY_RESEARCH.md`` §6 (W1); it never merges, only cuts, so
recall cannot
349         drop, and it is config-gated default-OFF until measured on the golden set.
350
351         ``window_hard_cap`` (default OFF) makes ``max_window_duration`` a TRUE cap: when
an
352         over-long window has NO internal inactivity gap to split on (a continuously-
active
353         trajectory ? the shuttle never stops moving, e.g. the real-footage
``mahadevpura-1``
354         blob where the gap-splitter alone left a single 174 s window), it is recursively
355         hard-chopped at its temporal midpoint until every piece is ? the cap. Still only
cuts
356         (never merges), so recall cannot drop; gated default-OFF until measured.
357
358         ``window_blob_fallback`` (default OFF ? the R5 last-resort blob splitter):
unlike
359         ``window_hard_cap`` (which midpoint-chops BEFORE the R4 trim, so it fires on
every gap-less
360         over-cap window and over-segments normal clips), this runs AFTER the R4
inactivity trim
361         and
362         force-chops ONLY a genuine *blob* ? a group still longer than
``window_blob_factor`` x
363         ``max_window_duration`` once the gap-split and the principled rally-end trim
have both had
364         their chance (e.g. mahadevpura-1's ~65 s residual ? 11x a 6 s cap; a 7 s rally
is left
365         alone). The blob is midpoint-chopped down to ? the cap, those forced cuts are
logged, and
366         each resulting window is flagged ``forced_cut=True`` ? surfacing the clip as one
that needs
367         rally-STATE cues (net-crossings / two-sided motion), not a bigger cap. Only cuts
(recall
368         can't drop). Requires ``max_window_duration`` > 0; gated default-OFF until
measured.
369
370         """
371         if fps <= 0:
372             fps = 30.0
373         if frame_width <= 0:
374             frame_width = 1280.0
375
376         active = TrackNetRunner._active_frames(
377             points, fps, frame_width, velocity_thresh
378         )

```

```

376
377         if not active:
378             return []
379
380         max_gap_frames = int(merge_gap * fps)
381
382         def _finalize(group: List[Tuple[int, float]]) -> Dict[str, Any]:
383             vels = [v for _, v in group]
384             return {
385                 "start": group[0][0] / fps + chunk_start,
386                 "end": group[-1][0] / fps + chunk_start,
387                 "active_frame_count": len(group),
388                 "peak_velocity": max(vels),
389                 "mean_velocity": sum(vels) / len(vels),
390                 "track_id_count": 1, # single shuttle; kept for window-shape parity
391                 "chunk_index": chunk_index,
392             }
393
394         groups: List[List[Tuple[int, float]]] = []
395         group = [active[0]]
396         for af in active[1:]:
397             if af[0] - group[-1][0] <= max_gap_frames:
398                 group.append(af)
399             else:
400                 groups.append(group)
401                 group = [af]
402         groups.append(group)
403
404         if max_window_duration > 0:
405             groups = TrackNetRunner._split_overlong_groups(
406                 groups,
407                 int(max_window_duration * fps),
408                 int(min_split_gap * fps),
409                 hard_cap=window_hard_cap,
410             )
411
412         # R4 ? rally-END inactivity trim (default OFF). Applied AFTER the cap split so
each rally
413         # piece has its own post-rally drift tail removed: the velocity windowing keeps
a window
414         # "active" while the shuttle drifts/gets-picked-up above threshold, over-
extending the END
415         # (the measured gap vs golden). Trim back to the last frame whose trailing
window is still
416         # dense with motion (rally on); a sustained low-density run = rally over. Only
cuts.
417         if window_inactivity_end > 0:
418             tw = int(window_inactivity_end * fps)
419             groups = [
420                 TrackNetRunner._trim_inactive_tail(
421                     g, tw, inactivity_min_density, inactivity_rally_thresh
422                 )
423                 for g in groups
424             ]
425
426         # R5 ? blob-fallback (default OFF). LAST resort: after the gap-split AND the R4
trim, any
427         # group still longer than the cap is a gap-less, rally-end-signal-less blob;
force-chop it to
428         # ? the cap at the midpoint and flag every piece, so the degenerate single-
window outcome is
429         # avoided and the clip is surfaced (logged) as needing rally-STATE cues, not a
bigger cap.
430         forced_flags = [False] * len(groups)
431         if window_blob_fallback and max_window_duration > 0:

```

```

432         groups, forced_flags = TrackNetRunner._blob_fallback_chop(
433             groups, int(max_window_duration * fps), window_blob_factor
434         )
435
436     windows = []
437     for g, forced in zip(groups, forced_flags):
438         w = _finalize(g)
439         if forced:
440             w["forced_cut"] = True
441         windows.append(w)
442     return [w for w in windows if (w["end"] - w["start"]) >= min_window_duration]
443
444     @staticmethod

```

55. user (2026-06-29T08:48:16.453Z)

```

255
256     # ---- R4: rally-END inactivity trim (drop the post-rally slow-drift tail)
257     ----- #
258     def test_trim_inactive_tail_drops_slow_drift():
259         # fast rally (vel 0.5) frames 0..60, then slow drift (vel 0.03, still > active 0.02)
260         61..200.
261         group = [(i, 0.5) for i in range(0, 61)] + [(i, 0.03) for i in range(61, 201)]
262         out = TrackNetRunner._trim_inactive_tail(
263             group, trail_frames=30, min_density=0.3, rally_thresh=0.2
264         )
265         assert out[0][0] == 0 # start untouched
266         assert out[-1][0] < 100 # the pure-drift tail (frames 100..200) is trimmed off
267         assert (
268             len(group) - len(out) >= 100
269         ) # ~all the slow drift cut (kept only a short decay tail)
270
271     def test_trim_inactive_tail_keeps_continuous_rally_and_is_noop_when_off():
272         allfast = [(i, 0.5) for i in range(0, 100)]
273         assert (
274             TrackNetRunner._trim_inactive_tail(allfast, 30, 0.3, 0.2) == allfast
275         ) # fully-fast: untouched
276         assert (
277             TrackNetRunner._trim_inactive_tail(allfast, 0, 0.3, 0.2) == allfast
278         ) # trail=0: OFF/no-op
279
280     def test_window_inactivity_end_trims_through_api():
281         # one window of fast motion then slow drift (both visible & > velocity_thresh ? one
282         group).
283         fast = [
284             TrajectoryPoint(frame=i, visible=True, x=100 + 40 * (i % 5), y=300)
285             for i in range(0, 90)
286         ]
287         slow = [
288             TrajectoryPoint(frame=i, visible=True, x=100 + (i % 2), y=300)
289             for i in range(90, 300)
290         ]
291         pts = fast + slow
292         off = TrackNetRunner.trajectory_to_action_windows(
293             pts,
294             fps=30.0,
295             frame_width=1280.0,
296             velocity_thresh=0.02,
297             merge_gap=2.0,
298             min_window_duration=0.3,
299         )
300         on = TrackNetRunner.trajectory_to_action_windows(
301             pts,

```



```

301         fps=30.0,
302         frame_width=1280.0,
303         velocity_thresh=0.02,
304         merge_gap=2.0,
305         min_window_duration=0.3,
306         window_inactivity_end=1.0,
307         inactivity_rally_thresh=0.2,
308         inactivity_min_density=0.3,
309     )
310     assert on and off
311     assert on[0]["end"] < off[0]["end"] # R4 trims the drift tail ? earlier END
312     assert on[0]["start"] == off[0]["start"] # START unchanged
313
314

```

56. assistant (2026-06-29T08:49:03.739Z)

Now I have what I need. Key insight that de-risks finding B: the `group` holds only *active* frames, and in *densely-tracked* clips the entry count equals the temporal frame span ? so switching the denominator to a temporal span leaves normal-tracking behavior (and the calibrated `inactivity_min_density=0.34``) unchanged, and only corrects the *sparse-tracking* case the bug is about. The existing tests use contiguous frames, so they'll still pass. Let me apply the fixes.

57. user (2026-06-29T08:49:06.401Z)

The file `c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\backend\pipeline\detectors\rally_gate.py` has been updated successfully. (file state is current in your context ? no need to Read it back)

58. user (2026-06-29T08:49:16.354Z)

The file `c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\backend\pipeline\detectors\rally_gate.py` has been updated successfully. (file state is current in your context ? no need to Read it back)

59. user (2026-06-29T08:49:22.680Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

60. assistant (2026-06-29T08:49:26.010Z)

Now finding B. Updating the docstring and the density denominator to be temporal:

61. user (2026-06-29T08:49:38.827Z)

The file `c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\backend\pipeline\detectors\tracknet_runner.py` has been updated successfully. (file state is current in your context ? no need to Read it back)

62. assistant (2026-06-29T08:50:04.451Z)

Both fixes applied. Now adding regression tests. Let me read the import header of the tracknet test file to match conventions, then add tests to both.

63. user (2026-06-29T08:50:05.478Z)

```

1     """Tests for the TrackNet runner's pure-Python pieces (path translation, CSV
2     parsing, trajectory -> action-window conversion). The WSL subprocess invocation
3     itself is not exercised here ? these run anywhere, no WSL/TensorFlow needed."""
4
5     import os

```

```

6     import subprocess
7     import textwrap
8     from unittest.mock import patch
9
10    from backend.pipeline.detectors.base import DetectorRunner
11    from backend.pipeline.detectors.tracknet_runner import (
12        TrackNetConfig,
13        TrackNetRunner,
14        TrajectoryPoint,
15        to_wsl_mnt_path,

```

64. assistant (2026-06-29T08:50:08.260Z)

Now adding the regression tests:

65. user (2026-06-29T08:50:14.220Z)

The file c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\tests\test_rally_gate.py has been updated successfully. (file state is current in your context ? no need to Read it back)

66. user (2026-06-29T08:50:22.557Z)

The file c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\tests\test_tracknet_runner.py has been updated successfully. (file state is current in your context ? no need to Read it back)

67. user (2026-06-29T08:50:29.282Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

68. assistant (2026-06-29T08:50:33.478Z)

Now updating the plan doc. Let me read the restored copies in the working tree.

69. user (2026-06-29T08:50:34.043Z)

```

1     <!--
2     PROJECT DOC TEMPLATE ? copy this file to docs/<YOUR_PLAN>.md and fill it in.
3     -->
4
5     # Refactor Rally Detection Edge Cases & Parser Stability
6
7     > **Status:** `IN PROGRESS` . **Owner:** `Agent` . **Created:** `2026-06-19` . **Last
updated:** `2026-06-19`
8     > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived`
9     > **Tracking anchors:** $7 progress tracker is the source of truth; indexed in
`docs/README.md`.
10    > **Relation to existing docs:** Extends `RALLY_DETECTION_QUALITY_REPORT.md`.
11
12    ---
13
14    ## 0. TL;DR
15    The rally detection pipeline is largely robust, but a deep code review highlighted three
minor edge cases/bugs: a potential divide-by-zero or zero-spread in `_spread` for tiny
sequences, a density calculation vulnerability for sparse tracks in `_trim_inactive_tail`, and
silent skipping of malformed rows in the CVAT parsing script. This plan outlines the fixes for
these issues to increase stability.
16
17    ## 1. Problem & goal
18    The current implementation of the candidate windowing handles normal inputs extremely
well (including handling blobs), but could fail or behave erratically under specific conditions:
19    - **Zero-spread calculation**: `_spread` in `rally_gate.py` evaluates to `0.0` for

```

arrays of length 2, which could cause a zero deadband if used on short clips.

```

20 - **Sparse trailing frames**: `_trim_inactive_tail` calculates frame density based on
*visible* frames. A long gap followed by a single active frame gives 100% density, artificially
inflating the rally end.
21 - **Silent failure on malformed CSV**: `load_canonical_csv` in `rally_cvat.py` silently
ignores rows without a valid `rally_number`.
22
23 **Goal**: Fix these three issues to harden the rally detection and ingestion pipelines
without changing the existing heuristic thresholds.
24
25 ## 4. Design / architecture
26 The changes will be entirely localized to the identified functions:
27 - `rally_gate.py`: Modify `_spread` to use `math.ceil()` or properly enforce a minimum
subset size before trimming percentiles, ensuring it never inadvertently calculates a spread of
0 for non-identical values.
28 - `tracknet_runner.py`: Update `_trim_inactive_tail` to factor the *temporal length* (or
total expected frames in the trailing window) into the density calculation, rather than purely
relying on `win_count` (the number of visible frames).
29 - `rally_cvat.py`: Introduce `logger.warning` before the `continue` statement in
`load_canonical_csv` to surface skipped rows during CVAT ingestion.
30
31 ## 6. Honest limits ? what this does NOT do
32 This will NOT change the core AI validation (Phase 3) or the existing `velocity_thresh`
values. It solely tightens up boundary calculation edge cases. It will not suddenly "fix"
rallies that the tracking model missed entirely due to lighting or occlusion.
33
34 ## 7. Deliverables & progress tracker ? **source of truth**
35 Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**,
branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.
36
37 | ID | Deliverable | Depends on | Gated? | Status | PR |
38 |----|-----|-----|-----|-----|----|
39 | P0 | Fix `_spread` and `_trim_inactive_tail` edge cases | ? | No | ? | ? |
40 | P1 | Add logging to `load_canonical_csv` for silent skips | ? | No | ? | ? |
41
42 ## 8. Open questions ? owner / external
43 1. Do we want `_trim_inactive_tail` to fall back to the existing behavior if there are
very few frames, or strictly enforce a minimum temporal density? (Assuming strict temporal
density is preferred for accuracy).
44
45 ## 9. Definition of done
46 - [ ] `_spread` bug fixed and tested.
47 - [ ] `_trim_inactive_tail` density calculation corrected.
48 - [ ] Warning log added for malformed rows in `rally_cvat.py`.
49 - [ ] Tests updated and passing.
50
51 ## 10. References
52 **Internal code anchors `[verified]`**
53 - `backend/pipeline/detectors/rally_gate.py`
54 - `backend/pipeline/detectors/tracknet_runner.py`
55 - `src/parsers/cvat/rally_cvat.py`
56 **Internal docs**
57 - `docs/RALLY_DETECTION_CODE_REVIEW.md`
58

```

70. user (2026-06-29T08:50:45.026Z)

The file c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\docs\RALLY_DETECTION_FIXES_PLAN.md has been updated successfully. (file state is current in your context ? no need to Read it back)

71. assistant (2026-06-29T08:50:48.953Z)

That diagnostic is just a spell-checker flagging "tracknet" ? not a real issue. Continuing the doc updates: design notes, tracker, open question, and DoD.

72. user (2026-06-29T08:50:59.411Z)

The file c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\docs\RALLY_DETECTION_FIXES_PLAN.md has been updated successfully. (file state is current in your context ? no need to Read it back)

73. user (2026-06-29T08:51:13.812Z)

The file c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\docs\RALLY_DETECTION_FIXES_PLAN.md has been updated successfully. (file state is current in your context ? no need to Read it back)

74. assistant (2026-06-29T08:51:17.566Z)

Spell-checker noise again (British "behaviour") ? ignoring. Now adding a short resolution banner to the code-review doc, then I'll run the suite.

75. user (2026-06-29T08:51:21.330Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

76. user (2026-06-29T08:51:24.988Z)

```
1      # Rally Detection Code Review
2
3      ## 1. How Rally Detection Works
4
5      The project identifies rallies through a combination of local trajectory tracking (e.g.,
TrackNet or WASB) and a series of heuristic gating/trimming rules, culminating in an AI-based
refinement (Phase 3). Here's a breakdown of the pipeline:
```

77. user (2026-06-29T08:51:31.809Z)

The file c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\docs\RALLY_DETECTION_CODE_REVIEW.md has been updated successfully. (file state is current in your context ? no need to Read it back)

78. assistant (2026-06-29T08:51:34.810Z)

Docs updated. Now running the targeted tests, then the full offline suite + lint per the pre-LGTM gate.

79. user (2026-06-29T08:51:37.441Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

80. user (2026-06-29T08:51:46.475Z)

```
..... [100%]
49 passed in 0.49s
```

81. assistant (2026-06-29T08:51:49.445Z)

Targeted tests pass (including the two new regression tests). Now the full offline suite and lint/type checks per the pre-LGTM gate.

82. user (2026-06-29T08:53:25.480Z)

```
.....SS..... [ 4%]
..... [ 9%]
..... [ 14%]
..... [ 19%]
```

```

..... [ 24%]
..... [ 28%]
..... [ 33%]
..... [ 38%]
..... [ 43%]
..... [ 48%]
..... [ 53%]
..... [ 57%]
..... [ 62%]
.....S.SS..... [ 67%]
.....SSSSS..... [ 72%]
..... [ 77%]
..... [ 81%]
..... [ 86%]
..... [ 91%]
..... [ 96%]
..... [100%]
1483 passed, 12 skipped in 88.93s (0:01:28)

```

```

=====
RESULT: PASS - all tests green
=====

```

83. assistant (2026-06-29T08:53:28.927Z)

Full suite green (1483 passed, 12 skipped). Now lint + type checks matching CI.

84. user (2026-06-29T08:54:18.106Z)

```

All checks passed!
=== ruff format check ===
4 files already formatted
=== mypy ===
backend\sports\__init__.py:9: note: By default the bodies of untyped functions are not checked,
consider using --check-untyped-defs [annotation-unchecked]
backend\providers\__init__.py:9: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
Success: no issues found in 2 source files

```

85. assistant (2026-06-29T08:54:24.813Z)

All checks pass ? ruff clean, format clean, mypy success, full suite green. Committing to the branch (which updates PR #393). Verifying I'm on the right branch first and staging explicit paths.

86. user (2026-06-29T08:54:30.157Z)

```

docs/preserve-rally-detection-review
---ahead of origin/master (should be only my commits)---
0a2dfea docs: preserve rally-detection code review + fixes plan
---staging explicit paths---
M backend/pipeline/detectors/rally_gate.py
M backend/pipeline/detectors/tracknet_runner.py
M docs/RALLY_DETECTION_CODE_REVIEW.md
M docs/RALLY_DETECTION_FIXES_PLAN.md
M tests/test_rally_gate.py
M tests/test_tracknet_runner.py

```

87. user (2026-06-29T08:54:46.702Z)

```

[docs/preserve-rally-detection-review dd598a2] fix(rally): correct _spread zero-collapse and
_trim_inactive_tail sparse density (P0)
6 files changed, 96 insertions(+), 17 deletions(-)
remote: This repository moved. Please use the new location:
remote: https://github.com/KheIsutra/badminton-highlight-indexer.git

```

To <https://github.com/avidullu/badminton-highlight-indexer.git>
0a2dfea..dd598a2 docs/preserve-rally-detection-review -> docs/preserve-rally-detection-review

88. assistant (2026-06-29T08:54:50.006Z)

Pushed. Now updating the PR title/body to reflect that it's now doc + code-fix.

89. user (2026-06-29T08:55:06.371Z)

<https://github.com/Khelsutra/badminton-highlight-indexer/pull/393>

90. user (2026-06-29T08:55:14.027Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

91. assistant (2026-06-29T08:55:27.402Z)

Done. PR #393 is now a single doc + code-fix PR.

What landed in [PR #393](<https://github.com/Khelsutra/badminton-highlight-indexer/pull/393>)

Two commits on `docs/preserve-rally-detection-review`:

1. `0a2dfea` ? the preserved review docs
2. `dd598a2` ? the P0 code fixes + tests + doc status updates

Fixes (both verified still-present before fixing):

- **A** ? [rally_gate.py `\_spread`](backend/pipeline/detectors/rally_gate.py): `math.ceil` on the p95 index ? no more `0.0` collapse on short arrays; large-array behavior unchanged.
- **B** ? [tracknet_runner.py `\_trim\_inactive\_tail`](backend/pipeline/detectors/tracknet_runner.py): temporal-length denominator ? a lone fast sample after a tracking gap no longer holds the rally tail open.

Dense-tracking behavior unchanged by construction, so the calibrated default isn't disturbed.

Verification (local, per the pre-LGTM gate):

- Full offline suite: **1483 passed, 12 skipped** (incl. 2 new regression tests)
- `ruff check` ? `ruff format --check` ? `mypy` ?

Two things to flag honestly

1. **Finding C is NOT in this PR** ? `load_canonical_csv` lives in the sibling `sports-data-collector` repo, so it can't go in #393. It's a small one (add `logger.warning` before the two silent `continue`'s). **Want me to open that separate PR in the sibling repo?**
2. **Owner-gated validation pending for B** ? I can't run the golden corpus locally (real video, your machine). Dense tracking is provably unaffected, but B's end-boundary effect on *sparse real footage* should get a golden-regression run before it's treated as quality-validated. I noted this in both the PR body and the plan doc, so it's not lost.

One more housekeeping note: the branch is still named `docs/preserve-...` even though it now carries code ? harmless, but I can rename it if you'd prefer it read as a fix branch.

92. user (2026-06-29T08:57:49.887Z)

ok so once we merge the PR even then the validation of B can be deferred to later and its recorded?

93. assistant (2026-06-29T08:58:07.642Z)

Good question ? let me verify two things rather than just assert them: (1) that deferring B's validation is genuinely *safe* (not just convenient), and (2) that the deferral is recorded somewhere that survives the merge, not just in the PR body.

94. user (2026-06-29T08:58:10.253Z)

```
backend\config\models.py:190:   window_inactivity_end: float = 1.5
backend\config\models.py:192:   inactivity_min_density: float = 0.30
```

95. user (2026-06-29T08:58:12.234Z)

```
Found 10 files
docs\RALLY_DETECTION_FIXES_PLAN.md
tests\test_tracknet_runner.py
backend\pipeline\detectors\tracknet_runner.py
backend\pipeline\segmenters\trajectory_hybrid.py
backend\eval\windowing.py
backend\config\models.py
docs\RALLY_QUALITY_RESEARCH.md
docs\RALLY_DETECTION_QUALITY_REPORT.md
docs\QUALITY_ITERATIONS.md
docs\R1_MILESTONE_LAUNCH_REPORT.md
```

96. assistant (2026-06-29T08:58:19.761Z)

Important ? I need to correct something. Let me check the actual config defaults and wiring, because this affects whether deferral is truly safe.

97. user (2026-06-29T08:58:20.732Z)

```
178         # HARD-CAP fallback for W1: when True, an over-long window with NO internal
inactivity gap
179         # (a continuously-active trajectory ? e.g. the real-footage mahadevpura-1 174s blob)
is
180         # chopped at its midpoint until ? max_window_duration, making the cap a true upper
bound.
181         # Only cuts (recall can't drop). OFF by default until measured/promoted.
182         window_hard_cap: bool = False
183         # R4 rally-END inactivity trim (s, 0 = OFF). After a rally the shuttle keeps moving
slowly
184         # (picked up / walked) above velocity_thresh, so the window over-extends its END
(the measured
185         # [?end| gap vs golden). Trim the post-rally tail back to the last frame whose
trailing
186         # window stays dense with FAST motion (velocity > inactivity_rally_thresh). Only
cuts.
187         # *** R1 MILESTONE DEFAULT-ON (1.5/0.20/0.30): on top of cap=6, R4 adds a small but
significant
188         # end-precision gain ? ?tolF1 +0.040, sign 9/1/3, p=0.022, [?end| 4.96?3.31s (n=13).
The W1 cap
189         # is the dominant lever; R4 is the marginal increment. See QUALITY_ITERATIONS.md
"R1". ***
190         window_inactivity_end: float = 1.5
191         inactivity_rally_thresh: float = 0.20
192         inactivity_min_density: float = 0.30
193         # R5 blob-fallback (default OFF). LAST-RESORT splitter for the continuously-active
blob: after
194         # the W1 gap-split AND the R4 inactivity trim have each had their chance, a group
STILL longer
195         # than window_blob_factor × max_window_duration (no internal gap, no rally-end
signal ? e.g.
196         # mahadevpura-1's ~65s residual ? 11× a 6s cap) is midpoint-chopped to ? the cap as
a FORCED cut
197         # (logged + flagged on the window) so the degenerate single-window outcome is
avoided and the
198         # clip is surfaced as needing rally-STATE cues, not a bigger cap. Differs from
window_hard_cap
199         # (which chops BEFORE R4 and over-segments normal clips): blob-fallback runs AFTER
R4 and only
```

98. user (2026-06-29T08:58:22.394Z)

```
247:         window_inactivity_end = idx_cfg.window_inactivity_end
248:         inactivity_rally_thresh = idx_cfg.inactivity_rally_thresh
249:         inactivity_min_density = idx_cfg.inactivity_min_density
301:         window_inactivity_end=window_inactivity_end,
302:         inactivity_rally_thresh=inactivity_rally_thresh,
303:         inactivity_min_density=inactivity_min_density,
```

99. assistant (2026-06-29T08:59:21.187Z)

I have to correct something I told you earlier ? and it changes the answer. Let me be precise:

****My earlier claim that "R4 is config-gated default-OFF" was wrong.**** I read the ***function signature*** `default (`window_inactivity_end=0.0`)`, but the ****production config sets it ON****:

```
- [models.py:190-192](backend/config/models.py#L190-L192): `window_inactivity_end = 1.5`,
`inactivity_rally_thresh = 0.20`, `inactivity_min_density = 0.30` ? and the comment says it was
**promoted DEFAULT-ON at the "R1 milestone" (tolF1 +0.040, p=0.022).
- [trajectory_hybrid.py:247-303](backend/pipeline/segmenters/trajectory_hybrid.py#L247-L303)
reads those config values and passes them straight through.
```

So ****R4 runs in production****, which means ****finding B's fix does change production rally end-boundaries**** for any clip with sparse shuttle tracking. It's behavior-neutral only for ***densely*-tracked clips**. So:

Answer to your question, honestly

```
- **Recorded? Yes, durably.** The deferral lives in the in-repo plan doc
(`RALLY_DETECTION_FIXES_PLAN.md` §8 Q2 + §9 DoD), which survives the merge and stays live (not
archived while items are open). I'll also add it to `docs/NEXT_STEPS.md` (the track's
prioritized pickup) so it's where people actually look.
- **Safe to defer *after* merging as-is? Not quite** ? because R4 is ON, merging B silently
shifts end-boundaries on sparse-tracking clips before anyone has measured it against the golden
corpus. That's exactly the kind of unmeasured rally-quality change your conventions say should
land *flag-gated, default-OFF, zero-regression until measured*.
```

Cleanest way to get what you want (merge now + defer validation + recorded)

Put B's new temporal-denominator behind a ****default-OFF flag**** (same pattern as ``window_hard_cap`/`window_blob_fallback``). Then:

- Merging changes ****nothing**** in production (zero regression) ? A's fix + docs land, B's code lands dormant.
- Validation becomes a ****recorded promotion gate****: owner runs the golden regression, and if B holds/improves, flips the flag default-ON in a tiny follow-up.

A's fix stays unconditional ? it's genuinely behavior-neutral in prod (``_spread`` only runs over whole-trajectory point sets, never the tiny-array path).

How do you want to handle B?

100. user (2026-06-29T08:59:52.401Z)

Your questions have been answered: "R4 is default-ON, so B's fix changes production behavior on sparse-tracking clips. How should B land in #393?"="Flag-gate B default-OFF". You can now continue with these answers in mind.

101. assistant (2026-06-29T09:00:10.359Z)

Flag-gating B default-OFF. Let me look at how the existing gated flags (``window_hard_cap``, ``window_blob_fallback``) are wired through config ? segmenter ? function, so I mirror the exact pattern.

102. user (2026-06-29T09:00:12.237Z)

```
240         velocity_thresh = getattr(idx_cfg, self.family).velocity_threshold
241         merge_gap = idx_cfg.dead_time_merge_gap
242         min_window_duration = idx_cfg.min_action_window_duration
243         # Bounded-windowing over-merge fix (W1): 0 = OFF (unchanged). See
IndexingConfig.
244         max_window_duration = idx_cfg.max_window_duration
245         window_min_split_gap = idx_cfg.window_min_split_gap
246         window_hard_cap = idx_cfg.window_hard_cap
247         window_inactivity_end = idx_cfg.window_inactivity_end
248         inactivity_rally_thresh = idx_cfg.inactivity_rally_thresh
249         inactivity_min_density = idx_cfg.inactivity_min_density
250         window_blob_fallback = idx_cfg.window_blob_fallback
251         window_blob_factor = idx_cfg.window_blob_factor
252         handoff_padding = idx_cfg.ai_handoff_padding
253         ai_max_workers = idx_cfg.ai_max_workers
254         log_candidates = idx_cfg.log_yolo_candidates
255         store_candidates = idx_cfg.store_yolo_candidates
256
257         detection_params = {
258             "detector": self.name,
259             "velocity_thresh": velocity_thresh,
260             "merge_gap": merge_gap,
261             "min_action_window_duration": min_window_duration,
262             **runner_params,
263         }
264
265         # --- Phase 1: env healthcheck (fail fast with a clear message) ---
266         ok, msg = runner.healthcheck()
267         if not ok:
268             logger.error("%s detector environment not ready: %s", label, msg)
269             return []
270         logger.info("%s detector env OK: %s", label, msg)
271
272         # --- Phase 2: trajectory -> candidate rally windows ---
273         # Trajectory detectors process the whole video in one pass (they carry
274         # their own frame buffering), so unlike yolo_hybrid we don't pre-chunk.
275         t_start = time.time()
276         out_dir = os.path.join(output_base(self.config), self.family)
277         csv_path = runner.run_predict(video_path, out_dir, video_id=video_id)
278         if not csv_path:
279             logger.error("%s produced no trajectory; aborting.", label)
280             return []
281
282         points = runner.parse_trajectory_csv(csv_path)
283         logger.info(
284             "Parsed %d trajectory points (%d visible).",
285             len(points),
286             sum(1 for p in points if p.visible),
287         )
288
289         all_candidate_windows = runner.trajectory_to_action_windows(
290             points,
291             fps=fps,
292             frame_width=frame_width,
293             chunk_start=0.0,
294             velocity_thresh=velocity_thresh,
295             merge_gap=merge_gap,
296             min_window_duration=min_window_duration,
297             chunk_index=0,
298             max_window_duration=max_window_duration,
299             min_split_gap=window_min_split_gap,
300             window_hard_cap=window_hard_cap,
301             window_inactivity_end=window_inactivity_end,
```

```

302         inactivity_rally_thresh=inactivity_rally_thresh,
303         inactivity_min_density=inactivity_min_density,
304         window_blob_fallback=window_blob_fallback,
305         window_blob_factor=window_blob_factor,
306     )
307     logger.info(
308         "Phase 2 (%s) completed in %.1fs. Candidate windows: %d",
309         label,

```

103. user (2026-06-29T09:00:13.673Z)

```

55:     knobs ``TrackNetRunner.trajectory_to_action_windows`` takes, so a preset *is* the
windowing.
73:     inactivity_min_density: float = 0.34
83:     ``trajectory_to_action_windows`` / ``distill_local`` call sites already speak. (The
97:         "inactivity_min_density": self.inactivity_min_density,
126:     hard_cap=bool(idx.window_hard_cap), # W1 hard-cap fallback (default OFF)
128:     idx.window_inactivity_end
131:     inactivity_min_density=float(idx.inactivity_min_density),
176:     hard_cap=bool(idx.get("window_hard_cap", SERVED.hard_cap)),
177:     inactivity_end=float(idx.get("window_inactivity_end", SERVED.inactivity_end)),
181:     inactivity_min_density=float(
182:         idx.get("inactivity_min_density", SERVED.inactivity_min_density)
208: A thin wrapper over ``TrackNetRunner.trajectory_to_action_windows`` that takes a named
preset
214: return TrackNetRunner.trajectory_to_action_windows(
222:     window_hard_cap=preset.hard_cap,
223:     window_inactivity_end=preset.inactivity_end,
225:     inactivity_min_density=preset.inactivity_min_density,

```

104. assistant (2026-06-29T09:00:20.440Z)

Let me set the plan and read the eval windowing preset module ? the owner validates via that harness, so the flag must reach it too.

105. user (2026-06-29T09:00:24.519Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

106. user (2026-06-29T09:00:28.495Z)

```

1     ""Single-sourced candidate-windowing presets ? the eval?serve foundation fix.
2
3     **The bug this closes.** The offline eval substrate (``distill_local.build_candidates``
?
4     ``fusion_golden`` ? the golden feature JSONs the ablation/compare path re-scores) was
built
5     on a *recall-oriented PROPOSAL* windowing (``vt=0.005, merge_gap=1.0, min_window=0.5``):
6     deliberately permissive so a learned scorer/gate can filter the junk. But the SERVED
pipeline
7     (``TrajectoryHybridSegmenter.process_video``) windows COARSELY (``vt=0.02,
merge_gap=2.0,
8     min_window=2.0`` ? the ``config.indexing`` defaults). A cue measured on the proposal set
is
9     measured on candidates **production never serves**, so a "win" need not transfer. This
is
10    exactly how Gate-A scored **+0.074** on the proposal set yet **?0.008** on the served
set
11    (``scratch/GATE_AB_NUMBERS_AND_IMPLICATIONS.md`` vs ``output/gateA_served_verify/``).
12
13    **The fix.** One module owns the two named windowings as :class:`WindowingPreset`s, and
the
14    SERVED preset is **single-sourced from the same Pydantic config defaults production

```

```

reads**
15     (:class:`backend.config.models.IndexingConfig`) ? eval and serve can no longer drift.
Helpers
16     build candidates under either preset, and :func:`served_windowing_from_config` lifts the
17     *actual* served windowing out of a live config (overrides included), so an A/B can
target the
18     operating point a given deployment truly serves.
19
20     The companion rule ? *a cue is promotable only if it does NOT regress at the SERVED
21     windowing* ? lives in :mod:`backend.eval.promotion` (which composes ``eval_stats`` +
22     ``per_user_gate``). This module is the **what windowing** ; that one is the **promotion
gate**.
23
24     Pure, offline, stdlib + config only. Nothing here changes existing eval behaviour until
a
25     caller opts in (``distill_local`` re-points its module-level presets here, byte-
identically).
26     """
27
28     from __future__ import annotations
29
30     from dataclasses import dataclass
31     from typing import Any, Dict, List, Mapping, Tuple
32
33     from backend.config.models import IndexingConfig
34
35     Interval = Tuple[float, float]
36
37     __all__ = [
38         "WindowingPreset",
39         "SERVED",
40         "PROPOSAL",
41         "PRESETS",
42         "served_windowing_from_config",
43         "build_windows",
44         "windows_to_intervals",
45     ]
46
47
48     @dataclass(frozen=True)
49     class WindowingPreset:
50         """A named (velocity_thresh, merge_gap, min_window_duration) windowing operating
point.
51
52         ``velocity_thresh`` ? normalised per-second shuttle velocity above which a frame
counts as
53         "in flight"; ``merge_gap`` ? seconds within which active frames coalesce into one
window;
54         ``min_window_duration`` ? windows shorter than this are dropped. These are exactly
the three
55         knobs ``TrackNetRunner.trajectory_to_action_windows`` takes, so a preset *is* the
windowing.
56         """
57
58         name: str
59         velocity_thresh: float
60         merge_gap: float
61         min_window_duration: float
62         #: 0 = OFF (default). When > 0, windows longer than this are split at their largest
internal
63         #: inactivity gap ? the bounded-windowing over-merge fix (W1,
RALLY_QUALITY_RESEARCH.md §6).
64         max_window_duration: float = 0.0
65         #: When True, ``max_window_duration`` is a HARD cap ? an over-long window with no
internal

```

```

66         #: inactivity gap is chopped at its midpoint until ? the cap (continuously-active
fix). OFF by default.
67         hard_cap: bool = False
68         #: R4 rally-END inactivity trim (s, 0 = OFF): trim a window's post-rally slow-drift
tail back to
69         #: the last frame whose trailing window stays dense with FAST motion. The measured
rally-end fix.
70         inactivity_end: float = 0.0
71         #: velocity above which motion counts as "rally" (vs slow drift) for the R4 trim;
min fast-fraction.
72         inactivity_rally_thresh: float = 0.08
73         inactivity_min_density: float = 0.34
74         #: R5 blob-fallback (default OFF). After the cap split + R4 trim, midpoint-chop any
group still
75         #: longer than ``blob_factor`` × ``max_window_duration`` (the gap-less, R4-survived
blob ?
76         #: mahadevpura-1's ~11× residual) and flag it. Runs AFTER R4 (vs ``hard_cap``
before) and the
77         #: factor keeps it surgical (a normal long rally is left alone).
78         blob_fallback: bool = False
79         blob_factor: float = 4.0
80
81         def as_tuple(self) -> Tuple[float, float, float]:
82             """(velocity_thresh, merge_gap, min_window_duration)`` ? the legacy tuple
shape the
83             ``trajectory_to_action_windows`` / ``distill_local`` call sites already speak.
(The
84             bounded-windowing knob ``max_window_duration`` is passed separately by
``build_windows``.)"""
85             return (self.velocity_thresh, self.merge_gap, self.min_window_duration)
86
87         def to_dict(self) -> Dict[str, Any]:
88             return {
89                 "name": self.name,
90                 "velocity_thresh": self.velocity_thresh,
91                 "merge_gap": self.merge_gap,
92                 "min_window_duration": self.min_window_duration,
93                 "max_window_duration": self.max_window_duration,
94                 "hard_cap": self.hard_cap,
95                 "inactivity_end": self.inactivity_end,
96                 "inactivity_rally_thresh": self.inactivity_rally_thresh,
97                 "inactivity_min_density": self.inactivity_min_density,
98                 "blob_fallback": self.blob_fallback,
99                 "blob_factor": self.blob_factor,
100             }
101
102
103         def _served_preset_from_defaults() -> WindowingPreset:
104             """The SERVED windowing, lifted from the SAME Pydantic config defaults production
reads.
105
106             ``TrajectoryHybridSegmenter.process_video`` reads exactly these three:
107             - velocity: ``idx_cfg[<family>].velocity_threshold`` (default 0.02 ?
wasb/fusion/tracknet)
108             - merge_gap: ``idx_cfg.dead_time_merge_gap`` (default 2.0)
109             - min window: ``idx_cfg.min_action_window_duration`` (default 2.0)
110
111             Sourcing them from :class:`IndexingConfig`'s defaults (not a hand-typed tuple) is
the whole
112             point: change a served default in ``config/models.py`` and the eval SERVED preset
follows,
113             so the two can never silently diverge again.
114             """
115             idx = IndexingConfig() # construct with defaults ? the served operating point
116             return WindowingPreset(

```

```

117         name="served",
118         velocity_thresh=float(
119             idx.wasb.velocity_threshold
120         ), # == fusion/tracknet default 0.02
121         merge_gap=float(idx.dead_time_merge_gap),
122         min_window_duration=float(idx.min_action_window_duration),
123         max_window_duration=float(
124             idx.max_window_duration
125         ), # W1 bounded-windowing (default 0=OFF)
126         hard_cap=bool(idx.window_hard_cap), # W1 hard-cap fallback (default OFF)
127         inactivity_end=float(
128             idx.window_inactivity_end
129         ), # R4 rally-end trim (default 0=OFF)
130         inactivity_rally_thresh=float(idx.inactivity_rally_thresh),
131         inactivity_min_density=float(idx.inactivity_min_density),
132         blob_fallback=bool(idx.window_blob_fallback), # R5 blob-fallback (default OFF)
133         blob_factor=float(idx.window_blob_factor),
134     )
135
136
137     #: The COARSE windowing production actually serves (config defaults; ~0.02/2.0/2.0). A
138     #: cue is
139     #: only promotable if it holds up HERE ? the operating point real users get.
140     SERVED: WindowingPreset = _served_preset_from_defaults()
141
142     #: The recall-oriented PROPOSAL windowing the offline substrate is built on
143     #: (deliberately
144     #: permissive: propose many, let the classifier/gate filter). NOT what production serves
145     #: ? a
146     #: win here must be re-confirmed on SERVED before promotion. Mirrors the historical
147     #: ``distill_local.PROPOSAL`` tuple (0.005/1.0/0.5) so the existing golden JSONs stay
148     valid.
149     PROPOSAL: WindowingPreset = WindowingPreset(
150         name="proposal", velocity_thresh=0.005, merge_gap=1.0, min_window_duration=0.5
151     )
152
153     #: Name ? preset, so a CLI / config string ("served" | "proposal") selects one.
154     PRESETS: Dict[str, WindowingPreset] = {SERVED.name: SERVED, PROPOSAL.name: PROPOSAL}
155
156
157     def served_windowing_from_config(config: Any, family: str = "wasb") -> WindowingPreset:
158         """The SERVED windowing a *specific* live config serves (overrides honoured).
159
160         Reproduces ``TrajectoryHybridSegmenter.process_video``'s knob reads exactly, against
161         an
162         arbitrary config (the typed :class:`AppConfig` or a plain dict) and segmenter
163         ``family``
164         (``"wasb"`` / ``"fusion"`` / ``"tracknet"``). Use this when a deployment overrode a
165         served
166         default in ``config.json`` ? the module-level :data:`SERVED` is the *default*
167         operating point;
168         this is *this config's* operating point. Falls back to the served defaults for any
169         absent key.
170
171         """
172         idx = _as_mapping(config, "indexing")
173         family_cfg = _as_mapping(idx, family)
174         return WindowingPreset(
175             name=f"served:{family}",
176             velocity_thresh=float(
177                 family_cfg.get("velocity_threshold", SERVED.velocity_thresh)
178             ),
179             merge_gap=float(idx.get("dead_time_merge_gap", SERVED.merge_gap)),
180             min_window_duration=float(
181                 idx.get("min_action_window_duration", SERVED.min_window_duration)
182             ),

```

```

173         max_window_duration=float(
174             idx.get("max_window_duration", SERVED.max_window_duration)
175         ),
176         hard_cap=bool(idx.get("window_hard_cap", SERVED.hard_cap)),
177         inactivity_end=float(idx.get("window_inactivity_end", SERVED.inactivity_end)),
178         inactivity_rally_thresh=float(
179             idx.get("inactivity_rally_thresh", SERVED.inactivity_rally_thresh)
180         ),
181         inactivity_min_density=float(
182             idx.get("inactivity_min_density", SERVED.inactivity_min_density)
183         ),
184         blob_fallback=bool(idx.get("window_blob_fallback", SERVED.blob_fallback)),
185         blob_factor=float(idx.get("window_blob_factor", SERVED.blob_factor)),
186     )
187
188
189     def _as_mapping(obj: Any, key: str) -> Mapping[str, Any]:
190         """`obj[key]` as a plain mapping, whether `obj` is a dict or a typed config
section
191         (`AppConfig.get` returns nested sections as dicts). Empty mapping if absent/non-
mapping."""
192         if obj is None:
193             return {}
194         if isinstance(obj, Mapping):
195             sub = obj.get(key, {})
196         elif hasattr(obj, "get"):
197             sub = obj.get(key, {})
198         else:
199             sub = getattr(obj, key, {})
200         return sub if isinstance(sub, Mapping) else {}
201
202
203     def build_windows(
204         points: List[Any], fps: float, frame_width: float, preset: WindowingPreset
205     ) -> List[Dict[str, Any]]:
206         """Trajectory `points` ? candidate windows under `preset` (the one windowing
call site).
207
208         A thin wrapper over `TrackNetRunner.trajectory_to_action_windows` that takes a
named preset
209         instead of three loose floats, so eval candidate-building and the served path
reference the
210         SAME knobs by name. Imported lazily to keep this module import-light (no detector
deps)."""
211         from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
212
213         vt, mg, mwd = preset.as_tuple()
214         return TrackNetRunner.trajectory_to_action_windows(
215             points,
216             fps=fps,
217             frame_width=frame_width,
218             velocity_thresh=vt,
219             merge_gap=mg,
220             min_window_duration=mwd,
221             max_window_duration=preset.max_window_duration,
222             window_hard_cap=preset.hard_cap,
223             window_inactivity_end=preset.inactivity_end,
224             inactivity_rally_thresh=preset.inactivity_rally_thresh,
225             inactivity_min_density=preset.inactivity_min_density,
226             window_blob_fallback=preset.blob_fallback,
227             window_blob_factor=preset.blob_factor,
228         )
229
230
231     def windows_to_intervals(windows: List[Dict[str, Any]]) -> List[Interval]:

```

```

232         """`[{start, end, ...}]` ? `[ (start, end)]` ? the interval shape the scorer/gate
speak."""
233         return [(w["start"], w["end"]) for w in windows]
234

```

107. assistant (2026-06-29T09:00:40.679Z)

This module is the promotion harness ? to let the owner validate, the flag must reach it too. I'll plumb `inactivity\_temporal\_density` (bool, default `False`) through all five layers, mirroring `window\_hard\_cap`. Starting with the core function ? making the temporal denominator conditional, defaulting to legacy behavior.

108. user (2026-06-29T09:00:42.793Z)

```

445     def _trim_inactive_tail(
446         group: List[Tuple[int, float]],
447         trail_frames: int,
448         min_density: float,
449         rally_thresh: float,
450     ) -> List[Tuple[int, float]]:
451         """R4: trim the post-rally drift tail. After a rally the shuttle keeps moving
*slowly*
452         (picked up / walked / knock-up) above ``velocity_thresh``, so the velocity
windowing
453         over-extends the END. A frame is "fast" (real rally motion) iff its velocity >
``rally_thresh``
454         (higher than the active threshold). The rally is "on" at frame ``i`` while its
trailing
455         ``trail_frames`` window holds >= ``min_density`` fraction of fast frames; trim
back to the
456         LAST such frame ? everything after is sustained slow drift (rally over). Only
cuts (recall
457         can't rise); a window whose tail never goes slow is untouched, and if no fast-
dense point
458         exists at all the group is kept whole (fail-safe, no over-trim). O(n) two-
pointer; the START
459         is left alone (already ~Gemini-accurate per the n=2 boundary-error finding).
460
461         Density is taken over the trailing window's TEMPORAL length (frames of footage),
not the
462         count of active samples in it: ``group`` carries only active frames, so under
sparse
463         tracking a single fast sample after a long detection gap would otherwise read as
100% dense
464         and hold the tail open. With dense tracking the active-sample count equals the
temporal
465         span, so well-tracked clips (and the calibrated ``min_density`` default) are
unaffected; the
466         correction only bites the sparse case."""
467         if trail_frames <= 0 or len(group) < 2:
468             return group
469         frames = [f for f, _ in group]
470         fast = [1 if v > rally_thresh else 0 for _, v in group]
471         lo = 0
472         fast_sum = 0
473         last_dense = -1
474         for i in range(len(frames)):
475             fast_sum += fast[i]
476             while frames[lo] <= frames[i] - trail_frames:
477                 fast_sum -= fast[lo]
478                 lo += 1
479             # Temporal window length: full ``trail_frames`` once that much footage has
elapsed,
480             # else the footage seen so far. NOT ``i - lo + 1`` (active-sample count),
which sparse

```

```

481         # tracking inflates.
482         win_span = min(trail_frames, frames[i] - frames[0] + 1)
483         if win_span > 0 and fast_sum / win_span >= min_density:
484             last_dense = i
485         return group[: last_dense + 1] if last_dense >= 0 else group
486

```

109. user (2026-06-29T09:01:03.237Z)

The file c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\backend\pipeline\detectors\tracknet_runner.py has been updated successfully. (file state is current in your context ? no need to Read it back)

110. assistant (2026-06-29T09:01:06.593Z)

Now the `trajectory\_to\_action\_windows` signature + call site for `\_trim\_inactive\_tail`:

111. user (2026-06-29T09:01:09.958Z)

The file c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\backend\pipeline\detectors\tracknet_runner.py has been updated successfully. (file state is current in your context ? no need to Read it back)

112. user (2026-06-29T09:01:16.371Z)

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113. assistant (2026-06-29T09:01:22.391Z)

Now the config model ? adding the flag in the R4 block (default-OFF):

114. user (2026-06-29T09:01:23.263Z)

```

190         window_inactivity_end: float = 1.5
191         inactivity_rally_thresh: float = 0.20
192         inactivity_min_density: float = 0.30
193         # R5 blob-fallback (default OFF). LAST-RESORT splitter for the continuously-active
blob: after
194         # the W1 gap-split AND the R4 inactivity trim have each had their chance, a group
STILL longer
195         # than window_blob_factor × max_window_duration (no internal gap, no rally-end
signal ? e.g.

```

115. user (2026-06-29T09:01:31.987Z)

The file c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\backend\config\models.py has been updated successfully. (file state is current in your context ? no need to Read it back)

116. assistant (2026-06-29T09:01:35.573Z)

Now wiring through `trajectory\_hybrid.py` (production read + pass-through):

117. user (2026-06-29T09:01:39.537Z)

The file c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\backend\pipeline\segmenters\trajectory_hybrid.py has been updated successfully. (file state is current in your context ? no need to Read it back)

118. user (2026-06-29T09:01:46.355Z)

The file c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\backend\pipeline\segmenters\trajectory_hybrid.py has been updated successfully. (file state is current in your context ? no need to Read it back)

119. assistant (2026-06-29T09:01:49.339Z)

Now the eval windowing preset ? field + `to\_dict` + both config-lift functions + `build\_windows`:

120. user (2026-06-29T09:01:54.055Z)

The file c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\backend\eval\windowing.py has been updated successfully. (file state is current in your context ? no need to Read it back)

121. user (2026-06-29T09:02:00.106Z)

The file c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\backend\eval\windowing.py has been updated successfully. (file state is current in your context ? no need to Read it back)

122. user (2026-06-29T09:02:09.841Z)

The file c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\backend\eval\windowing.py has been updated successfully. (file state is current in your context ? no need to Read it back)

123. user (2026-06-29T09:02:16.258Z)

The file c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\backend\eval\windowing.py has been updated successfully. (file state is current in your context ? no need to Read it back)

124. user (2026-06-29T09:02:22.993Z)

The file c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\backend\eval\windowing.py has been updated successfully. (file state is current in your context ? no need to Read it back)

125. assistant (2026-06-29T09:02:33.468Z)

Now updating the B test to reflect the flag (default-OFF = legacy holds tail; flag-ON = trims), and let me check the config-field tests that might assert an exact schema.

126. user (2026-06-29T09:02:44.743Z)

The file c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\tests\test_tracknet_runner.py has been updated successfully. (file state is current in your context ? no need to Read it back)

127. user (2026-06-29T09:02:50.052Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

128. assistant (2026-06-29T09:02:54.241Z)

Let me run the affected tests now to catch any schema-snapshot breakage from the new config field before I finalize docs.

129. user (2026-06-29T09:03:03.675Z)

..... [72%]

[100%]

99 passed in 3.05s

130. assistant (2026-06-29T09:03:08.366Z)

Passing. Let me check for any test that snapshots the windowing preset dict or config schema, then run the broader eval/windowing tests.

131. user (2026-06-29T09:03:09.620Z)

Found 11 files

```
tests\test_tracknet_runner.py
tests\test_variant_tuning.py
tests\test_windowing.py
tests\test_rally_seg_eval_report.py
tests\test_rally_seg_eval.py
tests\test_personalization_serving.py
tests\test_nightly_regression.py
tests\test_execution_policy.py
tests\test_experiment.py
tests\test_eval_harness.py
tests\test_ablation.py
```

132. user (2026-06-29T09:03:40.424Z)

Exit code 1

```
    "inactivity_min_density": 0.30,
    "blob_fallback": False,
    "blob_factor": 4.0,
}
```

E AssertionError: assert {'blob_factor...nd': 1.5, ...} == {'blob_factor...nd': 1.5, ...}

E

E Omitting 11 identical items, use -vv to show

E Left contains 1 more item:

E {'inactivity_temporal_density': False}

E Use -v to get more diff

tests\test_windowing.py:115: AssertionError

===== short test summary info =====

FAILED tests/test_windowing.py::test_preset_roundtrip_dict - AssertionError: ...

1 failed, 151 passed, 2 skipped in 22.22s

133. assistant (2026-06-29T09:03:44.612Z)

One snapshot test asserts the exact `to\_dict` keys. Let me update it to include the new field.

134. user (2026-06-29T09:03:45.617Z)

```
95
96
97     def test_served_windowing_from_config_falls_back_to_defaults_on_missing_keys():
98         assert W.served_windowing_from_config({}, "wasb").as_tuple() == W.SERVED.as_tuple()
99         assert (
100             W.served_windowing_from_config(None, "wasb").as_tuple() == W.SERVED.as_tuple()
101         )
102
103
104     def test_served_windowing_from_config_honors_r5_blob_fallback():
105         # The R5 blob-fallback + its magnitude gate lift out of a live config (so the
eventual default
106         # flip and any per-deployment override take effect on the served path).
107         d = {"indexing": {"window_blob_fallback": True, "window_blob_factor": 5.0}}
108         p = W.served_windowing_from_config(d, "wasb")
109         assert p.blob_fallback is True and p.blob_factor == 5.0
```

```

110
111
112     def test_preset_roundtrip_dict():
113         # R1 milestone FLIPPED ON (2026-06-18): the SERVED defaults now carry W1 cap=6 + R4.
114         d = W.SERVED.to_dict()
115         assert d == {
116             "name": "served",
117             "velocity_thresh": 0.02,
118             "merge_gap": 2.0,
119             "min_window_duration": 2.0,
120             "max_window_duration": 6.0,
121             "hard_cap": False,
122             "inactivity_end": 1.5,
123             "inactivity_rally_thresh": 0.20,
124             "inactivity_min_density": 0.30,

```

135. user (2026-06-29T09:03:49.524Z)

```

124         "inactivity_min_density": 0.30,
125         "blob_fallback": False,
126         "blob_factor": 4.0,
127     }
128
129
130     def test_served_preset_is_r1_milestone_default():
131         # R1 milestone FLIPPED ON (2026-06-18, PR #204): W1 bounded windowing (cap=6) + the
132         # rally-END inactivity trim are now the shipped SERVED defaults (the smallest safe
133         # flip, n=13 baseline?milestone ?tolF1 +0.222, sign 12/0, p=0.0005). The W1.5 hard-
134         cap, the

```

136. user (2026-06-29T09:03:55.895Z)

The file c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\tests\test_windowing.py has been updated successfully. (file state is current in your context ? no need to Read it back)

137. assistant (2026-06-29T09:04:03.030Z)

Let me add a test locking in the eval-harness override plumbing for the new flag (mirroring the blob-fallback test):

138. user (2026-06-29T09:04:07.226Z)

The file c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\tests\test_windowing.py has been updated successfully. (file state is current in your context ? no need to Read it back)

139. assistant (2026-06-29T09:04:10.949Z)

Now correcting the docs. The plan doc §4 B bullet currently has my wrong "default-OFF" claim and describes B as already done unconditionally ? both need fixing.

140. user (2026-06-29T09:04:20.114Z)

The file c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\docs\RALLY_DETECTION_FIXES_PLAN.md has been updated successfully. (file state is current in your context ? no need to Read it back)

141. assistant (2026-06-29T09:04:25.142Z)

Spell-check noise. Now updating the header note, the §7 tracker, and §8 Q2 to reflect the flag-gated promotion path.

142. user (2026-06-29T09:04:26.121Z)

```
12      > **2026-06-29 update:** the three findings were re-verified against current `master`  
(post the  
13      > 38-commit advance that touched both `rally_gate.py` and `tracknet_runner.py`) ? **all  
three were  
14      > still present and unfixed.** P0 (the two in-repo edge cases) is **implemented + unit-  
tested in this  
15      > same PR (#393)**. P1 lives in the sibling `sports-data-collector` repo, so it ships as  
a **separate  
16      > PR there** (cannot be in #393).  
17  
18      ---  
19  
20      ## 0. TL;DR  
21      The rally detection pipeline is largely robust, but a deep code review highlighted three  
minor edge cases/bugs: a potential divide-by-zero or zero-spread in `_spread` for tiny  
sequences, a density calculation vulnerability for sparse tracks in `_trim_inactive_tail`, and  
silent skipping of malformed rows in the CVAT parsing script. This plan outlines the fixes for  
these issues to increase stability.
```

143. user (2026-06-29T09:04:35.147Z)

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144. assistant (2026-06-29T09:04:38.574Z)

Now the §7 tracker and §8 Q2:

145. user (2026-06-29T09:04:39.731Z)

```
44      dense tracking the two denominators are identical ? so flipping the flag is a **no-op  
on  
45      well-tracked clips** and changes only the sparse-tracking case. **Correction  
(2026-06-29):** R4 is  
46      **default-ON in production** (`window_inactivity_end = 1.5`, promoted at the R1  
milestone ? an  
47      earlier draft of this doc wrongly called it default-OFF), so the fix is a *behaviour  
change* on  
48      sparse footage and ships behind the new `inactivity_temporal_density` flag (**default-  
OFF**, plumbed  
49      config ? `trajectory_hybrid` ? eval `WindowingPreset`), to be promoted only after a  
golden-set  
50      measurement at the SERVED operating point (§8 Q2).  
51      - `rally_cvat.py` ? **separate PR** (sibling `sports-data-collector` repo, not this  
one): introduce a  
52      `logger.warning` before each `continue` in `load_canonical_csv` so skipped/malformed  
rows are  
53      surfaced during ingestion.  
54  
55      ## 6. Honest limits ? what this does NOT do  
56      This will NOT change the core AI validation (Phase 3) or the existing `velocity_thresh`  
values. It solely tightens up boundary calculation edge cases. It will not suddenly "fix"  
rallies that the tracking model missed entirely due to lighting or occlusion.  
57  
58      ## 7. Deliverables & progress tracker    ? **source of truth**  
59      Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**,  
branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.  
60  
61      | ID | Deliverable | Depends on | Gated? | Status | PR |  
62      |----|-----|-----|-----|-----|----|  
63      | P0 | Fix `_spread` and `_trim_inactive_tail` edge cases | ? | No | ? | #393 |
```

```

64      | P1 | Add logging to `load_canonical_csv` for silent skips | ? | No | ? (sibling repo)
65      | ? |
66      ## 8. Open questions ? owner / external
67      1. ~~Do we want `_trim_inactive_tail` to fall back to the existing behavior if there are
very few
68      frames, or strictly enforce a minimum temporal density?~~ **Resolved (#393):** strict
temporal
69      density. Because `group` carries only active frames, the temporal denominator equals
the
70      sample-count denominator under dense tracking, so this is behaviour-preserving for
well-tracked
71      clips and only corrects the sparse case ? no separate few-frames fallback needed.
72      2. **Owner validation pending:** the P0 fixes preserve dense-tracking behaviour *by
construction* and
73      pass the offline unit suite, but the R4 trim's end-boundary effect on sparse real
footage has NOT
74      been measured against the golden corpus (owner-only, real-video). Recommend a golden-
regression run
75      before this is treated as quality-validated for the rally-detection track.
76
77      ## 9. Definition of done
78      - [x] `_spread` bug fixed and tested.

```

146. user (2026-06-29T09:04:55.015Z)

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147. user (2026-06-29T09:04:59.765Z)

```

80
81      ## 9. Definition of done
82      - [x] `_spread` bug fixed and tested.
83      - [x] `_trim_inactive_tail` density calculation corrected.
84      - [ ] Warning log added for malformed rows in `rally_cvat.py`. *(sibling-repo PR,
separate)*
85      - [x] Tests updated and passing *(offline suite; golden-corpus regression is owner-gated
? see Q2)*.
86
87      ## 10. References
88      **Internal code anchors `[verified]`:**
89      - `backend/pipeline/detectors/rally_gate.py`
90      - `backend/pipeline/detectors/tracknet_runner.py`
91      - `src/parsers/cvat/rally_cvat.py`

```

148. user (2026-06-29T09:05:07.954Z)

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149. assistant (2026-06-29T09:05:11.701Z)

Now the code-review banner correction and adding the promotion gate to NEXT_STEPS. Let me update the banner and check NEXT_STEPS structure.

150. user (2026-06-29T09:05:18.916Z)

The file c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\docs\RALLY_DETECTION_CODE_REVIEW.md has been updated successfully. (file state is current in your context ? no need to Read it back)

151. assistant (2026-06-29T09:05:22.111Z)

Now let me record the promotion gate in NEXT_STEPS.md (the track's prioritized pickup). Let me find where to add it.

152. user (2026-06-29T09:05:24.032Z)

```
1:# Next steps (live) ? refreshed 2026-06-21
3:> **How to read:** this top section is the **current cross-track state + prioritized pending
list** (refreshed 2026-06-21, replacing the stale 2026-06-13 header). The detailed **owned-model
& rally-quality track** plan (2026-06-13) is **preserved below, still valid** ? it is now *one
of several* active tracks. Live blow-by-blow: memory `current-state.md`. Doc lifecycle/hygiene:
`docs/DOC_STATUS.md`.
5:## ? Where we are (2026-06-21)
6:[Omitted long matching line]
8:## ? Prioritized pending ? all tracks
11:[Omitted long matching line]
37:## Owned-model & rally-quality track ? detail (2026-06-13 plan; still valid)
39:**Updated:** 2026-06-13 (**rally-detection quality track**: fusion + experiment-harness
design docs merged,
40:owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior: 2026-06-10
(audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**:
49:`current-state.md` / `session-handoff.md`. This file = the prioritized "what to do next,"
refreshed each session.
51:## ? Next-session / GPU-box pickup (2026-06-15)
55:1. **[owner / GPU box]** `python -m backend.eval.fusion_golden --manifest
output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-
data\features\2026-06-15-n6" ? produces `*_golden_features.json` for the 6 golden videos. (GPU
+ WSL + videos + trajectory CSVs + `rallies.csv` labels are owner-side only.)
56:2. Artifacts land in the **shared cloud folder** `C:\Users\avidu\OneDrive\rally-
annotator\golden-data\` (OneDrive auto-syncs both ways). Convention + workflow:
**[`GOLDEN_DATA_SHARING.md`](data_pipeline/GOLDEN_DATA_SHARING.md)**; tracked in **#175**.
66:## Where we are (one paragraph)
68:behind a CI-enforced import wall; the featurizer is single-sourced
(`backend/features/rally_seq.py`, train==serve);
75:## The quality table that now drives strategy (IoU 0.5 vs human golden labels)
84:wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game
pickups + dead-time merges,
90:## Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation in
progress)
91:**This is the track a desktop/local agent is slated to pick up.** Design docs landed (PR #112
+ session). All implementation is **owner-gated** (per explicit in the plan); each step = ONE PR
off `master`.
98:[Omitted long matching line]
107: 1. **"First-mile" paper framing (highest-value):** stroke-classifiers (BST/DSTA) + the big
datasets (ShuttleSet ~36k strokes, Fine-Badminton ~27k actions) all **assume a pre-trimmed rally
clip already exists** ? we solve the ignored prerequisite. Pitch the paper as "the missing
untrimmed-video preprocessing pipeline that lets BST/MonoTrack run on raw footage."
111: 5. **Court polygons ? pose ? the rally-END lever:** masking on-court players can resolve
*occluded end-of-rally states* (the #1 remaining gap vs Gemini), not just spectator noise.
116:## Prioritized next steps
150:## Pointers / artifacts
155: manifest reviewable). Featurizer: `backend/features/rally_seq.py` (FEATURE_SCHEMA v1).
```

153. user (2026-06-29T09:05:27.379Z)

```
8      ## ? Prioritized pending ? all tracks
9      **P0 ? highest leverage / unblocks the most**
10     0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few
recordings lying around ? highlights, from a page, no SSH".** Today the cloud flow is **SSH/CLI-
only** (`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12 does NOT fix it.** Needs (a separate khelsutra
`web/` SPA + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable
**service** (M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), **(ii)**
the **SPA ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ?
poll `/api/jobs` ? reel; ZERO engine code), **(iii)** flip **M9 edge-auth ON** (merged #290,
```

default-OFF) to gate alpha tenants, ***(iv)** a **compute PICKER in the UI**** ("your machine vs our cloud", D13/D15). The engine seems make it ***possible***; the SPA makes it ***usable***. Full record: **[[ui-driven-cloud-serving-gap]]**.

11 0b. ****? [packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable batteries-included app ? `docs/PACKAGING\_PLAN.md`.**** The LIGHT-tier engine is bundle-ready: **`pip install` ? `indexer --app` serves the **bundled SPA** single-origin, ****torch-free****, with app-home state / ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades (P0a?e, P1a/b, P2-launcher/P2a/P2d/P2e). ****? Next (fresh focused pushes):** ***(1)** P3 **GPU validation + screencast** (L4, ~\$0.50, DELETE-after, \$100 cap ? **`[[gcp-vm-always-delete]]`**; ? native-WASB 0-rally tuned-config blocker), ****(2)** P2b-install (uv/script + truly-local default config), ***(3)** P2c first-run wizard (cross-repo khelsutra SPA from `/api/capabilities`), then P3-cloud (Cloud Run) + P4 (ONNX/train/annotate = north star). Memory **`[[packaging-app-plan]]` / **`[[next-session-packaging-app]]`**.********

12 1. ? **``COMPUTE_DECOUPLED_SERVING` M0?M9 LARGELY SHIPPED**** (2026-06-21, ADR-014 ratified). M1?M4 (#279/#280/#282/#283), ****M5 #286, M6 #287, M8 #288, M9-auth #290**** all merged (default-OFF); the ****real M7 L4 run**** validated the worker (22 rallies, VM deleted, \$0). All \$8 gating Qs locked (D7?D17). ****Owner residual**** (not blocking Claude's default-OFF code): counsel ToS/DPA (M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12 real), CF team-domain/AUD + ingress-lock; real cloud verification authorized within a ****\$100/?10k cap**** (D17).

13 2. ****[owner, ongoing] Grow the golden corpus** ? multi-court badminton + ?3 matches/sport (TT first). The single highest-leverage move; the quality track ***and*** full-corpus training feed on it.**

14 3. ****[owner + Claude . GPU] Full-corpus training (n?5)** ? a real **`weights/<ver>/`** (the **`0.1.0-l4`** bundle is n=2 plumbing-only).**

15

16 ****P1 ? near-term, mostly Claude-doable (default-OFF, green-gated)****

17 4. ****[Claude] `COMPUTE\_DECOUPLED\_SERVING` batch ? REMAINING: M11 + M12**** (M5/M6/M8/M9-auth ? merged this session). ****M11** = **`backend/flywheel.py`** capture?store?shadow-eval?goldenize plumbing reusing **`workspace.py::for\_video`** + **`backend/eval/experiment.compare`** + **`backend/eval/promotion.py::promote\_if\_served\_safe`**, behind **`flywheel.\*`=OFF** + a faked **`consent\_attested`** (? privacy-sensitive ? give it fresh focus). ****M12** = **`backend/storage/gdrive\_api.py`** **`GDriveApiStorage(StorageBackend)`** + a **`gdrive-api://`** scheme (regex already allows the hyphen; add to **`\_KNOWN`** + **`\_backend\_class`** + the StorageConfig Literal) against an INJECTED fake Drive (mirror **`gcs.py`** DI); keep the mount-only **`gdrive://`** untouched. ? ****M9-consent cols = a new v7 migration**** (M8 took v6). M9-consent ToS/DPA text = owner/counsel.****

18 5. ****[Claude] Per-video workspace P1?P6**** (P1 v6 DB migration ? P2 worker ? P3 per-video routing of **`detector/`**+**`tmp/`** ***[the **`output/chunks\_`** de-hardcode base = ? done via #282]*** ? P4 ? P5 Launch-Report flip ? P6 GPU parity). ? P1's v6 migration coordinates with the M8 v6 cost-ledger migration ? author the version bump once.

19 6. ? ****Ops-Agent codified (#291)** ? **`deploy/gcp/create\_gpu\_vm.sh`** (the one-command VM helper) always bakes the Ops-Agent startup-script; provision.sh grants the APIs+SA roles. PROVEN active on a fresh VM 2026-06-21. (**[[gcp-ops-agent-enable]]**)**

20 7. ****[owner] Set the ship-gate target (#172)** ? needs corpus growth.**

21 8. ****[owner-side GPU] R-series: promote-or-reject **`R2`** tolF1 as the gate**** (the ablation) ? needs GPU-extracted golden features.

22

23 ****P2 ? later / gated / deferred****

24 9. Rally-quality ****P3/P4**** (person-fusion LOVO, audio cue) ? needs GPU golden features.

25 10. ****M0.4 scorer swap**** ? TemporalMaxer (per the 2026-06-17 external review).

26 11. ****Multisport**** (table-tennis first, then pickleball).

27 12. Serving ****M8/M9**** ? per-video cost ledger, edge auth, DPA/consent (within **`COMPUTE\_DECOUPLED\_SERVING`**; owner/legal-gated).

28 13. ****WASB-C3**** weight provenance (gates ***sharing*** a trained model) ? legal.

29 14. ****Rename repo off "badminton"**** (owner, BACKLOG).

30 15. Owner housekeeping: retire the shared ****DO droplet**** + ****DO token****; collector ****`md5`**** export (collector repo).

31 16. ****PMF field validation**** (owner-side, business track).

32

33 ****Doc-hygiene (tracked in **`docs/DOC\_STATUS.md`** \$4):** ? #1 per-video reconcile + ? #2 this refresh; ****pending:**** field/PMF/Roora ? archives + a data-layer **`docs/`** dir (owner boundary confirm); confirm the COMPLETED top-level stubs; verify **`HOSTING\_PLAN/`UI_PROPOSAL`** currency.**

34

35 ---

154. user (2026-06-29T09:05:40.410Z)

The file c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\docs\NEXT_STEPS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

155. user (2026-06-29T09:05:46.507Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

156. assistant (2026-06-29T09:05:49.346Z)

Now the full offline suite + lint/type checks before committing.

157. user (2026-06-29T09:07:12.472Z)

```
[mov,mp4,m4a,3gp,3g2,mj2 @ 000001fecb6bbac0] moov atom not found
[mov,mp4,m4a,3gp,3g2,mj2 @ 000001fecb6b4540] moov atom not found
[mov,mp4,m4a,3gp,3g2,mj2 @ 000001fecb6b4540] moov atom not found
=====
RESULT: PASS - all tests green
=====
```

158. assistant (2026-06-29T09:07:16.585Z)

Suite green (the `moov atom` lines are expected ffmpeg stderr from a mocked-video test). Now lint + types on all changed files.

159. user (2026-06-29T09:07:25.873Z)

```
All checks passed!
=== format check ===
Would reformat: backend\pipeline\detectors\tracknet_runner.py
Would reformat: tests\test_tracknet_runner.py
2 files would be reformatted, 6 files already formatted
=== mypy ===
backend\sports\__init__.py:9: note: By default the bodies of untyped functions are not checked,
consider using --check-untyped-defs [annotation-unchecked]
backend\providers\__init__.py:9: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
Success: no issues found in 5 source files
```

160. assistant (2026-06-29T09:07:28.788Z)

Two files need formatting. Let me apply ruff format and re-verify.

161. user (2026-06-29T09:07:34.665Z)

```
2 files reformatted
=== re-check ===
2 files already formatted
All checks passed!
```

162. assistant (2026-06-29T09:07:40.616Z)

Let me re-run the two reformatted test modules to confirm they still pass post-format, then commit.

163. user (2026-06-29T09:07:48.243Z)

```
..... [100%]
61 passed in 1.58s
```


164. assistant (2026-06-29T09:07:51.827Z)

All green. Committing the flag-gating work to the branch.

165. user (2026-06-29T09:07:56.955Z)

```
docs/preserve-rally-detection-review
M backend/config/models.py
M backend/eval/windowing.py
M backend/pipeline/detectors/tracknet_runner.py
M backend/pipeline/segmenters/trajectory_hybrid.py
M docs/NEXT_STEPS.md
M docs/RALLY_DETECTION_CODE_REVIEW.md
M docs/RALLY_DETECTION_FIXES_PLAN.md
M tests/test_tracknet_runner.py
M tests/test_windowing.py
```

166. user (2026-06-29T09:08:14.386Z)

```
[docs/preserve-rally-detection-review c1f17a4] fix(rally): gate finding-B density fix behind
default-OFF inactivity_temporal_density flag
 9 files changed, 121 insertions(+), 39 deletions(-)
remote:  https://github.com/Khelsutra/badminton-highlight-indexer.git
To https://github.com/avidullu/badminton-highlight-indexer.git
    dd598a2..c1f17a4 docs/preserve-rally-detection-review -> docs/preserve-rally-
detection-review
```

167. assistant (2026-06-29T09:08:27.608Z)

Pushed. Updating the PR body to reflect the flag-gating and the correction.

168. user (2026-06-29T09:08:48.576Z)

<https://github.com/Khelsutra/badminton-highlight-indexer/pull/393>

169. user (2026-06-29T09:08:54.862Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

170. assistant (2026-06-29T09:09:08.711Z)

To answer your original question directly: ****yes ?** once you merge, B's validation can safely be deferred, and it's recorded. ****** But that's only true ***because of the change I just made* ?** and I have to be straight with you about why.

The correction that matters

When I first said "R4 is default-OFF so deferral is safe," ****I was wrong.**** R4 is ****default-ON in production**** ([models.py:190](backend/config/models.py#L190), `window\_inactivity\_end=1.5`). So merging B as-written would have silently shipped an unmeasured rally-quality change. You picked "flag-gate B default-OFF," which is exactly the fix ? so now:

- ****Merging is a genuine zero-regression no-op.**** Production reads `inactivity\_temporal\_density=False`; B's code is dormant. A (the `\_spread` fix) is behavior-neutral in prod anyway.
- ****Deferral is safe****, not just convenient ? nothing changes until the flag is flipped.
- ****It's recorded in three durable, in-repo places**** (survive merge, won't be forgotten):
 - `RALLY\_DETECTION\_FIXES\_PLAN` ? ****P2**** tracker row + ****S8 Q2**** promotion gate
 - `docs/NEXT\_STEPS.md` ? ****P2 #9b**** (the track's prioritized pickup)
 - the config field + eval `WindowingPreset` carry the flag, so the owner A/B is a ready switch

What's in PR #393 now (3 commits)

| | |

```
|---|---|
| `0a2dfea` | preserved review docs |
| `dd598a2` | A + B fixes + tests |
| `c1f17a4` | **correction**: B moved behind default-OFF flag, plumbed config?segmenter?eval,
docs fixed |
```

****Verified locally:**** full offline suite ****1485 passed****, `ruff` + `ruff format` + `mypy` clean.

To actually promote B later (the deferred work, owner-only)

Run the golden-set A/B at the SERVED windowing with `inactivity\_temporal\_density` ON vs OFF ? if toIF1 doesn't regress, flip the config default to `True` in a one-line PR. That's the only step requiring your real-video corpus.

Two open items (your call)

1. ****Finding C**** (`load\_canonical\_csv` silent-skip warning) ? still needs a ****separate PR** in the sibling `sports-data-collector` repo******. Want me to do it?
2. The branch is still named `docs/preserve-...` though it now carries code ? harmless; I can rename if you prefer.

171. user (2026-06-29T09:12:28.228Z)

I just added the sibling repo (data collector) in this workspace (pointing to an existing repo on disk) can you use that to send and merge yourself the fix and yes please do rename the branch

172. assistant (2026-06-29T09:12:53.786Z)

On it ? two tasks. Let me first gather the state of the sibling repo and confirm the GitHub repo identity (needed to rename the branch without closing PR #393).

173. user (2026-06-29T09:12:59.904Z)

```
=== git root ===
C:/Users/avidu/Projects/khelsutra-guru/sports-data-collector
=== status ===
master...origin/master [behind 20]
M .gitignore
?? sports-data-collector/
=== remote ===
origin https://github.com/Khelsutra/sports-data-collector.git (fetch)
origin https://github.com/Khelsutra/sports-data-collector.git (push)
=== which rally_cvat.py are tracked ===
src/parsers/cvat/rally_cvat.py
```

174. user (2026-06-29T09:13:09.085Z)

Khelsutra/sports-data-collector

175. assistant (2026-06-29T09:13:47.778Z)

Key findings: the sibling repo's canonical file is `src/parsers/cvat/rally\_cvat.py` (the nested `sports-data-collector/` dir is untracked clutter). The repo has ****pre-existing uncommitted changes**** (`M .gitignore`, untracked `sports-data-collector/`) that aren't mine and is ****20 behind origin/master**** ? I'll branch off `origin/master` explicitly and stage only my file, never touching that WIP.

Let me check the logging setup in the target file and the repo's test/lint conventions.

176. user (2026-06-29T09:13:48.641Z)

```
1      """Adapter: CVAT rally exports -> our canonical per-rally records.
2
3      Why this exists
```

```

4 -----
5 Our annotation brief (``docs/annotation/R00RA_BRIEF.md``) asks vendors for a
6 per-rally CSV that imports directly via ``curate import-local``. A vendor may
7 instead do the work in CVAT and hand back the CVAT export: per-frame bounding
8 boxes, with the rally fields (``rally_number``, ``shots_count``,
9 ``ending_reason`` ...) carried as CVAT object attributes on each box. The box
10 geometry is irrelevant to us ? it is just a carrier the rater attached the rally
11 metadata to.
12
13 This module is the ingestion-side adapter that absorbs that vendor format so the
14 modeling repo never has to. It does two things, both deliberately on our side
15 so we fix mechanical problems rather than renegotiating them:
16
17 1. Recompute time from the frame range. A rally's interval is taken from the
18 first/last frame a box for that rally appears on:
19 ``seconds = frame / fps`` at the video's true fps. We ignore any
20 ``start_time``/``end_time`` attributes the vendor wrote ? a CVAT job that
21 samples every ``step`` frames and assumes 30 fps yields systematically wrong
22 times (we have seen ``start_time == frame / (step * 30)`` , i.e. off by the
23 sampling step at the wrong fps). The frame index is reliable; the vendor's
24 seconds are not.
25
26 2. Surface ? never silently fix ? content defects. Overlapping rallies,
27 zero-duration rallies, ending_reason values outside the agreed set, and
28 too-coarse sampling are reported as :class:`Issue`s. Guessing a corrected
29 rally boundary would poison ground truth, so that stays a human/vendor call.
30
31 Supported input: the single-file CVAT XML export, in either image mode
32 (``<image><box>...``) or track/interpolation mode (``<track><box frame=...>``).
33 The two PASCAL-VOC / COCO sibling exports encode the same data and can be added
34 later if a vendor sends only those.
35 """
36 from __future__ import annotations
37
38 import math
39 import os
40 import re
41 import zipfile
42 import xml.etree.ElementTree as ET
43 from collections import Counter
44 from dataclasses import dataclass, field
45 from typing import Any, Dict, List, Optional, Tuple
46
47 from src.core.parsing import safe_int
48 from src.core.downloader import probe_resolution_fps
49
50 # The agreed cross-sport ending_reason vocabulary for net-separated sports.

```

177. user (2026-06-29T09:13:57.588Z)

```

560         except ValueError:
561             return None
562
563
564 def load_canonical_csv(csv_path: str) -> List[RallyRecord]:
565     """Load a canonical per-rally CSV into RallyRecords, so a CSV delivery (not
566     just a CVAT export) can be validated. Decimal start/end preferred, mm:ss
567     fallback; rows with no usable start/end are skipped. (frame_* are -1 here, as a
568     CSV carries no frame info ? only the frame-derived checks become inapplicable.)
569     """
570     import csv as _csv
571     if not os.path.exists(csv_path):
572         raise FileNotFoundError(f"CSV not found: {csv_path}")
573     records: List[RallyRecord] = []
574     with open(csv_path, newline="", encoding="utf-8") as f:

```

```

575         reader = _csv.DictReader(f)
576         reader.fieldnames = [(n or "").strip().lower() for n in (reader.fieldnames or
[[]])]
577         for raw in reader:
578             row = {(k or "").strip().lower(): v for k, v in raw.items()}
579             rn = safe_int(row.get("rally_number"), None)
580             if rn is None:
581                 continue
582             start = _parse_secs(row.get("start_time"))
583             if start is None:
584                 start = _parse_secs(row.get("start_mmss"))
585             end = _parse_secs(row.get("end_time"))
586             if end is None:
587                 end = _parse_secs(row.get("end_mmss"))
588             if start is None or end is None:
589                 continue
590             records.append(RallyRecord(
591                 rally_number=rn, start_time=start, end_time=end,
592                 shots_count=safe_int(row.get("shots_count"), None),
593                 ending_reason=(row.get("ending_reason") or "").strip() or None,
594                 last_shot_outcome=(row.get("last_shot_outcome") or "").strip() or None,
595                 court=_norm_lower(row.get("court")),
596                 confidence=_norm_lower(row.get("confidence")),
597                 score_before=(row.get("score_before") or "").strip() or None,
598                 score_after=(row.get("score_after") or "").strip() or None,
599                 notes=(row.get("notes") or "").strip() or None,
600                 frame_min=-1, frame_max=-1, n_boxes=0,
601             ))
602         return records
603
604

```

178. user (2026-06-29T09:13:59.308Z)

No matches found

179. assistant (2026-06-29T09:14:12.817Z)

The repo's `src` uses **no logging at all** ? and this module's stated philosophy is "surface, never silently fix" via `Issue` objects. So I should match repo conventions, not impose a logging stack. Let me check the wider repo for how it surfaces warnings, the test file, and the test/lint runner.

180. user (2026-06-29T09:14:14.367Z)

No files found

181. user (2026-06-29T09:14:15.245Z)

```

tests\test_cvat_robustness.py
tests\test_cvat_parser.py
sports-data-collector\tests\test_cvat_parser.py
sports-data-collector\tests\test_cvat_robustness.py

```

182. user (2026-06-29T09:14:20.865Z)

```

Exit code 2
=== runner/config files ===
pyproject.toml

```

183. assistant (2026-06-29T09:14:35.674Z)

Let me find callers of `load\_canonical\_csv` and how the CLI surfaces messages (to pick the convention-matching approach), and check the test file.

184. user (2026-06-29T09:14:37.816Z)

```
=== callers (tracked only) ===
src/cli/curate.py:976:          convert, load_canonical_csv, detect_issues,
src/cli/curate.py:999:          records = load_canonical_csv(args.csv)
src/parsers/cvat/rally_cvat.py:564:def load_canonical_csv(csv_path: str) -> List[RallyRecord]:
=== how src surfaces user messages ===
src/cli/curate.py:35:          print(f"Warning: Configuration file not found at
{self.config_path}. Using defaults.")
src/cli/curate.py:44:          print("\nDatabase is currently empty. Run crawlers or import
local data.\n")
src/cli/curate.py:48:          print("\n=== Dataset Ingestion Status ===")
src/cli/curate.py:49:          print(tabulate(data, headers=["Sport", "Status", "Count"],
tablefmt="grid"))
src/cli/curate.py:50:          print()
src/cli/curate.py:76:          print(f"Error: duplicate rally/point identifier(s)
{sorted(dupes)} in the annotations ")
src/cli/curate.py:78:          print("  -> re-run with --normalize, or normalize first: ")
src/cli/curate.py:90:          print(f"Error: video_id '{video_id}' already has {n}
annotated segment(s); ")
```

185. assistant (2026-06-29T09:14:56.002Z)

The repo surfaces messages via `print` in the CLI, with parsers staying pure (no I/O). For a parser module, the idiomatic non-fatal signal is stdlib `warnings.warn` (testable, no return-type change, no stdout coupling). Let me confirm by checking the existing tests' style and whether `load\_canonical\_csv` is already tested.

186. user (2026-06-29T09:14:58.611Z)

```
=== load_canonical_csv in tests? ===
```

187. user (2026-06-29T09:15:04.070Z)

```
1      [build-system]
2      requires = ["setuptools>=61.0.0"]
3      build-backend = "setuptools.build_meta"
4
5      [project]
6      name = "sports-data-collector"
7      version = "0.1.0"
8      description = "A collector and curator for sports video datasets with sport-specific
schemas and license checks."
9      readme = "README.md"
10     requires-python = ">=3.9"
11     dependencies = [
12         "pydantic>=2.0.0",
13         "pyyaml>=6.0",
14         "tabulate>=0.9.0"
15     ]
16
17     [project.optional-dependencies]
18     # Video acquisition (`fetch` / crawler). Also requires the external binaries
19     # ffprobe (FFmpeg) and, for YouTube high-res, a JS runtime such as deno ?
20     # neither is pip-installable; see README "External requirements".
21     fetch = [
22         "yt-dlp>=2024.1.0"
23     ]
24     dev = [
25         "pytest>=7.0.0",
26         "pytest-cov>=4.0.0"
27     ]
28     # Per-video observability reports (SOFT dependency: ingestion degrades to a no-op
29     # if absent). Shared, permissively-licensed report toolkit. Pin a tag once the
```

```

30     # private repo is published, e.g.:
31     #   obsreport @ git+ssh://git@github.com/avidullu/sports-obsreport.git@v0.1.3
32     # For local development install it editable from a sibling checkout:
33     #   pip install -e ../sports-obsreport --no-build-isolation
34     report = [
35         "obsreport>=0.1.3",
36     ]
37

```

188. assistant (2026-06-29T09:15:43.399Z)

The sibling repo uses **pytest** (no ruff/mypy configured), Python 3.9, and surfaces messages via `print` from the CLI while keeping parsers pure. For a pure parser, the idiomatic non-fatal signal is `stdlib warnings.warn` ? testable, no return-type change, no stdout coupling.

One thing I must verify before any push/merge/rename: a PowerShell `gh repo view` earlier returned the *collector* repo, which is ambiguous. Let me ground the working directories explicitly so I never act on the wrong repo.

189. user (2026-06-29T09:15:49.023Z)

```

PWD: C:\Users\avidu\Projects\khelsutra-guru\sports-data-collector
--- primary remote ---
https://github.com/avidullu/badminton-highlight-indexer.git
--- sibling remote ---
https://github.com/Khelsutra/sports-data-collector.git
--- primary branch ---
docs/preserve-rally-detection-review

```

190. assistant (2026-06-29T09:16:07.123Z)

Critical catch: the PowerShell `cwd` is now the **sibling** repo (`sports-data-collector`), not the primary ? the workspace change shifted it. That's exactly why I verified. From here I'll use explicit `git -C / gh --repo` for every command so nothing lands in the wrong repo.

Let me confirm the primary repo's canonical name, then rename the branch via GitHub's API (which keeps PR #393 attached).

191. user (2026-06-29T09:16:12.467Z)

```
Khelsutra/badminton-highlight-indexer
```

192. user (2026-06-29T09:16:25.769Z)

```
fix/rally-detection-edge-cases
```

193. assistant (2026-06-29T09:16:31.682Z)

Remote branch renamed and PR #393 auto-retargeted. Now syncing the local branch name, then I'll set up a fresh todo list for the C fix.

194. user (2026-06-29T09:16:38.878Z)

```

- [deleted]      (none)      -> origin/refactor/approved-directory-structure
* [new branch]   fix/rally-detection-edge-cases -> origin/fix/rally-detection-edge-
cases
branch 'fix/rally-detection-edge-cases' set up to track 'origin/fix/rally-detection-edge-cases'.
--- verify ---

```

fix/rally-detection-edge-cases...origin/fix/rally-detection-edge-cases

195. assistant (2026-06-29T09:16:45.727Z)

Branch rename complete (PR #393 now on `fix/rally-detection-edge-cases`). Now the sibling-repo C

fix. Setting up a fresh task list.

196. user (2026-06-29T09:16:48.648Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

197. user (2026-06-29T09:17:00.793Z)

```
M      .gitignore
branch 'fix/cvat-load-canonical-warn' set up to track 'origin/master'.
Switched to a new branch 'fix/cvat-load-canonical-warn'
--- branch + head ---
fix/cvat-load-canonical-warn
75bb0fa docs: archive CODE_HEALTH_PLAN ? project DONE (#49)
```

198. assistant (2026-06-29T09:17:08.658Z)

Branch is on `origin/master` tip (the WIP `.gitignore` followed, untracked ? I'll leave it). Since this is 20 commits ahead of what I first read, let me re-read the target function on this branch before editing.

199. user (2026-06-29T09:17:09.619Z)

```
562
563     def render_issue_report(result: ConversionResult, source: str) -> str:
564         """A human-readable issue report (for review / sending back to the vendor)."""
565         lines = [
566             f"CVAT rally conversion report ? {source}",
567             f"  fps used      : {result.fps:.3f}",
568             f"  sampling step  : {result.step} frames (~{result.step / result.fps:.2f}s)",
569             f"  rallies parsed : {len(result.records)}",
570             f"  issues        : {len(result.issues)} ({result.n_blockers} blocker)",
571             "",
572         ]
573         if not result.issues:
574             lines.append("No issues detected.")
575         else:
576             for sev in ("blocker", "high", "medium", "low"):
577                 group = [i for i in result.issues if i.severity == sev]
578                 if not group:
579                     continue
580                 lines.append(f"[{sev.upper()}]")
581                 for i in group:
582                     lines.append(f"  - ({i.code}) {i.message}")
583                 lines.append("")
584         return "\n".join(lines)
585
586
587     # ----- #
588     # Validation reporting ? used by `curate validate-delivery`.      #
589     # These turn a (records, issues) pair into the artifacts a human reviews:  #
590     # a machine summary (JSON), a human report (md), and an annotatable sheet.  #
591     # ----- #
592
593     def _parse_secs(value: Optional[str]) -> Optional[float]:
594         """Decimal seconds or mm:ss / hh:mm:ss -> float; None if blank/unparseable."""
595         s = (value or "").strip()
596         if not s:
597             return None
598         if ":" in s:
599             try:
600                 total = 0.0
601                 for part in s.split(":"):
```

```
602         total = total * 60 + float(part)
603     return total
```

200. user (2026-06-29T09:17:10.582Z)

```
36     from __future__ import annotations
37
38     import math
39     import os
40     import re
41     import zipfile
42     import xml.etree.ElementTree as ET # noqa: F401 ? ET.Element is the type used in hints
43     # Parse vendor-supplied XML/zip with defusedxml to harden against entity-expansion /
44     # billion-laughs attacks (docs/CODE_HEALTH_PLAN.md §7 Tier C). defusedxml.ElementTree
45     # .ParseError is a subclass of the stdlib ParseError, so the except clause below is
46     # unchanged. The parsed Element objects are ordinary xml.etree Elements, so all
47     # .find/.findall calls work as-is.
```

201. assistant (2026-06-29T09:17:15.704Z)

The 20 commits shifted line numbers. Let me locate `load\_canonical\_csv` on this branch.

202. user (2026-06-29T09:17:16.803Z)

```
474:     and small deliveries (< 3 rallies) are skipped to avoid false positives.
612: def load_canonical_csv(csv_path: str) -> List[RallyRecord]:
615:     fallback; rows with no usable start/end are skipped. (frame_* are -1 here, as a
```

203. user (2026-06-29T09:17:19.971Z)

```
612     def load_canonical_csv(csv_path: str) -> List[RallyRecord]:
613         """Load a canonical per-rally CSV into RallyRecords, so a *CSV* delivery (not
614         just a CVAT export) can be validated. Decimal start/end preferred, mm:ss
615         fallback; rows with no usable start/end are skipped. (frame_* are -1 here, as a
616         CSV carries no frame info ? only the frame-derived checks become inapplicable.)
617         """
618         import csv as _csv
619         if not os.path.exists(csv_path):
620             raise FileNotFoundError(f"CSV not found: {csv_path}")
621         records: List[RallyRecord] = []
622         with open(csv_path, newline="", encoding="utf-8") as f:
623             reader = _csv.DictReader(f)
624             reader.fieldnames = [(n or "").strip().lower() for n in (reader.fieldnames or
625             [])]
626             for row in reader:
627                 row = {(k or "").strip().lower(): v for k, v in row.items()}
628                 rn = safe_int(row.get("rally_number"), None)
629                 if rn is None:
630                     continue
631                 start = _parse_secs(row.get("start_time"))
632                 if start is None:
633                     start = _parse_secs(row.get("start_mmss"))
634                 end = _parse_secs(row.get("end_time"))
635                 if end is None:
636                     end = _parse_secs(row.get("end_mmss"))
637                 if start is None or end is None:
638                     continue
639                 records.append(RallyRecord(
640                     rally_number=rn, start_time=start, end_time=end,
641                     shots_count=safe_int(row.get("shots_count"), None),
642                     ending_reason=(row.get("ending_reason") or "").strip() or None,
643                     last_shot_outcome=(row.get("last_shot_outcome") or "").strip() or None,
644                     court=_norm_lower(row.get("court")),
645                     confidence=_norm_lower(row.get("confidence")),
```



```

645         score_before=(row.get("score_before") or "").strip() or None,
646         score_after=(row.get("score_after") or "").strip() or None,
647         notes=(row.get("notes") or "").strip() or None,
648         frame_min=-1, frame_max=-1, n_boxes=0,
649     ))
650     return records
651
652
653     def build_summary(records: List[RallyRecord], issues: List[Issue],
654                       source: str = "", fps: Optional[float] = None) -> Dict[str, Any]:
655         """Machine-readable validation summary (JSON / CI)."""
656         by_sev = {s: [i for i in issues if i.severity == s]
657                  for s in ("blocker", "high", "medium", "low")}
658         flagged = sorted({i.rally_number for i in issues if i.rally_number is not None})
659         return {
660             "source": source,
661             "fps": fps,
662             "n_rallies": len(records),
663             "n_issues": len(issues),

```

204. assistant (2026-06-29T09:17:25.356Z)

The function is unchanged. Applying the fix ? first the `import warnings`, then the skipped-row tracking + summary warning.

205. user (2026-06-29T09:17:27.353Z)

The file c:/Users/avidu/Projects/khelsutra-guru/sports-data-collector/src/parsers/cvat/rally_cvat.py has been updated successfully. (file state is current in your context ? no need to Read it back)

206. user (2026-06-29T09:17:41.145Z)

The file c:/Users/avidu/Projects/khelsutra-guru/sports-data-collector/src/parsers/cvat/rally_cvat.py has been updated successfully. (file state is current in your context ? no need to Read it back)

207. assistant (2026-06-29T09:17:43.920Z)

Now the summary warning before `return records`:

208. user (2026-06-29T09:17:48.182Z)

The file c:/Users/avidu/Projects/khelsutra-guru/sports-data-collector/src/parsers/cvat/rally_cvat.py has been updated successfully. (file state is current in your context ? no need to Read it back)

209. assistant (2026-06-29T09:17:53.094Z)

Now a pytest test. Let me check the existing test file's style and imports.

210. user (2026-06-29T09:17:53.635Z)

```

1     """validate-delivery / convert-cvat must FAIL (not silently PASS or traceback) on a CVAT
2     export that is empty, corrupt, or zipped.
3
4     Regression for: an empty CVAT (0 rallies) printed GATE: PASS; corrupt XML raised an
5     uncaught ParseError; the standard .zip delivery artifact was unsupported.
6     """
7     import argparse
8     import zipfile
9
10    import pytest
11

```



```

60         ']>'
61         '<annotations>&lol4;</annotations>'
62     )
63     p = tmp_path / "bomb.xml"
64     p.write_text(bomb, encoding="utf-8")
65     # defusedxml raises (EntitiesForbidden / DTDForbidden), surfaced as ValueError.
66     with pytest.raises((ValueError, Exception)):
67         parse_cvat_boxes(str(p))
68
69
70     def test_zip_delivery_is_parsed(tmp_path):
71         inner = _xml(_box(30, {"rally_number": "1"}) + _box(60, {"rally_number": "1"}))
72         zpath = tmp_path / "delivery.zip"
73         with zipfile.ZipFile(zpath, "w") as zf:
74             zf.writestr("annotations.xml", inner)
75         meta, boxes = parse_cvat_boxes(str(zpath))
76         assert len(boxes) == 2
77
78
79     def test_validate_delivery_blocks_on_empty_cvat(tmp_path):
80         p = _write(tmp_path / "empty.xml", _box(30, {"wrong_name": "1"}))
81         cli = SportsDataCLI()
82         args = argparse.Namespace(
83             input=p, csv=None, video_id="d", fps=30.0, video_path=None,
84             out_dir=str(tmp_path),
85             )
86         with pytest.raises(SystemExit) as exc:
87             cli.validate_delivery(args)
88         assert exc.value.code == 2 # GATE: BLOCKED, not PASS

```

214. user (2026-06-29T09:18:28.136Z)

The file c:/Users/avidu/Projects/khelsutra-guru/sports-data-collector/tests/test_cvat_robustness.py has been updated successfully. (file state is current in your context ? no need to Read it back)

215. user (2026-06-29T09:18:33.818Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

216. assistant (2026-06-29T09:18:37.256Z)

Now run the sibling repo's test suite. Let me run the targeted file first, then the full suite.

217. user (2026-06-29T09:18:44.963Z)

```

..... [100%]
7 passed in 0.48s

```

218. user (2026-06-29T09:19:02.824Z)

```

..... [ 25%]
..... [ 50%]
..... [ 76%]
..... [100%]
283 passed, 2 skipped in 5.08s

```

219. assistant (2026-06-29T09:19:09.181Z)

Full suite green (283 passed). Let me check the repo's CI to make sure I match its gate (coverage floor, etc.) before merging.

220. user (2026-06-29T09:19:14.957Z)

```
=== workflows ===
ci.yml
=== pytest/cov config ===
=== merge method allowed ===
{"mergeCommitAllowed":true,"rebaseMergeAllowed":true,"squashMergeAllowed":true}
```

221. user (2026-06-29T09:19:18.432Z)

```
1      name: CI
2
3      on:
4        push:
5          branches: [master]
6        pull_request:
7
8      jobs:
9        test:
10         runs-on: ubuntu-latest
11         steps:
12           - uses: actions/checkout@v4
13           - uses: actions/setup-python@v5
14             with:
15               python-version: "3.11"
16           - name: Install dependencies
17             run: |
18               python -m pip install --upgrade pip
19               pip install -e ".[dev]"
20           - name: Run tests
21             run: python -m pytest -q
22
```

222. assistant (2026-06-29T09:19:25.272Z)

CI is just `pytest -q` on 3.11 ? my local full run mirrors it (green). Gate satisfied. Now committing only my two files (never the pre-existing `.gitignore` WIP), then push + PR.

223. user (2026-06-29T09:19:30.583Z)

```
fix/cvat-load-canonical-warn
--- ahead of origin/master (only mine) ---
--- staged ---
M .gitignore
M src/parsers/cvat/rally_cvat.py
M tests/test_cvat_robustness.py
?? sports-data-collector/
```

224. assistant (2026-06-29T09:19:35.194Z)

Only my two files staged; the `.gitignore` WIP and untracked dir are left untouched. Committing and pushing.

225. user (2026-06-29T09:19:45.519Z)

```
[fix/cvat-load-canonical-warn ae15222] fix(cvat): warn on rows silently skipped by
load_canonical_csv
2 files changed, 50 insertions(+), 2 deletions(-)
remote:
branch 'fix/cvat-load-canonical-warn' set up to track 'origin/fix/cvat-load-canonical-warn'.
To https://github.com/Khelsutra/sports-data-collector.git
* [new branch]      fix/cvat-load-canonical-warn -> fix/cvat-load-canonical-warn
```

226. assistant (2026-06-29T09:19:49.200Z)

Pushed. Now opening the PR and merging it (the user authorized self-merge). I'll create the PR, let me also link it back to the plan doc's P1.

227. user (2026-06-29T09:20:00.886Z)

<https://github.com/Khelsutra/sports-data-collector/pull/50>

228. assistant (2026-06-29T09:20:06.763Z)

PR #50 open. Per the pre-merge gate I won't merge red ? let me check CI status before merging.

229. user (2026-06-29T09:20:15.086Z)

```
test    pending 0      https://github.com/Khelsutra/sports-data-
collector/actions/runs/28361748170/job/84017819098
--- mergeable state ---
{"mergeable":"MERGEABLE","review":"","state":"UNSTABLE"}
```

230. assistant (2026-06-29T09:20:21.355Z)

CI is pending (it's just `pytest -q`, already green locally). I'll watch it to completion rather than poll.

231. user (2026-06-29T09:20:26.717Z)

```
test    pass    14s      https://github.com/Khelsutra/sports-data-
collector/actions/runs/28361748170/job/84017819098
test    pass    14s      https://github.com/Khelsutra/sports-data-
collector/actions/runs/28361748170/job/84017819098
```

232. assistant (2026-06-29T09:20:30.029Z)

CI passed. Merging PR #50 (squash, matching the indexer repo's convention) and deleting the remote branch.

233. user (2026-06-29T09:20:56.056Z)

(PowerShell completed with no output)

234. assistant (2026-06-29T09:20:59.727Z)

Let me verify the merge landed and sync local state.

235. user (2026-06-29T09:21:14.517Z)

```
--- PR state ---
{"mergedAt":"2026-06-29T09:20:54Z","state":"MERGED"}
--- switch local off the deleted branch + cleanup ---
Switched to branch 'master'
Deleted branch fix/cvat-load-canonical-warn (was ae15222).
--- master now behind by? (local not pulled) ---
21      0
```

236. assistant (2026-06-29T09:21:33.089Z)

Both tasks are done. Let me mark the todos complete and give you the summary.

237. user (2026-06-29T09:21:35.299Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

238. assistant (2026-06-29T09:21:49.191Z)

Both done. ?

1. Branch renamed (primary repo)

`docs/preserve-rally-detection-review` ? **fix/rally-detection-edge-cases**, done via GitHub's rename API so **PR #393** stayed attached (auto-retargeted, no close/reopen). Local branch renamed and re-pointed to the new upstream too.

2. Finding C ? fixed, merged into `sports-data-collector`

[PR #50](https://github.com/Khelsutra/sports-data-collector/pull/50) ? **MERGED** (squash, branch deleted).
- `load\_canonical\_csv` now collects skipped rows (line no. + reason) and emits a single `warnings.warn` summary instead of silently dropping them. Valid rows unaffected, return type unchanged.
- Chose `stdlib `warnings.warn`` over `logging`` ? the parser stays pure and it's `pytest.warns`-testable`; the repo has no logging infra and its CLI surfaces messages via `print``.
- Tests: malformed CSV warns + still loads good rows; clean CSV stays silent.
- **CI green** (`pytest -q``, 14s) before I merged ? I verified, didn't rubber-stamp. Local full suite: 283 passed, 2 skipped.

Two honesty notes on the sibling repo

- I **branched off `origin/master`** and **staged only my two files** ? your pre-existing uncommitted WIP there (`M .gitignore`` and the untracked `sports-data-collector/`` nested dir) was **left completely untouched**.
- After merge I switched local to `master`` and deleted the feature branch. **Local `master` is now 21 behind `origin/master`** (includes my merged fix). I did **not** auto-pull it, because a `--ff-only`` could interact with your dirty `.gitignore`` ? that's your WIP to resolve. Pull when ready, or say the word and I'll `ff-only`` it.

Where the whole effort stands

- **PR #393** (badminton repo): preserved review docs + finding A fix (unconditional) + finding B fix (flag-gated default-OFF) + the P2 owner promotion gate recorded. **Open**, awaiting your review/merge.
- **PR #50** (collector repo): finding C. **Merged**.
- Remaining owner-only item: the **P2 golden-set A/B** to promote finding B's flag ? recorded in `RALLY_DETECTION_FIXES_PLAN.md`` and `NEXT_STEPS.md``.

Want me to `ff-pull`` the sibling `master`` (flagging the `.gitignore`` WIP), or leave it for you?

239. user (2026-06-29T09:25:20.523Z)

yes pls auto pull and thats it! Thanks

240. user (2026-06-29T09:25:31.682Z)

```
--- WIP before pull (left untouched) ---
M .gitignore
?? sports-data-collector/
--- ff-only pull ---
create mode 100644 src/core/schemas/_mixins.py
create mode 100644 tests/test_cleanup.py
create mode 100644 tests/test_crawler_tempfile.py
create mode 100644 tests/test_license_consistency.py
create mode 100644 tests/test_manifests.py
create mode 100644 tests/test_reporting_noraise.py
--- after ---
```

master...origin/master

```
M .gitignore
?? sports-data-collector/
```

241. assistant (2026-06-29T09:25:36.731Z)

Done. ?

The sibling `master` fast-forwarded cleanly to `origin/master` (now in sync, includes your merged finding-C fix), and your WIP was **preserved exactly** ? `.gitignore` still shows as modified and the untracked `sports-data-collector/` dir is still there, both untouched.

That wraps up everything:

- **PR #393** (badminton) ? preserved docs + fixes A & B (B flag-gated default-OFF) + recorded P2 promotion gate ? open for your review.
- **PR #50** (collector) ? finding C ? merged, both repos in sync.

Thanks ? happy to pick up the P2 golden-set promotion whenever you're ready!

242. user (2026-06-29T09:48:22.397Z)

<ide_opened_file>The user opened the file \temp\readonly\PowerShell tool output (ibz1gk) in the IDE. This may or may not be related to the current task.</ide_opened_file>
Seems #393 is closed?

243. assistant (2026-06-29T09:51:22.491Z)

Let me check the actual state of PR #393.

244. user (2026-06-29T09:51:32.341Z)

claude-opus-4-8 is temporarily unavailable, so auto mode cannot determine the safety of PowerShell right now. Wait briefly and then try this action again. If it keeps failing, continue with other tasks that don't require this action and come back to it later. Note: reading files, searching code, and other read-only operations do not require the classifier and can still be used.

245. user (2026-06-29T09:51:45.532Z)

claude-opus-4-8 is temporarily unavailable, so auto mode cannot determine the safety of PowerShell right now. Wait briefly and then try this action again. If it keeps failing, continue with other tasks that don't require this action and come back to it later. Note: reading files, searching code, and other read-only operations do not require the classifier and can still be used.

246. assistant (2026-06-29T09:51:50.815Z)

The command classifier is temporarily unavailable for the PowerShell tool. Let me try via the Bash tool instead.

247. user (2026-06-29T09:52:01.236Z)

claude-opus-4-8 is temporarily unavailable, so auto mode cannot determine the safety of Bash right now. Wait briefly and then try this action again. If it keeps failing, continue with other tasks that don't require this action and come back to it later. Note: reading files, searching code, and other read-only operations do not require the classifier and can still be used.

248. assistant (2026-06-29T09:52:11.286Z)

Both shell tools are temporarily blocked by a transient classifier outage (not something on our end). The note says to wait briefly and retry. Let me try once more.

249. user (2026-06-29T09:52:20.137Z)

claude-opus-4-8 is temporarily unavailable, so auto mode cannot determine the safety of Bash right now. Wait briefly and then try this action again. If it keeps failing, continue with other tasks that don't require this action and come back to it later. Note: reading files, searching code, and other read-only operations do not require the classifier and can still be used.

250. assistant (2026-06-29T09:52:40.486Z)

Still blocked ? the safety classifier for shell commands is having a transient outage, so I can't run `gh` to inspect #393 this moment. Read-only file tools still work, but those won't tell me a PR's GitHub state.

While I wait for it to recover, here's my read on what likely happened and the fix:

****Most likely cause:**** the branch rename. I renamed the head branch `docs/preserve-rally-detection-review` ? `fix/rally-detection-edge-cases` via GitHub's rename API. That API is ***supposed*** to retarget open PRs automatically, but if GitHub instead treated the old head as deleted, it would have ****auto-closed #393****. Good news either way: ****nothing is lost**** ? the branch `fix/rally-detection-edge-cases` still exists on the remote with all three commits (preserved docs, A+B fixes, the flag-gating correction).

****Remedy once the shell tool is back****, depending on what I find:

- If it's ****closed (not merged)**** ? `gh pr reopen 393` (works since the head branch still exists); if GitHub won't reopen, `gh pr create` a fresh PR from `fix/rally-detection-edge-cases` ? `master` and I'll re-paste the body.
- If it's ****merged**** ? nothing to do.
- If it's ****open**** ? it was just a stale view on your end.

Let me retry the check now.

251. user (2026-06-29T09:52:49.499Z)

claude-opus-4-8 is temporarily unavailable, so auto mode cannot determine the safety of Bash right now. Wait briefly and then try this action again. If it keeps failing, continue with other tasks that don't require this action and come back to it later. Note: reading files, searching code, and other read-only operations do not require the classifier and can still be used.

252. assistant (2026-06-29T09:53:02.829Z)

The shell-command classifier is still down, so I can't query GitHub right now (every `git`/`gh` call routes through it). I'll stop hammering it to avoid wasting cycles.

Here's where things stand and what I'll do the moment it recovers:

- ****Nothing is lost regardless of #393's state.**** The branch `fix/rally-detection-edge-cases` is on the remote with all three commits intact.
- ****As soon as the tool is back****, I'll check #393 and: reopen it if it's closed-but-not-merged, or open a fresh PR from the branch if GitHub refuses the reopen (I have the full body ready to re-paste).

Two options for you:

1. ****I retry shortly**** ? just say "check now" (or send anything) in a minute or two and I'll re-run the check.
2. If you're on the page already and it shows ****Closed****, you can click ****Reopen**** directly ? the head branch still exists, so it'll work; no need to wait on me.

Either way, the work is safe. Want me to retry now, or shall I wait for your nudge?

253. user (2026-06-29T09:55:23.120Z)

<local-command-caveat>Caveat: The messages below were generated by the user while running local commands. DO NOT respond to these messages or otherwise consider them in your response unless the user explicitly asks you to.</local-command-caveat>

254. user (2026-06-29T09:55:23.120Z)

<command-name>/model</command-name>
<command-message>model</command-message>
<command-args>default</command-args>

255. user (2026-06-29T09:55:23.120Z)

```
<local-command-stdout>Set model to claude-opus-4-8[1m]</local-command-stdout>
```

256. user (2026-06-29T09:55:39.106Z)

you can retry and check out #394 which i made as a roll forward

257. user (2026-06-29T09:55:57.653Z)

```
Exit code 1
=== PR 393 ===
Unknown JSON field: "merged"
Available fields:
  additions
  assignees
  author
  autoMergeRequest
  baseRefName
  baseRefOid
  body
  changedFiles
  closed
  closedAt
  closingIssuesReferences
  comments
  commits
  createdAt
  deletions
  files
  fullDatabaseId
  headRefName
  headRefOid
  headRepository
  headRepositoryOwner
  id
  isCrossRepository
  isDraft
  labels
  latestReviews
  maintainerCanModify
  mergeCommit
  mergeStateStatus
  mergeable
  mergedAt
  mergedBy
  milestone
  number
  potentialMergeCommit
  projectCards
  projectItems
  reactionGroups
  reviewDecision
  reviewRequests
  reviews
  state
  statusCheckRollup
  title
  updatedAt
  url
=== PR 394 ===
Unknown JSON field: "merged"
Available fields:
  additions
  assignees
```

author
autoMergeRequest
baseRefName
baseRefOid
body
changedFiles
closed
closedAt
closingIssuesReferences
comments
commits
createdAt
deletions
files
fullDatabaseId
headRefName
headRefOid
headRepository
headRepositoryOwner
id
isCrossRepository
isDraft
labels
latestReviews
maintainerCanModify
mergeCommit
mergeStateStatus
mergeable
mergedAt
mergedBy
milestone
number
potentialMergeCommit
projectCards
projectItems
reactionGroups
reviewDecision
reviewRequests
reviews
state
statusCheckRollup
title
updatedAt
url

258. assistant (2026-06-29T09:56:02.876Z)

The classifier is back. Let me re-query with valid fields.

259. user (2026-06-29T09:56:08.542Z)

```
=== PR 393 ===
{"base":"master","closedAt":"2026-06-29T09:16:28Z","head":"docs/preserve-rally-detection-
review","mergedAt":null,"state":"CLOSED"}
=== PR 394 ===
{"base":"master","head":"fix/rally-detection-edge-cases","mergedAt":null,"ncommits":3,"state":"O
PEN","url":"https://github.com/Khelsutra/badminton-highlight-indexer/pull/394"}
```

260. assistant (2026-06-29T09:56:19.506Z)

Confirmed ? and I owe you a straight correction.

****#393 is CLOSED (not merged), closed at 09:16:28Z ? exactly when I renamed the branch.**** So my branch rename closed it. I was wrong earlier to assume GitHub's rename API would keep the PR

attached; it instead treated the old head ref as gone and auto-closed #393. That's on me.

****Your #394 is the correct live PR****: open, head `fix/rally-detection-edge-cases` ? `master`,
****3 commits**** ? which matches all the work. Let me verify it contains exactly the right content.

261. user (2026-06-29T09:56:27.232Z)

```
=== #394 commits ===
0a2dfea docs: preserve rally-detection code review + fixes plan
dd598a2 fix(rally): correct _spread zero-collapse and _trim_inactive_tail spa?
c1f17a4 fix(rally): gate finding-B density fix behind default-OFF inactivity_?
=== #394 files changed ===
backend/config/models.py
backend/eval/windowing.py
backend/pipeline/detectors/rally_gate.py
backend/pipeline/detectors/tracknet_runner.py
backend/pipeline/segmenters/trajectory_hybrid.py
docs/NEXT_STEPS.md
docs/RALLY_DETECTION_CODE_REVIEW.md
docs/RALLY_DETECTION_FIXES_PLAN.md
tests/test_rally_gate.py
tests/test_tracknet_runner.py
tests/test_windowing.py
=== local branch matches PR head? ===
c1f17a4 fix(rally): gate finding-B density fix behind default-OFF inactivity_temporal_density
flag
dd598a2 fix(rally): correct _spread zero-collapse and _trim_inactive_tail sparse density (P0)
0a2dfea docs: preserve rally-detection code review + fixes plan
```

262. assistant (2026-06-29T09:56:39.892Z)

#394 is verified complete and correct ? it's the full roll-forward:

- ****3 commits****, identical to the work: `0a2dfea` (preserved docs) ? `dd598a2` (A + B fixes) ? `c1f17a4` (B flag-gating correction)
- ****All 11 files**** present (5 backend, 3 docs, 3 tests)
- Your local `fix/rally-detection-edge-cases` HEAD (`c1f17a4`) matches the PR head exactly

So nothing was lost ? ****#394 fully supersedes #393****, and #393 can stay closed. Thanks for rolling it forward.

Let me record this failure mode so I don't repeat it.

263. assistant (2026-06-29T09:56:46.720Z)

```
---
name: branch-rename-closes-open-pr
description: Renaming a branch that has an open PR closes that PR ? don't rename branches with
open PRs
metadata:
  type: feedback
---
```

Renaming the head branch of an ****open PR**** closes the PR ? it does NOT retarget it.
On 2026-06-29 I renamed `docs/preserve-rally-detection-review` ? `fix/rally-detection-edge-cases`
via the GitHub rename API (`POST /repos/{o}/{r}/branches/{b}/rename`) expecting the open PR #393 to follow; instead GitHub auto-closed #393 (closedAt == rename time). The owner had to roll it forward as a fresh PR #394 from the renamed branch (work was intact ? only the PR object died).

****Why:**** I assumed the rename API preserves open PRs. It does not, at least here ? the old head ref is treated as gone and the PR is closed (not merged).

****How to apply:**** Do NOT rename a branch that already has an open PR. If a rename is requested:

either (a) do it BEFORE opening the PR, or (b) warn that the PR will close and plan to re-create it (`gh pr create` from the renamed branch) ? reopen may be refused. Prefer just leaving the branch name and noting it's cosmetic. Commits/branch survive a rename; the PR object does not.

264. user (2026-06-29T09:56:47.760Z)

File created successfully at: C:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru-badminton-highlight-indexer-badminton-highlight-indexer\memory\branch-rename-closes-open-pr.md (file state is current in your context ? no need to Read it back)

265. user (2026-06-29T09:56:51.258Z)

File does not exist. Note: your current working directory is C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer.

266. assistant (2026-06-29T09:56:56.822Z)

Memory index

- [Branch rename closes open PR](branch-rename-closes-open-pr.md) ? don't rename a branch that has an open PR; it closes the PR.

267. user (2026-06-29T09:56:57.849Z)

File created successfully at: C:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru-badminton-highlight-indexer-badminton-highlight-indexer\memory\MEMORY.md (file state is current in your context ? no need to Read it back)

268. assistant (2026-06-29T09:57:03.837Z)

All sorted. Summary:

- **#393** ? closed by my branch rename (my mistake; the rename API closes an open PR rather than retargeting it). Leave it closed.
- **#394** ? your roll-forward, **verified complete**: 3 commits, all 11 files, matches the local branch. This is the live PR to review/merge.
- **Collector PR #50** (finding C) ? merged; both repos in sync.
- I saved a memory so I won't rename a branch with an open PR again.

Net: no work lost, everything's on #394, and the only open thread remaining is your review/merge of #394 (plus the owner-gated P2 golden-set promotion of finding B's flag, recorded in the docs). Thanks for the catch and the roll-forward!

269. user (2026-06-29T10:03:36.949Z)

`<ide_opened_file>`The user opened the file c:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\badminton-highlight-indexer\.gitattributes in the IDE. This may or may not be related to the current task.`</ide_opened_file>`
#394 merged and maybe you can also complete the P2 golden set promotion and that would close the session loop.

270. assistant (2026-06-29T10:04:02.422Z)

Let me first sync the merge, then I'll be straight with you about what "completing P2" actually requires.

271. user (2026-06-29T10:04:10.476Z)

```
=== #394 state ===
{"mergeCommit":"cc60ebe","mergedAt":"2026-06-29T09:59:15Z","state":"MERGED"}
=== sync master ===
Switched to branch 'master'
12 files changed, 319 insertions(+), 8 deletions(-)
```

```
create mode 100644 docs/RALLY_DETECTION_CODE_REVIEW.md
create mode 100644 docs/RALLY_DETECTION_FIXES_PLAN.md
=== delete merged local branch ===
Deleted branch fix/rally-detection-edge-cases (was c1f17a4).
=== flag present on master? ===
backend/config/models.py:199:    inactivity_temporal_density: bool = False
```

272. assistant (2026-06-29T10:04:33.418Z)

#394 is merged (`cc60ebe`), `master` synced, the flag is on `master`
(`inactivity\_temporal\_density: bool = False`), and the merged branch is cleaned up.

Now the honest part: ****I can't actually complete the P2 promotion**** ? and flipping the flag anyway would defeat the reason we built it. Let me explain precisely why, then look at the harness so I can hand you a turnkey runbook (which I *can* do).

273. user (2026-06-29T10:04:35.359Z)

```
10: windows under a :class:`~backend.eval.windowing.WindowingPreset` (so it tracks the SERVED
31: from backend.eval.windowing import PRESETS, SERVED, WindowingPreset
216:     video: Dict[str, Any], preset: WindowingPreset, min_crossings: int = 0
276:     golden: List[Dict[str, Any]], preset: WindowingPreset, min_crossings: int = 0
315:     preset: WindowingPreset,
345:     base: WindowingPreset,
346:     cand: WindowingPreset,
403:     preset: WindowingPreset,
568: def _resolve_preset(name: str) -> WindowingPreset:
574: def _build_preset(args: argparse.Namespace, side: str) -> WindowingPreset:
618: def main() -> None:
622:     ap.add_argument("--manifest", default=DEFAULT_MANIFEST)
623:     ap.add_argument("--features-dir", default=DEFAULT_FEATURES_DIR)
624:     ap.add_argument(
627:     ap.add_argument("--vs", default=None, help="second preset to diff against --preset")
628:     ap.add_argument("--iou", type=float, default=DEFAULT_IOU)
629:     ap.add_argument(
635:     ap.add_argument(
641:     ap.add_argument(
647:     ap.add_argument(
654:     ap.add_argument(
660:     ap.add_argument(
665:     ap.add_argument(
671:     ap.add_argument(
676:     ap.add_argument(
682:     ap.add_argument(
688:     ap.add_argument(
694:     ap.add_argument(
700:     ap.add_argument(
706:     ap.add_argument(
```

[Showing results with pagination = limit: 30]

274. user (2026-06-29T10:04:39.966Z)

```
=== eval entry points ===
c:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/badminton-highlight-
indexer/run_eval.py
c:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/badminton-highlight-
indexer/run_regression.py
=== who reads the flag / preset A-B knobs ===
backend/eval/rally_seg_eval.py
backend/eval/serve_contrast.py
backend/eval/windowing.py
```

275. assistant (2026-06-29T10:04:49.745Z)

`rally\_seg\_eval.py` is the A/B harness (`--preset` vs `--vs`). Let me check whether the new flag is actually togglable from its CLI ? if not, the owner *can't even run the A/B* without a code change, and exposing that toggle is something I can do offline.

276. user (2026-06-29T10:04:50.872Z)

```
568     def _resolve_preset(name: str) -> WindowingPreset:
569         if name in PRESETS:
570             return PRESETS[name]
571         raise SystemExit(f"unknown preset '{name}'; choices: {sorted(PRESETS)}")
572
573
574     def _build_preset(args: argparse.Namespace, side: str) -> WindowingPreset:
575         """Resolve the ``base`` (`--preset`) or ``cand`` (`--vs`) windowing from
576         ``args`` by
577         applying the ``dataclasses.replace`` overrides for cap / hard-cap / blob-fallback /
578         inactivity. ``cand`` falls back to the base override when its ``--vs-*`` variant is
579         unset
580         (so a baseline-vs-candidate promotion check runs in one shot)."""
581         from dataclasses import replace
582
583         if side == "base":
584             preset = _resolve_preset(args.preset)
585             cap = args.max_window_duration
586             hard_cap = args.hard_cap
587             blob_fallback = args.blob_fallback
588             inactivity_end = args.inactivity_end
589         else:
590             preset = _resolve_preset(args.vs)
591             cap = (
592                 args.vs_max_window_duration
593                 if args.vs_max_window_duration is not None
594                 else args.max_window_duration
595             )
596             hard_cap = args.vs_hard_cap or args.hard_cap
597             blob_fallback = args.vs_blob_fallback or args.blob_fallback
598             inactivity_end = (
599                 args.vs_inactivity_end
600                 if args.vs_inactivity_end is not None
601                 else args.inactivity_end
602             )
603         if cap is not None:
604             preset = replace(preset, max_window_duration=cap)
605         if hard_cap:
606             preset = replace(preset, hard_cap=True)
607         if blob_fallback:
608             preset = replace(preset, blob_fallback=True)
609         if args.blob_factor is not None:
610             preset = replace(preset, blob_factor=args.blob_factor)
611         if inactivity_end is not None:
612             preset = replace(preset, inactivity_end=inactivity_end)
613         if args.inactivity_rally_thresh is not None:
614             preset = replace(preset, inactivity_rally_thresh=args.inactivity_rally_thresh)
615         if args.inactivity_min_density is not None:
616             preset = replace(preset, inactivity_min_density=args.inactivity_min_density)
617         return preset
618
619     def main() -> None:
620         ap = argparse.ArgumentParser(
621             description="Rally-segmentation eval vs golden labels (offline)."
622         )
623         ap.add_argument("--manifest", default=DEFAULT_MANIFEST)
624         ap.add_argument("--features-dir", default=DEFAULT_FEATURES_DIR)
625         ap.add_argument(
```

```

625         "--preset", default=SERVED.name, help="windowing preset (default: served)"
626     )
627     ap.add_argument("--vs", default=None, help="second preset to diff against --preset")

```

277. assistant (2026-06-29T10:05:00.659Z)

Found the real gap: `\_build\_preset` applies overrides for cap, hard-cap, blob-fallback, inactivity-end/thresh/density ? but **not** `inactivity\_temporal\_density`. So the A/B harness can't toggle finding B's flag at all; the owner couldn't run the promotion A/B even with the corpus. **That** wiring is offline code I can do. Let me read the rest of the argparse to mirror the existing boolean-flag pattern.

278. user (2026-06-29T10:05:01.590Z)

```

627     ap.add_argument("--vs", default=None, help="second preset to diff against --preset")
628     ap.add_argument("--iou", type=float, default=DEFAULT_IΟΥ)
629     ap.add_argument(
630         "--min-crossings",
631         type=int,
632         default=0,
633         help="net-crossings rally-state gate on --preset (0=off; W2 Gate-A)",
634     )
635     ap.add_argument(
636         "--vs-min-crossings",
637         type=int,
638         default=0,
639         help="net-crossings gate on --vs (default: same as --min-crossings)",
640     )
641     ap.add_argument(
642         "--max-window-duration",
643         type=float,
644         default=None,
645         help="override bounded-windowing cap (s) on BOTH presets (W1; e.g. 12)",
646     )
647     ap.add_argument(
648         "--vs-max-window-duration",
649         type=float,
650         default=None,
651         help="cap (s) on --vs ONLY (default: same as --max-window-duration); lets a "
652         "BASELINE (no cap) vs CANDIDATE (capped) promotion check run in one shot",
653     )
654     ap.add_argument(
655         "--hard-cap",
656         action="store_true",
657         help="enforce the cap as a HARD cap on --preset (chop continuously-active "
658         "windows at the cap when no inactivity gap exists; W1 hard-cap fallback)",
659     )
660     ap.add_argument(
661         "--vs-hard-cap",
662         action="store_true",
663         help="hard-cap fallback on --vs (default: same as --hard-cap)",
664     )
665     ap.add_argument(
666         "--blob-fallback",
667         action="store_true",
668         help="R5 blob-fallback on --preset (after R4, force-chop any group still over
the "
669         "cap at its midpoint + flag/log it; the continuously-active blob last resort)",
670     )
671     ap.add_argument(
672         "--vs-blob-fallback",
673         action="store_true",
674         help="R5 blob-fallback on --vs (default: same as --blob-fallback)",
675     )
676     ap.add_argument(

```

```

677         "--blob-factor",
678         type=float,
679         default=None,
680         help="R5 magnitude gate (both presets): chop only groups > this × cap (default
4.0)",
681     )
682     ap.add_argument(
683         "--inactivity-end",
684         type=float,
685         default=None,
686         help="R4 rally-end inactivity trim (s) on --preset (0=off; trims post-rally
drift tail)",
687     )
688     ap.add_argument(
689         "--vs-inactivity-end",
690         type=float,
691         default=None,
692         help="R4 inactivity trim on --vs (default: same as --inactivity-end)",
693     )
694     ap.add_argument(
695         "--inactivity-rally-thresh",
696         type=float,
697         default=None,
698         help="R4 velocity above which motion counts as 'rally' (both presets; default
0.08)",
699     )
700     ap.add_argument(
701         "--inactivity-min-density",
702         type=float,
703         default=None,
704         help="R4 min trailing fast-fraction to stay 'in rally' (both presets; default
0.34)",
705     )
706     ap.add_argument(
707         "--tau-start",
708         type=float,
709         default=None,
710         help="R2 tolerance-match start window (s); default 2.0 (forgives the service
window)",
711     )
712     ap.add_argument(
713         "--tau-end",
714         type=float,
715         default=None,
716         help="R2 tolerance-match end window (s); default 1.5 (keeps the rally-end
honest)",
717     )
718     ap.add_argument(
719         "--guard-weight-merge",
720         type=float,
721         default=None,

```

279. assistant (2026-06-29T10:05:25.775Z)

Now I have the full picture. Let me set up tasks and check for existing `\_build\_preset` tests to extend.

280. user (2026-06-29T10:05:28.337Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

281. user (2026-06-29T10:05:33.896Z)

No matches found

282. assistant (2026-06-29T10:05:46.887Z)

No existing `\_build\_preset` test. Let me make the source edits first ? the two CLI flags (mirroring `--hard-cap`/`--vs-hard-cap`):

283. user (2026-06-29T10:05:52.702Z)

The file c:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/badminton-highlight-indexer/backend/eval/rally_seg_eval.py has been updated successfully. (file state is current in your context ? no need to Read it back)

284. assistant (2026-06-29T10:05:55.707Z)

Now the override logic in `\_build\_preset` (base + cand, mirroring `hard\_cap`/`blob\_fallback`):

285. user (2026-06-29T10:06:04.173Z)

The file c:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/badminton-highlight-indexer/backend/eval/rally_seg_eval.py has been updated successfully. (file state is current in your context ? no need to Read it back)

286. user (2026-06-29T10:06:09.637Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

287. assistant (2026-06-29T10:06:14.688Z)

Now a test. Let me check the test file's imports/style to add a focused `\_build\_preset` test.

288. user (2026-06-29T10:06:15.396Z)

```
1      """Tests for backend.eval.rally_seg_eval ? the pure scoring/aggregation core (no golden
files,
2      no trajectory I/O). The windowing + trajectory parsing it composes are tested in their
own modules.
3      """
4
5      import pytest
6
7      from backend.eval.rally_seg_eval import (
8          aggregate,
9          compare,
10         court_type_for,
11         evaluate,
12         filter_by_min_duration,
13         score_intervals,
14         stratified_report,
15         windows_for,
16     )
17     from backend.eval.windowing import WindowingPreset
18
19     GTS = [(0.0, 10.0), (20.0, 30.0), (40.0, 50.0)]
20     # A permissive preset so a short synthetic trajectory still forms a window.
21     _PRESET = WindowingPreset(
22         name="t", velocity_thresh=0.02, merge_gap=2.0, min_window_duration=0.5
23     )
24
25
26     def _write_oscillating_traj(path, fps=30, secs=4, width=1280):
27         """A fast shuttle oscillating smoothly across the net (x=width/2) via a triangle
wave ?
```

```

28         continuous high velocity (active) + many net crossings ? one gated-survivable
window. """
29         lo, hi, period = 200, width - 200, 20 # frames per full oscillation
30         rows = ["Frame,Visibility,X,Y"]

```

289. user (2026-06-29T10:06:23.732Z)

```

30         rows = ["Frame,Visibility,X,Y"]
31         for i in range(int(fps * secs)):
32             phase = (i % period) / period # 0..1
33             tri = 2 * phase if phase < 0.5 else 2 * (1 - phase) # triangle 0?1?0
34             rows.append(f"{i},1,{lo + (hi - lo) * tri:.1f},300")
35         path.write_text("\n".join(rows))
36         return str(path)
37
38
39     def test_score_intervals_perfect():
40         row = score_intervals(GTS, GTS)
41         assert row["f1"] == pytest.approx(1.0)
42         assert row["f1@0.25"] == pytest.approx(1.0)
43         assert row["merge_rate"] == 0.0 and row["split_rate"] == 0.0
44         assert row["merges"] == 0 and row["num_pred"] == 3 and row["num_gt"] == 3
45
46
47     def test_score_intervals_surfaces_over_merge():
48         # One mega-window swallowing two rallies: tIoU-F1 is hurt AND merge_rate makes it
explicit.
49         preds = [(0.0, 30.0), (40.0, 50.0)]

```

290. user (2026-06-29T10:06:31.402Z)

The file c:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/badminton-highlight-indexer/tests/test_rally_seg_eval.py has been updated successfully. (file state is current in your context ? no need to Read it back)

291. user (2026-06-29T10:06:42.981Z)

The file c:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/badminton-highlight-indexer/tests/test_rally_seg_eval.py has been updated successfully. (file state is current in your context ? no need to Read it back)

292. user (2026-06-29T10:06:48.391Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

293. assistant (2026-06-29T10:06:51.785Z)

Now run the targeted tests + lint, then the full suite. Let me also update the plan doc to note the A/B harness is now wired (so the runbook is accurate).

294. user (2026-06-29T10:06:58.418Z)

```

=== targeted tests ===
..... [100%]
15 passed in 1.25s
=== CLI smoke: flags parse + help ===
    [--inactivity-temporal-density]
    [--vs-inactivity-temporal-density]
    [--tau-start TAU_START] [--tau-end TAU_END]
--
--inactivity-temporal-density
    R4 density over the trailing window's TEMPORAL length,
--

```

```
--vs-inactivity-temporal-density
    R4 temporal-density on --vs (default: same as
    --inactivity-temporal-density); use '--preset served
    --vs served --vs-inactivity-temporal-density' for the
    P2 promotion A/B
```

295. assistant (2026-06-29T10:07:01.627Z)

Tests pass and the CLI flags parse. Let me update the plan doc's promotion gate to reflect the harness is now wired with the exact command, then run the full suite + lint.

296. user (2026-06-29T10:07:02.670Z)

```
58      ## 7. Deliverables & progress tracker  ? **source of truth**
59      Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**,
branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.
60
61      | ID | Deliverable | Depends on | Gated? | Status | PR |
62      |----|-----|-----|-----|-----|----|
63      | P0a | Fix `_spread` zero-collapse (behaviour-neutral, unconditional) | ? | No | ? |
#393 |
64      | P0b | `_trim_inactive_tail` temporal density behind `inactivity_temporal_density` flag
(default-OFF) | ? | No | ? | #393 |
65      | P2 | **Promote P0b**: golden-set A/B at SERVED windowing ? flip flag default-ON if no
regression | P0b | ? owner | ? | ? |
66      | P1 | Add logging to `load_canonical_csv` for silent skips | ? | No | ? (sibling repo)
| ? |
67
68      ## 8. Open questions ? owner / external
69      1. ~~Do we want `_trim_inactive_tail` to fall back to the existing behavior if there are
very few
70      frames, or strictly enforce a minimum temporal density?~~ **Resolved (#393):** strict
temporal
71      density. Because `group` carries only active frames, the temporal denominator equals
the
72      sample-count denominator under dense tracking, so this is behaviour-preserving for
well-tracked
73      clips and only corrects the sparse case ? no separate few-frames fallback needed.
74      2. **Owner promotion gate (P2) ? pending.** P0b ships flag-OFF, so the merge is a
**zero-regression
75      no-op in production**. To promote it: run the golden-set A/B with
`inactivity_temporal_density`
76      ON vs OFF at the **SERVED** operating point (the eval `WindowingPreset` already
carries the flag).
77      R4 is default-ON in prod, so this measures B's real end-boundary effect on sparse
footage. Flip
78      the `inactivity_temporal_density` config default to True (a one-line PR) only if
tolF1 does not
79      regress. This is owner-only (real video / golden corpus).
```

297. user (2026-06-29T10:07:21.031Z)

The file c:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/badminton-highlight-indexer/docs/RALLY_DETECTION_FIXES_PLAN.md has been updated successfully. (file state is current in your context ? no need to Read it back)

298. user (2026-06-29T10:07:33.450Z)

The file c:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/badminton-highlight-indexer/docs/RALLY_DETECTION_FIXES_PLAN.md has been updated successfully. (file state is current in your context ? no need to Read it back)

299. assistant (2026-06-29T10:07:46.466Z)

Now the full offline suite + lint, then commit on a fresh branch off `origin/master` (no rename this time).

300. user (2026-06-29T10:09:13.452Z)

```
=== ruff + format + mypy ===
All checks passed!
Would reformat: tests\test_rally_seg_eval.py
1 file would be reformatted, 1 file already formatted
backend\sports\__init__.py:9: note: By default the bodies of untyped functions are not checked,
consider using --check-untyped-defs [annotation-unchecked]
backend\providers\__init__.py:9: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
Success: no issues found in 1 source file
=== full offline suite ===
1487 passed, 12 skipped in 77.65s (0:01:17)
=====
RESULT: PASS - all tests green
=====
```

301. assistant (2026-06-29T10:09:17.387Z)

Suite green (1487 passed), mypy clean. Test file needs formatting ? applying and re-verifying.

302. user (2026-06-29T10:09:23.341Z)

```
1 file reformatted
2 files already formatted
=== re-run targeted after format ===
..... [100%]
15 passed in 1.26s
[This command modified 1 file you've previously read: tests\test_rally_seg_eval.py. Call Read
before editing.]
```

303. assistant (2026-06-29T10:09:27.788Z)

All green and formatted. Now branch off `origin/master` (no rename this time), commit, push, open PR.

304. user (2026-06-29T10:09:30.042Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

305. user (2026-06-29T10:09:41.028Z)

```
Switched to a new branch 'feat/rally-seg-eval-temporal-density-ab'
feat/rally-seg-eval-temporal-density-ab
--- only my commits expected after commit ---
```

306. user (2026-06-29T10:09:55.931Z)

```
M backend/eval/rally_seg_eval.py
M docs/RALLY_DETECTION_FIXES_PLAN.md
M tests/test_rally_seg_eval.py
--- commit ---
caabb04 feat(eval): expose inactivity_temporal_density in the rally_seg_eval A/B CLI (P2a)
```

307. user (2026-06-29T10:10:04.779Z)

branch 'feat/rally-seg-eval-temporal-density-ab' set up to track 'origin/feat/rally-seg-eval-temporal-density-ab'.
To <https://github.com/avidullu/badminton-highlight-indexer.git>

```
* [new branch]      feat/rally-seg-eval-temporal-density-ab -> feat/rally-seg-eval-temporal-density-ab
```

308. user (2026-06-29T10:10:22.876Z)

<https://github.com/Khelsutra/badminton-highlight-indexer/pull/395>

309. user (2026-06-29T10:10:28.321Z)

Todos have been modified successfully. Ensure that you continue to use the todo list to track your progress. Please proceed with the current tasks if applicable

310. assistant (2026-06-29T10:10:48.525Z)

Here's the honest status on closing the loop.

****P2 has two halves, and I can only do one of them:****

```
| | What | Who | Status |
|---|---|---|---|
| **P2a** | Make the promotion A/B *runnable* (wire the flag into the eval CLI) | me ? offline
code | ? **done ? [PR #395](https://github.com/Khelsutra/badminton-highlight-indexer/pull/395)**
|
| **P2b** | Actually *run* the A/B and flip the default if no regression | **owner only** | ?
still gated |
```

****Why I can't finish P2b myself**** ? and why flipping the flag anyway would be wrong:

- It needs the ****golden corpus + GPU/WSL-extracted trajectory features****, which don't exist offline (your machine/labelled videos only).
- Finding B is a ****no-op under dense tracking**** by construction ? so on synthetic/offline fixtures, flag-ON vs flag-OFF produce **identical** output. There's literally ****no signal to measure**** without real footage where the shuttle tracker drops out. Flipping the default without that measurement is exactly the unmeasured rally-quality change the flag exists to prevent.

****What I removed:**** the real blocker was that ``rally_seg_eval``'s A/B harness couldn't toggle the flag at all ? so even **you** couldn't have run the comparison. Now it's one command (verified the flags parse, tests cover the toggle, full suite green at 1487):

```
...
python -m backend.eval.rally_seg_eval \
    --manifest <golden_manifest.json> --features-dir <golden_features_dir> \
    --preset served --vs served --vs-inactivity-temporal-density
...
```

? if tolF1 doesn't regress, flip ``IndexingConfig.inactivity_temporal_density`` to ``True`` (a one-line PR). The runbook is recorded in the plan doc (P2 split into P2a done / P2b pending).

****So the loop is as closed as it can be from my side.**** Remaining:

1. ****PR #395**** ? ready for your review/merge (I left it unmerged; say the word and I'll merge it like #50).
2. ****P2b**** ? your golden-corpus run, whenever you have the GPU box.

Net session tally: #394 merged (A + B flag-gated), collector #50 merged (C), #395 open (P2a harness). The only thing that truly requires you is the golden measurement.

311. user (2026-06-29T10:13:24.182Z)

<ide_opened_file>The user opened the file `\temp\readonly\command (rvp4k4)` in the IDE. This may or may not be related to the current task.</ide_opened_file>
so pls go ahead and merge it and then this repo will be in a complete fresh state i.e. as fresh as a new clone from the remote head? Or are there still any pending items here

312. assistant (2026-06-29T10:14:02.151Z)

Good question ? "fresh as a new clone" is a stricter bar than "git status clean," so I'll merge

#395 and then do a real audit against that bar.

(Note: I'm doing this solo rather than via a multi-agent workflow on purpose ? it's a strictly sequential git flow with shared-working-tree write hazards, where parallel agents would be unsafe and add nothing.)

First, let me check #395's CI before merging (I won't merge red), and snapshot current state.

313. user (2026-06-29T10:14:08.325Z)

```
=== #395 checks ===
Built wheel smoke (install + launch)    pass    1m15s    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28364589874/job/84027207506
Dependency license scan (no copleft in runtime deps)    pass    35s
https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28364589874/job/84027207515
Full test suite (offline, mocked)        pass    2m47s    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28364589874/job/84027207483
Golden fixtures cross-OS (macos-latest)    pass    19s    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28364589874/job/84027207465
Golden fixtures cross-OS (ubuntu-latest)    pass    25s
https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28364589874/job/84027207474
Golden fixtures cross-OS (windows-latest)    pass    56s
https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28364589874/job/84027207567
Leak scan (PII / attribution guard)        pass    7s    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28364589874/job/84027207491
Mypy (lenient baseline)    pass    31s    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28364589874/job/84027207543
Ruff lint        pass    12s    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28364589874/job/84027207541
Syntax sweep + import smoke (no GPU stack)    pass    21s
https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28364589874/job/84027207532
=== #395 mergeable ===
{"mergeable":"MERGEABLE","review":"","state":"CLEAN"}
```

314. assistant (2026-06-29T10:14:11.864Z)

All 10 CI checks green and mergeable is CLEAN. Merging (squash, matching repo convention) and deleting the remote branch.

315. user (2026-06-29T10:14:31.071Z)

```
=== verify merged ===
{"mergeCommit":"0c9c693","mergedAt":"2026-06-29T10:14:29Z","state":"MERGED"}
```

316. assistant (2026-06-29T10:14:35.537Z)

#395 merged (`0c9c693`). Now syncing local `master`, cleaning up the local branch, and running the full freshness audit against the "fresh clone" bar.

317. user (2026-06-29T10:14:49.613Z)

```
Switched to branch 'master'
  tests/test_rally_seg_eval.py          | 48 ++++++-----
  3 files changed, 87 insertions(+), 11 deletions(-)
Deleted branch feat/rally-seg-eval-temporal-density-ab (was caabb04).
===== FRESHNESS AUDIT =====
--- [1] HEAD vs origin/master (want: 0 0) ---
0      0
--- [2] HEAD sha == remote master sha? ---
local : 0c9c6933887e3d6f87ed6f796e862540425d0838
```

```

remote: 0c9c6933887e3d6f87ed6f796e862540425d0838
--- [3] working tree status (want: empty) ---
--- [4] local branches (a fresh clone has only master) ---
* master 0c9c693 [origin/master] feat(eval): expose inactivity_temporal_density in the
rally_seg_eval A/B CLI (P2a) (#395)
--- [5] stashes (want: empty) ---
--- [6] in-progress op? ---
(none above = clean)
--- [7] worktrees ---
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer/badminton-highlight-indexer
0c9c693 [master]
--- [8] untracked INCLUDING ignored (cruft a clone would NOT have) ---
!! .mypy_cache/
!! .pytest_cache/
!! .ruff_cache/
!! __pycache__/
!! backend/__pycache__/
!! backend/annotations/__pycache__/
!! backend/api/__pycache__/
!! backend/compute/__pycache__/
!! backend/config/__pycache__/
!! backend/eval/__pycache__/
!! backend/eval/audio/__pycache__/
!! backend/features/__pycache__/
!! backend/infrastructure/__pycache__/
!! backend/pipeline/audio/__pycache__/
!! backend/pipeline/detectors/__pycache__/
!! backend/pipeline/personalization/__pycache__/
!! backend/pipeline/segmenters/__pycache__/
!! backend/pipeline/stitcher/__pycache__/
!! backend/pipeline/validators/__pycache__/
!! backend/providers/__pycache__/
!! backend/reporting/__pycache__/
!! backend/sports/__pycache__/
!! backend/storage/__pycache__/
!! backend/tools/__pycache__/
!! backend/utils/__pycache__/
!! backend/worker/__pycache__/
!! scripts/__pycache__/
!! tests/__pycache__/
!! tools/__pycache__/
!! training/__pycache__/
!! training/gen0/__pycache__/
--- [8b] count of ignored entries ---
31

```

318. assistant (2026-06-29T10:15:43.440Z)

Merged and audited. Here's the verdict against the "fresh clone" bar:

? Git state ? identical to a fresh clone at remote HEAD

```

| Check | Result |
|---|---|
| HEAD vs `origin/master` | **0 ahead / 0 behind** ? same commit `0c9c693` |
| Working tree (`git status`) | **clean** ? nothing modified, staged, or untracked-tracked |
| Local branches | **only `master`** ? all feature branches (`proposal/?`,
`docs/preserve?`/`fix/?`, `feat/?`) deleted |
| Stashes | none |
| In-progress merge/rebase/cherry-pick | none |
| Worktrees | single, on `master` |
| No unpushed commits / no pending PRs from me | confirmed |

```

So from git's perspective the repo is **pristine** and exactly at remote HEAD ? no pending items.

?? One delta from a *literal* fresh clone: gitignored build caches

The test/lint runs I did this session left ****31 gitignored cache dirs**** ? ``.pytest_cache/`,
`.mypy_cache/`, `.ruff_cache/`, and __pycache__` throughout. A brand-new `git clone` wouldn't have these. They're **gitignored** (hence `git status` is clean) and **fully regenerable**, so they don't affect correctness ? but they're the only reason this checkout isn't byte-for-byte a fresh clone.`

I verified that's the ***complete*** set of ignored-and-present files (nothing precious like a local ``config.json``, ``.venv``, DB, or ``output/`` artifacts is in there ? just caches).

****Want me to wipe them**** with ``git clean -dX`` (removes only gitignored files) to make it literally clone-fresh? I held off because it deletes files and you asked a state question, not explicitly to delete ? but it's zero-risk and one command if you say go.

Session-wide, everything else is closed: ****#394**** (fixes A+B, B flag-gated) merged, ****collector #50**** (C) merged, ****#395**** (P2a A/B harness) merged. The only open thread anywhere is the owner-only ****P2b**** golden-corpus promotion run ? which lives in your court, not this repo's tree. (Unrelated: the sibling ``sports-data-collector`` still has ***your*** pre-existing ``.gitignore`` WIP ? untouched, not mine to clear.)

319. user (2026-06-29T10:28:26.925Z)

Correct. so there is a different clone of this repo which is also on this machine at "C:\Users\avidu\Projects\badminton-highlight-indexer" and it has a lot more in it's scratch/ and also the F:\ drive which holds all the data we have used to build the repo's quality is also live on this machine now. So I plan to delete this repo and complete the P2b in that repo and the GPU, golden labels etc. are all available now, so that in the next session you can run all the tests and I can do a final validation signoff on whether to "launch" it. If that makes sense, can you give me a good prompt which I can use in the next new session which starts off this this item and helps me close it out fully

320. assistant (2026-06-29T10:29:24.728Z)

That plan makes sense ? and it's the right move: P2b genuinely needs the GPU + golden corpus + F:\ data that only exist on that other clone. Let me ground a few specifics so the handoff prompt is accurate (default manifest/features paths, the comparison metric, the promotion gate), then I'll write you the prompt.

321. user (2026-06-29T10:29:31.463Z)

```
=== rally_seg_eval defaults ===
backend/eval/rally_seg_eval.py:33:DEFAULT_MANIFEST = os.path.join("output",
"human_lovo_manifest.json")
backend/eval/rally_seg_eval.py:34:DEFAULT_FEATURES_DIR = "output"
backend/eval/rally_seg_eval.py:35:DEFAULT_IOU = 0.5
backend/eval/rally_seg_eval.py:84: iou: float = DEFAULT_IOU,
backend/eval/rally_seg_eval.py:184: manifest_path: str = DEFAULT_MANIFEST, features_dir: str
= DEFAULT_FEATURES_DIR
backend/eval/rally_seg_eval.py:316: iou: float = DEFAULT_IOU,
backend/eval/rally_seg_eval.py:347: iou: float = DEFAULT_IOU,
backend/eval/rally_seg_eval.py:404: iou: float = DEFAULT_IOU,
backend/eval/rally_seg_eval.py:628: ap.add_argument("--manifest", default=DEFAULT_MANIFEST)
backend/eval/rally_seg_eval.py:629: ap.add_argument("--features-dir",
default=DEFAULT_FEATURES_DIR)
=== promotion gate API ===
backend/eval/promotion.py:57: ``eps`` is the regression tolerance: 0.0 = "must not drop at
all on the served mean". A tiny
backend/eval/promotion.py:64: # Regression if the served mean dropped beyond tolerance, OR
the whole CI is net-negative.
backend/eval/promotion.py:117:def promote_if_served_safe(
backend/eval/promotion.py:147: served_regress_eps: served regression tolerance (0.0 = "no
served-mean drop allowed").
=== fusion_golden entrypoint (feature gen) ===
```



```

backend/eval/fusion_golden.py:233: def main() -> None:
backend/eval/fusion_golden.py:237:     ap.add_argument("--manifest",
default="output/human_lovo_manifest.json")
backend/eval/fusion_golden.py:238:     ap.add_argument("--out-dir", default="output")
backend/eval/fusion_golden.py:239:     ap.add_argument("--frame-skip", type=int, default=3)
backend/eval/fusion_golden.py:240:     ap.add_argument("--iou", type=float, default=0.5)
backend/eval/fusion_golden.py:241:     ap.add_argument(
backend/eval/fusion_golden.py:244:     ap.add_argument(
=== how --vs reports (compare output keys) ===
backend/eval/rally_seg_eval.py:13: diff two windowings with a paired sign-test + CI (the per-tier
`delta`).
backend/eval/rally_seg_eval.py:95:     Also carries the **R2** task-faithful block
(`tolerance_metrics`, the headline going forward;
backend/eval/rally_seg_eval.py:97:     tolerance-match `tol_f1/tol_precision/tol_recall` (over-
production costs precision) + the
backend/eval/rally_seg_eval.py:104:     from backend.eval import tolerance_metrics as tm
backend/eval/rally_seg_eval.py:113:     tol_matches, tol_p, tol_r, tol_f1 = tm.match_1to1(preds,
gts, ts, te)
backend/eval/rally_seg_eval.py:137:     # --- R2 (tolerance-match) ? the task-faithful
headline block ---
backend/eval/rally_seg_eval.py:140:     "tol_f1": tol_f1,
backend/eval/rally_seg_eval.py:141:     "tol_precision": tol_p,
backend/eval/rally_seg_eval.py:142:     "tol_recall": tol_r,
backend/eval/rally_seg_eval.py:143:     "tol_matched": float(len(tol_matches)),

```

322. assistant (2026-06-29T10:30:45.290Z)

I have the precise hooks now. Here's a ready-to-paste prompt for your next session. It's written to be self-contained (a fresh session has none of this context), starts in the *other* clone, and drives P2b ? full validation ? a launch go/no-go for you to sign off.

```

---
...

```

We're closing out the rally-detection quality work ? specifically P2b: the golden-corpus promotion decision for the "finding-B" rally-end density fix ? and then producing a launch-readiness signoff for me to approve.

WHERE TO WORK

- Use the clone at C:\Users\avidu\Projects\badminton-highlight-indexer (NOT any khelsutra-guru... nested path ? that clone was deleted).
- FIRST: `git fetch origin && git pull --ff-only` on master so you're at remote HEAD. master already contains: PR #394 (fixes A+B), #395 (the A/B CLI), and the collector fix. Then branch off origin/master for any change (one PR, no stacking).
- GPU + WSL, the golden labels/videos/trajectory CSVs, and the F:\ data drive are ALL live on this machine now ? so the real-video / golden / GPU test paths that used to be owner-only are runnable this session. Locate the golden manifest + features (likely on F:\ or under .\output); confirm the actual paths before running ? don't assume defaults.

RAMP UP (read before touching code)

- docs/RALLY_DETECTION_FIXES_PLAN.md ? S7 tracker (P2a ? done, P2b pending) and S8 Q2 has the exact A/B runbook + decision rule. This is the source of truth.
- docs/RALLY_DETECTION_CODE_REVIEW.md ? the original 3 findings (A/B/C).
- docs/HOW_RALLY_DETECTION_WORKS.md, docs/CODE_MAP.md ? mental model + navigation.
- Code anchors: backend/pipeline/detectors/tracknet_runner.py::_trim_inactive_tail (the fix, gated on `temporal\_density`); backend/config/models.py (`inactivity\_temporal\_density: bool = False` ? the flag to flip); backend/eval/rally_seg_eval.py::_build_preset (+ the --[vs]-inactivity-temporal-density CLI flags); backend/eval/windowing.py (WindowingPreset carries the flag); backend/eval/promotion.py ::promote_if_served_safe (the formal served-safety gate).

WHAT FINDING B IS / WHY THE FLAG

- _trim_inactive_tail's R4 rally-end trim measured "fast-frame density" over the count of active SAMPLES, so under sparse shuttle tracking one fast sample after a tracking gap read as 100% dense and held the rally tail open (inflated end). The fix uses the

trailing window's TEMPORAL length instead. It is a NO-OP under dense tracking, so it only moves on clips where the tracker actually drops out. R4 is default-ON in prod, so the fix shipped behind `inactivity\_temporal\_density` (default-OFF) = a zero-regression no-op until measured. P2b is that measurement + the promote/reject decision.

P2b PROCEDURE

1. Ensure golden features exist (regenerate if needed; GPU/WSL):
`python -m backend.eval.fusion_golden --manifest <golden_manifest.json> --out-dir <features_dir>`
2. Run the A/B at the SERVED operating point (flag OFF vs ON):
`python -m backend.eval.rally_seg_eval --manifest <golden_manifest.json> \`
`--features-dir <features_dir> --preset served --vs served \`
`--vs-inactivity-temporal-density --iou 0.5`
Headline metric = tol_f1 (R2 tolerance-match F1); also inspect |?end| / rally-end precision and the paired sign-test + CI delta. Apply promote_if_served_safe (served_regress_eps=0.0 ? no served-mean drop allowed).
3. DECIDE:
 - If tol_f1 does NOT regress (neutral or better) at SERVED ? PROMOTE: flip IndexingConfig.inactivity_temporal_density default to True (one-line change) + a test update; open one PR.
 - If it regresses ? DO NOT promote: leave the flag OFF (the dormant code stays), and record the numbers as the rejection rationale.Either way: update docs/RALLY_DETECTION_FIXES_PLAN.md §7 (flip P2b to ? DONE or mark REJECTED with the measured deltas) and follow docs/DOC_STATUS.md to honestly update ? archive the plan doc (P1, P2a already done; P2b is the last open row).

FULL VALIDATION (for my signoff)

- Run the complete offline suite: `python run_all_tests.py` (expect ~1487+ passing).
- Now that GPU/WSL/data are live, ALSO run the real-video / golden / native-WASB / GPU paths that were previously skipped ? enumerate what's runnable and run them; report what passed and anything still not verifiable.
- ruff check . and mypy must be clean (match CI); keep coverage at/above the CI floor.
- Cross-reference the quality table (docs/RALLY_DETECTION_QUALITY_REPORT.md / NEXT_STEPS ship target) so the launch call is evidence-based.

DELIVERABLE = a LAUNCH-READINESS REPORT, then stop for my signoff:

- ? P2b result (the A/B numbers + promote/reject decision + PR link)
- ? full test-suite status (offline + the now-unlocked GPU/golden paths) ? what passed, what couldn't run and why
- ? any open guardrail items (licensing, WASB weight provenance, etc.)
- ? a clear GO / NO-GO recommendation with the reasoning.

Recommend the launch call ? do NOT make it; I sign off.

GUARDRAILS: branch off origin/master; one PR per item; explicit `git add <paths>` (a sibling/owner session may share the checkout); never push to master directly. Honor CLAUDE.md (commercial-clean licensing; paid LLMs = inference only; owner-gated work).

...

A couple of notes on the prompt:

- **It deliberately makes the agent *recommend*, not decide, the launch** ? matching your "final validation signoff" intent.
- **It tells the agent the data paths are on F:\ and to confirm them** rather than trusting the code defaults (`output/human\_lovo\_manifest.json`, `--features-dir output`), since the real corpus lives elsewhere now.
- **It folds in "run all the tests"** including the GPU/real-video paths that were previously owner-only.

Two optional add-ons I can do right now if you want:

1. **Persist this as `session-handoff.md`** in the **other** clone's memory namespace (`\projects\c--Users-avidu-Projects-badminton-highlight-indexer\memory\`, which I can already write to) with a "? Start here" pointer in its `MEMORY.md` ? so the next session auto-ramps even if you don't paste the prompt.

2. ****Wipe the gitignored caches**** in this (soon-to-be-deleted) clone ? moot if you're deleting it, so I'll skip unless you say otherwise.

Want me to write the handoff file into the other clone's memory?

323. user (2026-06-29T10:31:42.208Z)

session handoff definitely and no need to wipe

324. assistant (2026-06-29T10:31:52.272Z)

On it. Let me first check what's already in the other clone's memory namespace so I don't clobber an existing index or handoff.

325. user (2026-06-29T10:31:56.830Z)

=== dir exists? ===

YES: c:/Users/avidu/.claude/projects/c--Users-avidu-Projects-badminton-highlight-indexer/memory

=== contents ===

lrwxrwxrwx 1 avidu 197609 80 Jun 15 11:01 c:/Users/avidu/.claude/projects/c--Users-avidu-Projects-badminton-highlight-indexer/memory -> /c/Users/avidu/OneDrive/Documents/claude-sync/memory/badminton-highlight-indexer

=== existing MEMORY.md (if any) ===

Memory Index

? Start here

- [? START HERE ? multi-shuttle / prominent-court rally detection](next-session-multishuttle-rally.md) ? ****icebreaker for the rally-DETECTION design journey (2026-06-19/20).**** WASB's detector sees every shuttle; its single-target tracker discards all but one (sometimes the wrong one). Built a multi-track + court-aware selector, ****measured vs GOLDEN truth (GX010128): it does NOT beat the single-track baseline as a general producer ? NO PR.**** It's a TARGETED lever (separable-adjacent-court + specific-rally recovery), not an F1 win; Gemini-reference oversold it. The 21-25s recovery is real + visually confirmed. ****? NEXT = the one experiment that could flip the verdict: a separable-adjacent-court golden clip (`Boxhill\_Doubles`, needs owner GPU run) with recall-preserving (court-membership-only) selection.**** Full detail [[next-session-multishuttle-rally]] + [[current-state]] top block. *(Design/scratch only; distinct from packaging + UX/serving tracks.)*

- [? START HERE ? batteries-included downloadable app](next-session-packaging-app.md) ? ****icebreaker + ramp-up for the packaging track.**** ****? P3 + P2b + P2c ALL DONE 2026-06-22:**** P3 GPU validation (\$0, L4?22 rallies?73.7MB reel, proof `gs://?/highlights/p3\_`); ****P2b-install ? engine [#310] MERGED**** (`f8b2c3b` ? seeds truly-local + ****motion**** default on first `indexer --app`, gate ****08**** resolved; `scripts/install.py`/`uninstall.py`); ****P2c wizard ? khelsutra [#88] MERGED**** (`7ae9cee` ? 3-step `SetupWizard` from `/api/capabilities`, 6 langs); ****bundle ? engine [#312] MERGED**** (`293e5c0` ? re-bundled `backend/ui/` so ****indexer --app` SERVES the wizard****, live-boot proven). *(The earlier "RECONCILE" note is RESOLVED.)* ****? v1 LIGHT is FEATURE-COMPLETE ? only OWNER steps remain: clean-Windows/macOS-box acceptance + ffmpeg-acquisition UX.**** North-star (native-WASB-in-default-download + training + annotation) stays future + counsel(03)/owner-GPU-gated. Tracked doc `docs/PACKAGING\_PLAN.md` (\$7); full detail [[packaging-app-plan]]. *(Distinct from the UX/l10n + serving tracks.)*

- [? START HERE ? khelsutra UX/l10n/runnable-tool](next-session-ux-localization.md) ? ****refreshed 2026-06-22.**** The runnable tool is ****BUILT + fully localized (6 langs en/kn/hi/es/da/id)****: in-UI upload (T4) ? process ? pick ? ****reel watermarked in the user's language**** ? download, + PMF locale analytics. DONE this session: A1/A4/A5/A8/A9, T4, MULTILINGUAL-A, B3(#70), T11(#299), ****locale?watermark+analytics (LW1 #304 / LW2 #73 / LW3 #306, verified via real watermark video for all 6 locales)****. ****? ALPHA IS LIVE + REAL-USE-PROVEN (2026-06-22) ? see [[current-state]] top "SESSION WRAP" for the full ramp-up.**** Use ****`https://api.khelsutra.guru`**** (SINGLE-ORIGIN: the engine serves the latest SPA + API behind one CF-Access cookie ? the split-origin `app.`+Pages topology broke the cross-origin cookie; `app.` needs a DNS edit the token can't make). Owner uploaded a real match in a browser ? ****26 rallies ? reel**** (the "build failed" was a Cloudflare 524, reel is fine).

Deploy/Finding-B/cross-origin-fix all done. ****? NEXT = the 13-issue triage backlog from real use**** (in current-state SESSION WRAP): top = ****async `/api/compile` [engine #329]**** (kills the 524) + ****M7 Cloud-Run L4 GPU**** (kills the CPU perf wall, ~\$0.08/match) + ****NSFW litmus [engine**

#332]** (hard gate before dropping the allowlist); plus UX polish (#98/#99/#101/#102/#103) + obs (#326/#328). Launch gate also still needs native-speaker review (es/da/id + watermark) + kn/hi font eyeball. *(Engine/serving track = own [session-handoff](session-handoff.md); don't conflate.)*

- [? START HERE ? golden regression fixtures (optional next items)](next-session-golden-fixtures.md) ? **icebreaker for the GOLDEN-FIXTURES track ? implementation COMPLETE 2026-06-22** (8 PRs #186/#189/#191/#192/#193/#194/#314/#317). A clean clone runs rally-seg regression (synthetic + **6 de-attributed REAL fixtures `fx\_r0X`**) with no video/GPU/DB, **cross-OS-enforced** (Win/mac/Linux) + leak-guarded + adversarially audited. Tracker `M1.M2.M3.MX.P1.P2.DOC.P0.P3` all ?. **? NEXT (optional, prioritized):** (1) **M4 quality-floor** on the real fixtures (per-slug F1-floor in `eval\_baselines/regression\_baseline.json` + a windowing fingerprint, CONTROL-only; real F1 is 0.0?0.59 so the floor is per-fixture `current?tol`); (2) the **deferred functional video-name rename** (`eval\_baselines` keys / `MULTICOURT\_CLIPS` ? coupled to the gitignored manifest+data+training-floor ? coordinated, before open-sourcing); (3) 01 protobuf / 02 Tier-1-key-gated (forward-looking). Locked: option (a) per-frame CSV · CONTROL-only · pinned preset. Full detail [[next-session-golden-fixtures]]. *(SEPARATE thread from packaging/UX/serving/quality.)*
- [Current state](current-state.md) ? **live micro-checkpoint** (updated every checkpoint): open PRs, what just landed, next steps, open decisions. **Read this first.** Now LIVE-only (handoff + today's checkpoints); kept read-first-short.
- [Current-state archive](current-state-archive.md) ? chronological checkpoint history (2026-06-06 ? 06-18, 96 blocks) split out of current-state 2026-06-19. Deep history only; NOT read each session.
- [Session handoff](session-handoff.md) ? current state · next steps · open decisions · ramp-up kit. AUTO-refreshed at meaningful checkpoints (PR merged, key decision, phase shift, session wrap) ? see working-agreement-checkpoints staleness tripwire.
- [Code map artifact](code-map-artifact.md) ? for CODE/architecture ramp-up, read `docs/CODE\_MAP.md` FIRST (dense anchor-rich map; replaces a file crawl). How to regenerate it.
- [File-placement conventions](file-placement-conventions.md) ? **where new files go** (scripts/ vs backend/eval/ vs tools/ vs scratch/ vs training/ vs artifacts/ ?); **read before committing a new file.** Authoritative table = `docs/CODE\_MAP.md` §0 (PR #138); scratch/ gitignored (#136).
- [Working agreement: attribution](working-agreement-attribution.md) ? never assert fabricated facts; NEVER invent a provenance (a discussion/review/source that didn't happen). The silver-bullet rule.
- [Working agreement: checkpoints](working-agreement-checkpoints.md) ? user prefers frequent small PRs + a Claude-consumable current-state file updated at every micro-checkpoint.
- [Feedback: tail long-job logs](feedback-tail-long-job-logs.md) ? proactively tail progress/summaries from long-running background jobs while they run, not just the final result.
- [Working agreement: merging](working-agreement-merging.md) ? owner authorized Claude to MERGE its own good-engineering-fix PRs on verified-green CI (2026-06-12); product/strategy/licensing stay owner-gated.
- [Working agreement: tracked project docs](working-agreement-tracked-project-docs.md) ? substantial design/project work gets a **tracked project doc** (template + §7 progress tracker + definition-of-done); freeze ? track across sessions ? mark done ? archive. Owner-loved framing (2026-06-16); codified in `CLAUDE.md` + `docs/PROJECT\_DOC\_TEMPLATE.md` (PR #186).
- [Doc-status register + archival due-diligence](doc-status-register.md) ? owner directive (2026-06-21): `docs/DOC\_STATUS.md` is the **living doc-lifecycle register** ; **archive foundational docs only AFTER honestly updating them** (verify?flip status+completion note?update refs?keep lineage?move). Worked example = DECOUPLED #270. Codified in `CLAUDE.md` archive policy.
- [Lesson: verify engine-surface before scoping](lesson-verify-engine-surface-before-scoping.md) ? **anti-spiral (2026-06-21):** before scoping a doc's `[gap]`/`net-new`/"X-only" **engine** claim as work, RE-VERIFY against engine code at a cited `file:line` ? the engine ships PRs between sessions so its surface drifts. Born from B-P1 ("gcs ingest is net-new" ? but `api/process` already accepted `gs://` since M2/#280 + tested). `[verified]` without a `file:line` = a hypothesis. Docs fixed ? khelsutra [#55].
- [Launch Report convention](launch-report-convention.md) ? owner directive (2026-06-17): EVERY default-config/quality-bit flip ships with a documented **Launch Report** (total-quality-impact table on both rulers + over-seg guard + honest caveats), recorded for posterity BEFORE the flip merges. Born from the Gate-A #146 flip that regressed (#151). Pairs with [[eval-dual-set-regression]].
- [Feedback: launch quality bar](feedback-launch-quality-bar.md) ? approaching launch (2026-06-20): tightly **isolated one-item PRs + rising coverage**, and **serious observability** (full log analysis across every NEW code path; correlation/job IDs; loud failures, no silent

drops). Make Observability a first-class doc section + DoD item.

- [Review: trust CI, don't re-run it](review-trust-ci-over-rerun.md) ? owner feedback (2026-06-25): ****don't re-run the exact CI checks locally during review**** (CI already ran them green on that commit ? wasted time); rely on green CI + a documented PR ****`## Verification`**** section, and spend local effort only on what CI can't cover (merge-safety, env-noise vs real defect, findings, machine-side paths). Refines the global pre-LGTM gate. Proposed the Verification convention on #378 (+offered a PR template).
- [Review: sign-off policy](review-signoff-policy.md) ? owner's settled rule (2026-06-25): ****sign off (written LGTM) on pure-refactoring / zero-behavior PRs once CI is green**** (no coverage bump needed; confirm semantics-free via AST-identity, not by re-running CI); ****HOLD logic/behavior changes for the owner**** unless explicitly authorized per-PR. Red CI blocks sign-off regardless (e.g. #379 mypy: `ruff format` moved a `# type: ignore` off its line ? AST-identity can't catch it). gh=avidullu ? written LGTM, never a formal Approve.
- [Gotcha: shared-checkout branch collisions](gotcha-shared-checkout-branch-collisions.md) ? spawned tasks + sibling sessions SHARE this checkout; ****branch from `origin/master` explicitly, re-verify HEAD before commit, stage explicit paths, serialize repo writes.**** The #116/#117 collision (owner upset). Global form in ~/.claude/CLAUDE.md.
- [Gotcha: long GPU/WSL runs](gotcha-long-runs-gpu-wsl.md) ? running the WASB pipeline E2E on a big 4K video: Modern Standby kills it (keep-awake), cv2 extraction is the bottleneck (ffmpeg pre-extract), proxy-index/4K-stitch recipe + how to observe each phase.
- [GCP Ops Agent: always enable](gcp-ops-agent-enable.md) ? owner directive (missed TWICE): every rally GPU VM needs the Ops Agent for Observability/GPU graphs ? ****3 pieces**** (logging+monitoring APIs ON + SA metric/log-writer roles + agent install via startup-script). BAKE into `deploy/gcp` provisioning so it's automatic.
- [GCP VM: always DELETE, never just stop](gcp-vm-always-delete.md) ? owner cost rule (2026-06-20): a STOPPED VM still bills its disk (~\$34/mo for 200GB pd-ssd); ****DELETE**** ? \$0 (boot disk autoDelete=true). `teardown.sh` now defaults to `delete`; audit `gcloud compute disks list` for orphans. DO droplet `claude-do-rally-test` (~\$24/mo) is a shared testbed (owner-gated).
- [Compute = GCP only; DO dropped](compute-stack-gcp-only.md) ? ****standing decision (ADR-013, 2026-06-20):**** GCP is the SOLE compute stack (training + serving); ****DigitalOcean dropped for compute**** (GPU not enabled/credit-funded; Paperspace subscription-gated). Provision via `deploy/gcp/` only; don't relitigate DO/Paperspace without a new ADR; \$200 DO credits ? non-compute only.
- [GCS = corpus source of truth](gcs-data-source-of-truth.md) ? ****owner goal (2026-06-22):**** run ALL train+infer from cloud with ALL data in `gs://khehsutra-rally-corpus` ? remove the local box as a blocker. ****Pull data from the bucket, not C:/F:/Drive.**** Authoritative map = `docs/DATA\_IN\_GCS.md`. Cloud engine already built (`backend.worker infer|train` by URI). Gap = ~25 MB corpus metadata (trajectories+manifest+labels) is local-only + trivially migratable; videos go Drive?bucket from a cloud box. Pairs with [[compute-stack-gcp-only]].
- [Cache-hygiene policy](cache-hygiene-policy.md) ? retention table (purge frames/staged vs keep CSVs/DB/reels) + the shipped tooling: self-cleaning runners (keep_frames default false), tools/cleanup_caches.py (dry-run-first), disk guard, SessionStart nudge hook. The `for e in` WSL footgun.
- [Tool: trim_video.py](tool-trim-video.md) ? reusable `scratch/trim\_video.py` for keyframe-aware LOSSLESS video trims (drop first/last N sec), `--exact` re-encode with NVENC?libx264 fallback. Reach for it instead of hand-rolling ffmpeg trims.
- [Decision-rigor norm](decision-rigor-norm.md) ? ****NUMBERS INFORM, FOUNDATIONS DECIDE**** (owner directive 2026-06-14, set by the Gate-A flip). Every signal/architecture change gets ablation-layer rigor (per-signal marginal ?F1 + honest CI/sign-test) AND a foundational lens (extensible/orthogonal/reversible), never numbers alone. Ties [[weak-signals-retained]] + [[paper-coauthor-aim]].

Project: rally segmentation pipeline

- [Candidate segments & fine-tuning](candidate-segments-finetuning.md) ? why raw detector candidates are persisted (detector-agnostic table); golden-eval-set goal
- [TrackNetV4/WASB via WSL2](tracknet-wasb-wsl2-decision.md) ? self-hosted shuttle trajectory models (not LLM APIs); WASB/TrackNetV3 = MIT, commercial-clear
- [Collector?indexer bridge](collector-indexer-bridge.md) ? 4-phase plan to train/eval a generic temporal rally segmenter; Phases 1?2 done, 3 in progress, 4 next
- [Golden-set vendoring](golden-set-vendoring.md) ? generate golden rally labels via 3rd-party vendors (Bangalore-local) + extensible regression flow; Roora selected for a 3-sport pilot, guideline + collector ingestion built, GS-1/2/4 landed, GS-3 + gold key pending
- [Roora guidelines Drive doc](roora-guidelines-drive-doc.md) ? the CANONICAL vendor-facing

Brief+Guide Google Doc (supersedes v3/v5/v6/v7 Drive copies; v7 anomaly RESOLVED 2026-06-12);
 TODO markers still to fill before sending

- [Audio (E1) findings](audio-e1-findings.md) ? quantitative gains/losses of the classical-onset audio cue (recall 100% but discrimination ~2.2x, band-pass null) + reusable lessons for adding the NEXT signal. Audio NOT adopted for fusion; detector kept (PR #35).
- [Weak signals retained](weak-signals-retained.md) ? owner directive: non-significant cues (person/audio/court-bounds/future **shuttle-drop**) stay **default-OFF + retained** as scorer feature columns (no special-casing) for a growing (Roora) corpus ? never deleted on a small-n null. Measured: person ?F1 ?0.021, +audio +0.079 (both CIs span 0 at n=6).
- [Eval boundary tolerance](eval-boundary-tolerance.md) ? owner directive (2026-06-16): rally scoring must tolerate the ~1?1.5s **service-window** start-fuzziness ($\pm 2s$, not strict tIoU 0.5 which only allows $\pm 1.9s$ on a 5.7s rally). KEY measured finding (mp2+GX010128, n=2): local's |?start|?Gemini but |?end| 2?3x worse ? **the gap is rally-END detection**, not generic boundary precision ? next lever (R2 metric ? R4 end-rule). Reusable `scratch/at\_par.py`.
- [Golden corpus grew 9?15](golden-corpus-grew-9-to-15.md) ? **2026-06-22:** the #316 GCS migration RESTORED the original-6 single-court labels, so the local golden eval set is now **15 clips** (was 9 multicourt). Trips any 9-clip-pinned litmus (e.g. `test\_served\_gate\_a\_regression`, skipped-in-CI so master CI stays green while the local run breaks) ? re-grounded in PR #322 (merged); check corpus size before suspecting a code regression. Pairs with [[eval-dual-set-regression]] [[gcs-data-source-of-truth]].
- [Eval dual-set regression](eval-dual-set-regression.md) ? owner directive (2026-06-16, proven practice): keep a **RAW eval set** (currently-bad videos, random sample) + a **GROWING GOLDEN-finalized eval set** (promote raw?golden when at-par) + a separate **golden TRAINING set**;
 every quality launch must NOT regress on EITHER eval set (don't lose a graduated win). The **ablation layer A/B**s these. Checkpointed video folder; wire dual-set aggregate into `run\_regression.py` when R2 lands.
- [Decision-snapshot regression](decision-snapshot-regression.md) ? **tracked project, decisions LOCKED, #383 merged (2026-06-25).** A deterministic **quality-neutral proof**: freeze+compare the pipeline's DECISION output (segments + stitch plan) off a cached trajectory, in CI ? the behavioral proof that lets a *genuine* (non-AST-identical) refactor be auto-called quality-neutral. **NOT a reel-MD5** (non-deterministic). D10 = double-run determinism guard (deterministic segmenters only); D11 = advisory?gating staging; B1/#385 is the enabler. **? next = §7 P1.** Doc `docs/DECISION\_SNAPSHOT\_REGRESSION.md`.

Packaging / distribution

- [? Next session: packaging app](next-session-packaging-app.md) ? **the icebreaker/handoff to resume this track** (read first; also under ? Start here).
- [Batteries-included app plan](packaging-app-plan.md) ? downloadable batteries-included Python app (local web service for inference/training/annotation, local+cloud). **11 PRs merged 2026-06-21/22** ? LIGHT-tier engine bundle-ready (`pip install`?`indexer --app`?bundled SPA, torch-free). Tracked doc = `docs/PACKAGING\_PLAN.md` (§7 + ? NEXT SESSION block). **01 (cross-platform, no native component) + 02 (LIGHT-first, WASB-complete north star) LOCKED**; 03+ (WASB-C3 counsel, hosted-vs-BYO-cloud) gated. Per-PR detail inside. Pairs with [ui-driven-cloud-serving-gap](ui-driven-cloud-serving-gap.md). Don't relitigate the locked calls.

Monetization (hosted API)

- [Monetization plan](monetization-plan.md) ? hosted video?highlights API on a GPU VM; licensing & per-video cost are first-class; user is Bangalore-based
- [Monetization research status](monetization-research-status.md) ? audit done: Ultralytics AGPL swapped out (PR #8); MonoTrack/YOLO-NAS weights avoided; codec=watch; full report in docs/
- [Machine-pooling idea](machine-pooling-idea.md) ? owner "food for thought" (2026-06-20): pool GPU machines (queue-on-one-host vs Cloud-Run scale-to-zero) instead of spin-up/kill per task ? born from a single-GPU-slot collision between two sessions (quota=1). Infra-track design item.
- [UI-driven cloud-serving gap](ui-driven-cloud-serving-gap.md) ? **owner directive (2026-06-21): the cloud highlight flow must be UI-driven + ALPHA-SHAREABLE, not SSH/CLI; surface compute options (local vs cloud) in the UI at start.** Today = SSH-only; M12 ? the fix. Needs: M7 deploy (persistent service) + SPA ingest screen (khelsutra B-P2) + M9 auth ON + a UI compute-picker (D13/D15). Engine-makes-possible vs SPA-makes-usable. **? The compute-picker (both halves) + a real Cloud Run L4 GPU e2e validation LANDED 2026-06-22 ? see [[cloudrun-gpu-serving-gotchas]] + current-state top block.**
- [Cloud Run GPU serving gotchas](cloudrun-gpu-serving-gotchas.md) ? **reusable knobs to run the WASB worker on Cloud Run L4 GPU** (region-limited to asia-southeast1/etc NOT Mumbai · `--no-gpu-zonal-redundancy` · GCS volume-mount for weights · PowerShell-not-Bash for `mnt` flags · `--set skip\_ai\_handoff=true` for no-key · digest-pin ? `jobs update` after rebuild · `[storage-gcs]`

in the image · control plane needs google-cloud-run+ADC). Verified on the 2026-06-22 e2e (proof = `artifacts/cloudrun\_e2e\_validation/`).

- [khelsutra site droplet parity](khelsutra-site-droplet-parity.md) ? ****RETIRED 2026-06-25.**** The DO droplet `claude-do-rally-test` was ****dual-purpose**** (site mirror :8787 + alpha engine `api.khelsutra.guru` :8000) and kept drifting from CF ? nightly `parity.yml` failures. Owner chose ****retire****: data archived to `gs://khelsutra-rally-corpus/archive/droplet-?-20260625/` (+ local), droplet ****deleted**** (\$0, no orphans), monitor removed via khelsutra ****PR #117****. Follow-ups: `api.` CNAME/CF-Access cleanup + stale droplet runbook docs. Re-sync recipe kept as history. Pairs with [[compute-stack-gcp-only]] [[ui-driven-cloud-serving-gap]].

Forward-looking

- [Sibling internship repo](sibling-internship-repo.md) ? khelacharya shot-intelligence layer builds ON TOP of our rally segmentation; kept off master at branch level (NOT a separate GitHub repo ? corrected 2026-06-06)
- [Paper co-author aim](paper-coauthor-aim.md) ? ****standing side-aim****: co-author a paper on "local, cheap & efficient rally detection via multiple clever ideas". Keep `docs/QUALITY\_ITERATIONS.md` as the ablation-ready empirical backbone; honest measured results ONLY.

=== existing session-handoff.md (if any) ===
EXISTS ? will update in place

326. assistant (2026-06-29T10:32:34.576Z)

This is the main clone's canonical memory (symlinked to OneDrive, shared across sessions) ? far more advanced than the clone I worked in, with several per-track "? Start here" icebreakers and a `session-handoff.md` that the index explicitly scopes to the ****engine/serving track**** ("don't conflate"). So I must ****not**** clobber `session-handoff.md`; the right move is a dedicated icebreaker for this track plus a "? Start here" pointer ? matching the established `next-session-\*.md` pattern. Let me read the existing handoff and the live current-state to ground the new file accurately.

327. user (2026-06-29T10:32:36.816Z)

=== session-handoff.md (first 40 lines ? confirm it's serving-track) ===

```
---
name: session-handoff
description: "Living \"you are here\" handoff ? current state, next steps, open decisions, ramp-up kit. AUTO-REFRESHED at meaningful checkpoints (PR merged, key decision, phase shift, session wrap). \"update the handoff\" forces a full rewrite."
metadata:
  node_type: memory
  type: project
  originSessionId: 45ea3a91-654d-4972-9cd3-65f35db54caf
---
```

Session handoff ? badminton-highlight-indexer (rename PENDING) + siblings: sports-

****Last updated:**** 2026-06-25 (****review-automation + audit-completion session**** ? see ? You are here (2026-06-25) below. Maintenance/PR-review phase, NOT serving/alpha: 0 open PRs, all CI green, no GCP VMs, PR-review watch loop OFF. The serving/alpha track below is UNCHANGED + owner-gated.) Prior: 2026-06-22 (****M11+M12 serving-code session**** ? owner: "build 3 then 4 then we pair on the alpha (2)". ****Both Claude-doable engine items DONE + merged:**** ****M11**** data-flywheel skeleton [#336](https://github.com/Khelsutra/badminton-highlight-indexer/pull/336) (`334ad0f` ? default-OFF + consent FAKED-DENY, inert; reuses workspace/eval/promotion; NO migration) + ****M12**** `gdrive-api://` OAuth StorageBackend [#338](https://github.com/Khelsutra/badminton-highlight-indexer/pull/338) (`819e395` ? idempotent put, lazy Google imports, injected-fake tests; live per-user OAuth owner-gated). Both adversarially reviewed, full-suite+cross-OS green, merged on green, in an isolated worktree. ****NEXT** = pair with owner on the ALPHA GO-LIVE (option 2) ? owner brings CF dashboard + box + rotated Gemini key; the counsel ToS/DPA inputs (the 7-pt checklist) gate M11 *live* + M9-consent. ? schema ladder drifted under me this session: v7=`locale`, v8=compute-picker ? the future M9-consent migration is ****v9**** (check `PRAGMA user\_version` before authoring). Prior same-day: ****maintenance / green-PR merge-sweep session**** ? owner asked to merge the green PRs + finish loose ends. 6 PRs now on master: worker

dated-label fix **##319**, nightly Cloud-Run scheduler **##320**, DATA_IN_GCS map **##316**, doc **##321**, gate-A litmus re-ground **##322**, golden-fixtures backlog **##323**. engine `origin/master` = `9b37269`, suite **1428?**, **0** open engine PRs; docs-archive had no terminal candidate (owner-gated). Full per-PR detail = `[[current-state]]` top. ? ~10 concurrent worktrees/sessions ? owner is trimming concurrency before new work. The serving/alpha plan below is UNCHANGED. Prior **2026-06-21** ALPHA-SHAREABLE session ? Claude's half of "make it UI-driven + alpha-shareable" is **DONE**: 3 PRs merged ? engine **294** exposure-safety boot posture + khelsutra **64** SPA `VITE\_API\_BASE` + **65** deploy kit. **KEY CORRECTION**: B-P2 (SPA ingest screen) was ALREADY shipped (#56) ? alpha-shareable was a DEPLOY/AUTH task, not an SPA build. Owner chose the **SPLIT** topology (SPA on CF Pages + `api.` cloudflared tunnel + CF Access). Remaining = OWNER stand-up via `khelsutra/deploy/DEPLOY\_ALPHA.md`; then engine M11/M12. Full record `[[ui-driven-cloud-serving-gap]]`. Prior: HUGE serving session ? M1?M9-auth shipped (#279?#291) + real M7 L4 run. **Repo** is under the `Khelsutra` GitHub org.

> **Read order**: this file ? `[[current-state]]` top (dense per-checkpoint detail) ?
`docs/NEXT\_STEPS.md` (the prioritized cross-track plan, refreshed 2026-06-21) ?
`docs/DOC\_STATUS.md` (doc lifecycle/hygiene).

? You are here (2026-06-25)

Review-automation + audit-completion session. engine `origin/master` = **0b4f492** . all CI green . **0** open PRs . no GCP VMs . **PR-review** watch loop OFF (cancelled). NOT serving/alpha (that track is the "Earlier" block below, unchanged + owner-gated).
- **Repo tidy**: synced master, pruned ~92 stale local branches + 7 stale worktrees; **fixed** a corrupted (UTF-16-tail) `.gitignore` ? `artifacts/` ignored + `KHELSUTRA\_PRODUCT\_OVERVIEW.md` tracked (**377**); owner un-hid the hidden `C:\Users\avidu\Projects` folder.
- **khelsutra site-droplet RETIRED** (`[[khelsutra-site-droplet-parity]]`): the dual-purpose `claude-do-rally-test`/`168.144.126.176` (marketing-site mirror + alpha engine) kept drifting from CF ? **deleted** (\$0; data archived to `gs://khelsutra-rally-corpus/archive/droplet-?-20260625/` + local), parity monitor removed (khelsutra #117), Node-20 actions bumped (#118). ? This means the prior "LIVE alpha on the droplet" in the Earlier block is **GONE** ? alpha is paused pending the Cloud-Run move; `api.khelsutra.guru` CNAME/CF-Access need cleanup.
- **CODE_CLEANUP_AUDIT** advanced ? 3 HIGH items + infra all merged: **B2/P17** surface-Gemini-failures (**378**) . **O1/P11** Gemini remote-file GC (**380**) . **B1/P9** remove-fabricated-motion-data (**385**); + ruff-format split (#379), audit-doc reconcile/accuracy (#375/#376), Tier-1 nightly `nightly.yml` (#381), per-job CI timeouts (#382). **Remaining audit items**: D1 (split `main.py` ~1450 lines), D13 (ByteTrack?`trackers`), Phase-3 (B3/B4/B6/B7/B8/D5/D6/D8-D10/O3/O4).
- **NEW tracked project: Decision-Snapshot Regression** (`[[decision-snapshot-regression]]` . `docs/DECISION\_SNAPSHOT\_REGRESSION.md` . **383** merged) ? a deterministic **quality-neutral** proof (freeze+compare the DECISION output = segments + stitch plan off a cached trajectory; **NOT** a reel-MD5). **Decisions D1?D11 LOCKED** (D10 = double-run determinism guard, deterministic segmenters only; D11 = advisory?gating staging). **?? NEXT = \$7 P1** (canonical snapshot schema/serializer) ? P2 (`build\_stitch\_plan`) extract ? P3 (extend `test\_golden\_fixtures` + guard). B1/#385 makes `motion` snapshot-eligible.
- **Review-automation patterns established** (`[[review-signoff-policy]]` + `[[review-trust-ci-over-rerun]]`): spin off **multi-agent review Workflows** (Ultracode; agents READ-ONLY) ? adversarially verify ? post synthesized findings; **sign-off policy** = LGTM pure-refactors on green (AST-identity vs master), HOLD logic/CI-infra for owner; **written LGTM only** (gh=`avidullu` = owner's own account ? GitHub blocks self-Approve); **trust green CI + a PR** `## Verification` section, don't re-run identical CI checks ? reserve local effort for what CI can't cover (AST-identity, merge-safety, env-noise-vs-real-defect, findings-actually-fixed).

? Likely next pick-up: start the Decision-Snapshot project at **\$7 P1**, or continue the audit (D1 split-main.py / D13 ByteTrack). Serving/alpha remains owner-gated (now needs the Cloud-Run move since the droplet is gone).

Earlier ? serving/alpha track (2026-06-22/23, UNCHANGED + owner-gated; note the droplet alpha is now retired per above)

? You are here (2026-06-22)

(2026-06-23 #342) engine `origin/master` = **688d28f** ? merged a small compute-path fix: `GoogleCloudRunJobClient.run\_job` now guards an unset/empty `GOOGLE\_CLOUD\_PROJECT` (was passing

`Optional[str]` into `JobsClient.job\_path(project: str,?)` ? latent mypy `arg-type` seen only with `google-cloud-run` installed + a bogus `projects/None/?` path at runtime). Guard runs before the lazy SDK import (CI-testable, SDK-absent) and raises a clear `RuntimeError`; +`test\_real\_client\_rejects\_missing\_project`. Full suite 1488?, cov 85.86%, all 10 CI green, merged on green per [[working-agreement-merging]]. Follow-up doc [#345](https://github.com/Khelsutra/badminton-highlight-indexer/pull/345) (`9329473`) added the operator note to `deploy/cloudrun/README.md` Gotchas. (Originated as a spawned chip; the live cloud-toggle work above is the sibling track ? see [[current-state]] top.)

(2026-06-22 M11+M12) engine `origin/master` = **`819e395`** (past #336 M11 + #338 M12; also sibling #333 M6?/api/process, #334 Cloud-Run env, #6d3d054 SPA compute-picker) · khelsutra `origin/master` = `ba38f15` (past #96 Privacy Policy, #100). **The two Claude-doable serving-code items are DONE** (M11 flywheel skeleton #336 + M12 gdrive-api #338 ? both default-OFF/inert, adversarially reviewed, merged on green). **Claude's serving + alpha-shareable code is now complete.** **?? THE ONLY REMAINING ALPHA STEP IS OWNER-DRIVEN: pair on the go-live** (cloudflared tunnel + CF Pages + CF Access + Gemini-key rotation per `khelsutra/deploy/DEPLOY\_ALPHA.md`) ? Claude guides, owner drives the CF dashboard/box. M11 going *live* + M9-consent need the counsel ToS/DPA inputs (the 7-pt checklist in [[ui-driven-cloud-serving-gap]] / this session). Other future engine code: M7 real cloud-serving e2e (owner GCP spend), per-video workspace P1?P6, the rally-quality track (corpus-gated).

(2026-06-22 sweep) engine `origin/master` = **`9b37269`** (+ today's #316/#319/#320/#321/#322/#323) · engine **~1428** tests green (combined-master verified w/ corpus manifest) · **0 open engine PRs** · no GCP VMs. This was a cross-track maintenance sweep (worker fix, nightly scheduler, GCS-data doc, gate-A litmus re-ground, golden-fixtures backlog) ? the serving/alpha forward-plan below is unchanged. ? a sibling's redundant gate-A branch `fix/served-gate-a-regrounded-15clip` sits in the MAIN checkout; concurrency being trimmed.

(2026-06-21 prior) engine `origin/master` = `0c0e064`+ (past #292/#293/#294) · khelsutra `origin/master` = `4627830` (past #63/#64/#65) · **no GCP VMs (audited \$0)** · engine ~1259 tests green.
?? ALPHA-SHAREABLE SESSION (2026-06-21) ? 3 PRs MERGED; Claude's half of the owner's top priority is DONE:
- **KEY CORRECTION (anti-spiral [[lesson-verify-engine-surface-before-scoping]]):** the handoff said "do the B-P2 SPA ingest screen + M7 Cloud Run *service*" ? but **B-P2 was already shipped** (khelsutra #56, live e2e in `DRY\_RUN\_CUJ.md` §9) and recon showed alpha-shareable is a **DEPLOY/AUTH** task, not an SPA build. CORS is already env-driven; the engine needed no CORS code.
- **Owner decisions (this session):** (1) pursue alpha-shareable via topology **(i) Tunnel-to-a-box** ; (2) sub-topology = **SPLIT** (SPA on CF **Pages** + engine on `api.` **cloudflared tunnel** + CF Access over both ? engine `HOSTING\_PLAN.md` §2), single-origin Caddy kept as the alternative.

=== current-state.md (first 35 lines ? the read-first live file) ===

name: current-state

description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not only on request). What just landed, what's next, open decisions."

metadata:

node_type: memory

type: project

originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301

Current state ? updated continuously (see [[working-agreement-checkpoints]])

> **Scope of this file = LIVE only:** the current handoff + *today's* checkpoints + open PRs/decisions. Keep it read-first-short (?1 page). Older checkpoints are archived in [[current-state-archive]] ? when today's checkpoints age out (next clean-slate day), move them there, don't let them pile up here.

?? SESSION WRAP ? 2026-06-23 (owner called a clean slate; out of budget, "human runs the loop" until quota refresh). READ FIRST on return.

- **THE WIN:** owner demoed a small clip ? highlights to a friend = **"pure bliss."** The whole

cloud/serving/docs arc this session shipped + merged.

- **WHAT SHIPPED (all merged):** cloud compute-picker **validation method** (#341 ? ``result.compute.path`` proves which backend ran + the GCP-verifiable Cloud Run execution; ``scripts/verify_compute_path.py``; live-proven cloud?cloud_run+22 rallies, local?in_process) . `async `/api/compile` (#339/#108) . #342 GOOGLE_CLOUD_PROJECT guard . user guide`
- docs/CREATING_REELS.md** (#343) . operator runbook
- docs/CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md** + committed ``deploy/cloudrun/cloudbuild.yaml`` (#346) . **khelsutra/deploy/GPU_VM_SERVING.md** (#113, owner-merged) = move the live engine off the droplet onto an L4 GPU VM + cut the token-managed tunnel over, with a **Gemini?WASB detection switch**. Droplet full-install suite GREEN (1461 passed).
- **OPEN THREADS for return (prioritized):** (1) **khelsutra** #114(<https://github.com/avidullu/khelsutra/issues/114>) ? CF Access: an ALLOWLISTED guest email still couldn't sign in (blocks sharing the alpha; likely the email isn't in the right Access app's policy / selector-type / OTP delivery ? full triage in the issue). (2) Owner wants a **DOC-CONSOLIDATION pass** (deferred ? budget). (3) Backlog (``docs/BACKLOG.md`` ? **Batch / declarative orchestration**): a **one-YAML `indexer --batch`** driver + a **one-shot GPU-VM `serving-startup.sh`** (design hinge = secrets out of plaintext metadata ? Secret Manager). (4) Make the cloud toggle live on the alpha (owner deploy ? GPU VM per #113, or the droplet stays).
- **STATE:** alpha LIVE on the droplet (CPU, ``gemini`` segmenter) at ``api.khelsutra.guru``; GPU-VM-serving is documented not deployed; 0 of my PRs open; GCP \$0 (all validation resources torn down); only sibling worktrees remain. `[[cloudrun-gpu-serving-gotchas]]` `[[gotcha-shared-checkout-branch-collisions]]`

Checkpoint: 2026-06-23 (**CLOUD TOGGLE LIT + VALIDATION METHOD + LIVE E2E PROVEN + user guide + droplet full-suite**). master ``688d28f``. Owner: "light up the cloud compute toggle with Cloud Run + a good validation method to ensure the intended path was taken" ? then "run the test on the droplet with a full install" ? "write a doc to help users set up + create reels." ALL DONE:

- **Validation method ? engine** [#341](<https://github.com/Khelsutra/badminton-highlight-indexer/pull/341>) MERGED (``d92cc10``). **compute-path PROVENANCE** ? the job result carries ``compute={path:"in_process"|"cloud_run"|"not_run", requested, execution, job, region, outputs_uri}`` (the cloud ``execution`` = the real Cloud Run execution name, GCP-verifiable). ``ComputeResult.execution`` threaded via ``run_infer``; honest failure records (path=cloud_run only when it actually dispatched+ran, else not_run); ``[COMPUTE]`` audit log;
- scripts/verify_compute_path.py** asserts the path + cross-checks the execution in GCP.
- Also forwards `indexing.skip\_ai\_handoff`** to the cloud job so the picker's cloud path yields rallies without a Gemini key. Adversarial review caught 2 honesty bugs (false "report carries it" claim; misleading log on a never-dispatched reject) ? fixed. Suite 1480?.
- **LIVE E2E PROVEN** (\$0 after teardown): rebuilt the Cloud Run L4 image + Job; ran a local **cloud-serving** engine (``deployment.profile=cloud-serving`` + ``skip_ai_handoff=true`` override + ``CLOUD_RUN_JOB/REGION=asia-southeast1/PROJECT`` + ``storage.output_root``) ? ``/api/capabilities`` reported ``cloud_available:true`` (toggle lit). ``verify_compute_path.py --compute cloud`` ? ``path=cloud_run``, execution ``rally-worker-qbcqb`` (GCP cross-check: `succeededCount=1`) ? **22 rallies** (skip_ai_handoff forward works). ``--compute local`` ? ``path=in_process`` ? Both directions provably correct. Proof logs in ``artifacts/cloudrun_lightup/``. Job+image+AR repo+prefix deleted.
- **Verify-script Windows bug** ? [#344](<https://github.com/Khelsutra/badminton-highlight-indexer/pull/344>) OPEN: ``--verify-gcp`` spawned bare ``gcloud`` (Windows = ``gcloud.cmd``, WinError 2) ? resolve via ``shutil.which``. Found during the live run (path assertion passed first).
- **[#342]**(<https://github.com/Khelsutra/badminton-highlight-indexer/pull/342>) MERGED (``688d28f``) ? ``GoogleCloudRunJobClient.run_job`` now guards an unset ``GOOGLE_CLOUD_PROJECT`` (originated as a spawned background-task chip). It passed ``project`` (``Optional[str]``, from the ``GOOGLE_CLOUD_PROJECT`` env in ``get_compute_backend``, can be None) straight to ``JobsClient.job_path(project: str, ?)`` ? a real latent bug: mypy ``arg-type`` visible ONLY with ``google-cloud-run`` installed (CI green because the SDK is deliberately NOT a CI/test dep), and at runtime a None/empty project builds a bogus ``projects/None/locations/?`` path on the real dispatch. Fix = guard BEFORE the lazy SDK import (so it's CI-testable, SDK-absent) ? raise a clear ``RuntimeError`` naming the env var; narrows ``Optional[str]`` ? ``str`` for ``job_path``. + ``test_real_client_rejects_missing_project`` (None + ""); compute suite 14?15, full suite **1488?**, cov **85.86%**, all 10 CI green. Verified the error reproduced at ``cloud_run.py:74`` pre-fix (the earlier broad-``mypy backend`` false-pass was a stale ``mypy_cache`` ? caught via ``--no-incremental``). Follow-up doc [#345](<https://github.com/Khelsutra/badminton-highlight-indexer/pull/345>) (``9329473``) noted ``GOOGLE_CLOUD_PROJECT`` is required in ``deploy/cloudrun/README.md`` Gotchas. `[[working-agreement-merging]]`
- **Droplet FULL-install suite GREEN:** pushed master? ``root@168.144.126.176:~/bhi-fulltest``,

fresh venv + `pip install -e .[dev,auth,storage-gcs,storage-s3]` + torch CPU + ffmpeg/cv2 libs ?
1461 passed, 12 skipped (the skips = the private `obsreport` git+ssh dep, not installed).
Droplet restored (dir removed, 1.6G freed).
- **? User guide ? [#343](https://github.com/Khelsutra/badminton-highlight-indexer/pull/343)
OPEN:** new **`docs/CREATING\_REELS.md`** (end-user: prereqs?`indexer --app`?wizard?reel via web
UI AND the API chain
capabilities?upload/gs:///process[202+poll]?query?compile[202+poll]?download; +cloud toggle
+troubleshooting). Researched via a verify-then-write workflow (every command/endpoint/flag
checked vs the shipped engine; LIGHT-tier-honest). Indexed in docs/README.md + DOC_STATUS.md
§3d.
- **? NEXT:** merge #343+#344 on green; clean worktrees (bhi-computeval/bhi-userdoc/bhi-
verifyfix). Owner follow-up to make the toggle live on the ALPHA = deploy a persistent Cloud Run
Job + flip the droplet engine to cloud-serving (a deploy decision, not code). [[cloudrun-gpu-
serving-gotchas]] [[gcp-vm-always-delete]] [[working-agreement-merging]]

Checkpoint: 2026-06-22 (**ASYNC `/api/compile` [#329] ? engine [#339] + SPA [#108] BUILT;
kills the Cloudflare 524 on slow reel stitches**). engine PR
[#339](https://github.com/Khelsutra/badminton-highlight-indexer/pull/339) (branch `feat/async-
compile`, CI running?merge on green), khelsutra
[#108](https://github.com/avidullu/khelsutra/pull/108) **MERGED** (`ed33a51`). The owner's first
real reel 524'd (origin timeout) while the reel actually built ? `/api/compile` ran the ffmpeg
stitch SYNC (~3min on the 2-core droplet > CF's ~100s edge). **Fix = make it async, mirroring
`/api/process`:** submit ? **202 + compile job id**; cheap checks (video/segments/floor/safe-
name) fast-fail 4xx at submit; the stitch runs on the worker; the client **polls
`/api/jobs/{id}`**.** **Engine #339:** schema **v9** (`jobs.kind` 'process'|'compile' NULL?process
+ `jobs.params` JSON; additive/idempotent) + `create\_compile\_job` + `\_run\_claimed\_job`
dispatches on `kind` (compile?`\_execute\_compile`, else unchanged process path; cost/flywheel
process-only) + `\_resolve\_compile` shared submit/worker + a stitch error ? clean
`done`+`result.status=failed` (no crash) + the in-repo `backend/api/static/app.js` polls + **re-
bundled `backend/ui/`** from #108. Suite **1473**?, ruff+mypy clean, cov **85.60%**; adversarial
review clean (its one finding ? "clients must poll" ? fixed via app.js + the re-bundle). **SPA
#108:** `compileReel` submit?`pollCompileJob`?`CompileResult`; a `done`+`failed` result throws
an ApiError carrying the engine message (so classifyError surfaces it, not "engine
unreachable"); `onBuild` unchanged (still `highlightUrl`, split-origin-safe); 228 web + 95 site
tests green. ? **schema head is now v9 = compile job (NOT compute-picker ? that's v8);
M9-consent migration ? v10** (the prior checkpoint's "M9-consent=v9" is now stale ? check
`PRAGMA user\_version`). **Deploy:** #339 ships the API + the poll-aware bundle together ? deploy
as ONE unit (don't deploy async engine vs an old sync bundle). [[gotcha-shared-checkout-branch-
collisions]] [[working-agreement-merging]]******

Checkpoint: 2026-06-22 (**M11 + M12 serving-code DONE ? 2 PRs merged; next = pair on the
alpha go-live**). Owner: "build 3 (M11) then 4 (M12) then we pair on the alpha (2); tell me the
counsel inputs and I'll have them ready." Both Claude-doable serving items merged on green
(isolated worktree off origin/master): **M11** data-flywheel skeleton
[#336](https://github.com/Khelsutra/badminton-highlight-indexer/pull/336) (`334ad0f`) ?
`backend/flywheel.py` capture/shadow_eval/propose_promotion + `FlywheelConfig` (default-OFF) +
gated `\_maybe\_run\_flywheel` hook; **INERT via TWO guards** (`enabled=False` AND
`consent\_attested` faked-deny ? captures nothing in prod); reuses
workspace/eval/promotion/storage; **NO migration**; +14 tests incl. a guardrail-lock;
adversarial review (`wf\_1ee5ffdf`) inert_guarantee=yes. **M12** `gdrive-api`://` OAuth
StorageBackend [#338](https://github.com/Khelsutra/badminton-highlight-indexer/pull/338)
(`819e395`) ? `backend/storage/gdrive\_api.py` (path?Drive-id resolve, **idempotent** put, lazy
Google imports, injected-fake hermetic tests) + scheme wiring (ref/`\_\_init\_\_`/models +
hyphen?underscore settings); adversarial review (`wf\_ae8c9b0f`) no_regression=yes, folded the
idempotent-put fix. Suite 1467? cross-OS, ruff+mypy clean. ? **schema ladder drifted mid-
session: v7=`locale`, v8=compute-picker ? M9-consent migration is now v9** (check `PRAGMA
user_version` first). **? NEXT = OWNER-DRIVEN alpha go-live pairing** (CF tunnel+Pages+Access +
Gemini-key rotation per `khelsutra/deploy/DEPLOY\_ALPHA.md`); M11-live + M9-consent need the
counsel ToS/DPA 7-pt checklist ([[ui-driven-cloud-serving-gap]]). Sibling tracks advanced in
parallel: M6?/api/process (#333), Cloud-Run env (#334), the SPA compute-picker + real Cloud-Run-
GPU validation (below). [[lesson-verify-engine-surface-before-scoping]]**************

Checkpoint: 2026-06-22 (**LOCAL-APP ? CLOUD RUN path ? ? ALL 4 ITEMS DONE (5 PRs merged +
real-GPU e2e VALIDATED + proof on the local box + \$0 teardown)**). engine master `334ad0f`,
khelsutra master `f67dfe7`. Owner: **"build the cloud Run path in the order + ensure due-****

diligence testing; I approve the GPU VM budget until you get a clean e2e screencast on my local machine."* APPROVED ORDER = (1) API?compute-seam wiring ? (2) P3a micromamba Dockerfile ? (3) SPA compute-picker (engine 3a + SPA 3b) ? (4) real Cloud Run validation + screencast. **PRs: #333 (1), #334 (2), #335 (3a), khelsutra #104 (3b), #337 (the cloud-dep fix item-4 found).**
- **? #104 ? compute-picker SPA half (3b)** (khelsutra master `f67dfe7`): IngestPanel probes `/api/capabilities`, shows a "On my computer"/"In the cloud" toggle ONLY when `deployment.cloud\_available`, gates CLOUD behind a required cost-ack (disabled controls + early-return), threads `compute` ? `POST /api/process`. Default-OFF (no picker / `compute` omitted when cloud unavailable). New `ingest.compute.\*` strings x6 locales + site mirror (kn/hi/es/da/id AI-drafted ? native review pending). typecheck + 226 web tests + build + coverage + 89 site tests green; adversarial review clean.
- **? ITEM 4 DONE ? real Cloud Run L4 GPU e2e VALIDATED, both directions, \$0 teardown.** Proof bundle on the local box = `artifacts/cloudrun\_e2e\_validation/`** (`RUN\_TRANSCRIPT.md` + `cloudrun\_e2e\_reel.mp4` 22-rally 1080p reel + `contact\_sheet.png` + Half-A/B logs + intervals + `TEARDOWN.md`). **Half A** (direct execute) and **Half B** (the LOCAL control plane dispatching `worker infer` in cloud-serving mode ? bucket-poll ? ingest) BOTH produced **22 rallies** on `mahadevpura\_smoke\_180s.mp4` (deterministic vs the prior local-L4 run), `cuda True`, 325s detect; the local box then stitched the cloud-detected intervals into the reel (the M6 split proven). **Total spend ? \$0.60, \$0 ongoing** (Job + image + AR repo + validation prefixes all deleted; corpus untouched; worktrees removed).

328. assistant (2026-06-29T10:33:36.840Z)

This is the main clone's canonical, OneDrive-synced memory ? actively maintained, far ahead of where I worked, with a `session-handoff.md` explicitly scoped to the **serving/maintenance track** ("don't conflate"). So I will **not** touch that file. The right pattern here (matching the existing `next-session-\*.md` icebreakers under "? Start here") is a dedicated icebreaker for this track + a pointer. I'll also port the branch-rename lesson here, since the clone that currently holds it is being deleted and its sessions will never load that namespace again.

Let me write the icebreaker first.

329. assistant (2026-06-29T10:34:09.269Z)

```
---
name: next-session-rally-finding-b-promotion
description: "? START HERE ? rally-detection finding-B promotion (P2b) + launch signoff. The R4 inactivity_temporal_density density fix shipped flag-gated default-OFF (zero-regression no-op); P2b = run the golden A/B at SERVED, promote-or-reject, then full validation + a GO/NO-GO for owner signoff. GPU + golden labels + F:\ data are now live."
metadata:
  node_type: memory
  type: project
---
```

? START HERE ? rally-detection finding-B promotion (P2b) + launch signoff

****Owner's plan (2026-06-29):**** complete ****P2b**** here (in the main clone at `C:\Users\avidu\Projects\badminton-highlight-indexer`), now that ****GPU + WSL + golden labels + the F:\ data drive are all live on this machine****, then run the full test suite and produce a ****launch-readiness GO/NO-GO**** for the owner to sign off. The owner makes the launch call; the agent recommends.

You are here (what's already merged)

The rally-detection code-review (A/B/C) work is DONE on remote `master` (HEAD ****`0c9c693`****; the main clone was at `0b4f492` on 2026-06-25 ? ****git pull --ff-only` first****):
- ****Finding A**** ? `rally\_gate.\_spread` ceil-index fix (behaviour-neutral, unconditional). PR ****#394****.
- ****Finding B**** ? `tracknet\_runner.\_trim\_inactive\_tail` density denominator fix, shipped behind the ****`inactivity\_temporal\_density` flag (default-OFF)**** ? a ****zero-regression no-op in prod**** until measured. PR ****#394****. (R4 itself is ****default-ON**** in prod: `window\_inactivity\_end=1.5`.)

```
- **A/B harness wiring (P2a)** ? `--[vs-]inactivity-temporal-density` exposed in
`rally_seg_eval`. PR #395.
- **Finding C** ? sibling `sports-data-collector` `load_canonical_csv` skipped-row warning.
#sdc PR #50 (merged).
- ? PR #393 is CLOSED, not merged ? a branch-rename closed it; #394 is the real roll-
forward. See
[[lesson-branch-rename-closes-open-pr]].
```

What finding B is / why the flag

R4's rally-end trim measured "fast-frame density" over the **count of active samples**, so under **sparse** shuttle tracking one fast sample after a tracking gap read as 100% dense and held the rally tail open (inflated end). The fix uses the trailing window's **temporal length** instead. It's a **no-op under dense tracking**, so it only moves on clips where the tracker drops out ? which is exactly why it can't be measured on synthetic/offline fixtures and needs the real golden corpus.

P2b procedure (the actual promotion)

- Ensure golden features exist** (regenerate if needed; GPU/WSL). Confirm the real paths ? the golden manifest + features are on **F:** (or **.\output**), NOT necessarily the code defaults (**output/human_lovo_manifest.json**, **--features-dir output**). Corpus is **15 clips** now (**[[golden-corpus-grew-9-to-15]]**).

```
python -m backend.eval.fusion_golden --manifest <golden_manifest.json> --out-dir
<features_dir>
```
- Run the A/B at the SERVED operating point** (flag OFF vs ON):

```
python -m backend.eval.rally_seg_eval --manifest <golden_manifest.json> \
--features-dir <features_dir> --preset served --vs served \
--vs-inactivity-temporal-density --iou 0.5
```

Headline = **tol_f1** (R2 tolerance-match F1); also inspect **|?end|** / rally-end precision (that's what B targets) and the paired sign-test + CI delta. Apply the formal gate **backend/eval/promotion.py::promote_if_served_safe** (**served_regress_eps=0.0** = no served-mean drop allowed). Honor the **dual-set** rule (**[[eval-dual-set-regression]]**): no regression on EITHER eval set, and ship a **Launch Report** (**[[launch-report-convention]]**) before any default flip.
- DECIDE**:
 - No regression** (neutral/better) ? **PROMOTE**: flip **IndexingConfig.inactivity_temporal_density** default to **True** (one-line change + test) in ONE PR.
 - Regresses** ? **DO NOT promote**: leave the flag OFF (dormant code stays); record the numbers.
 - Either way: update **docs/RALLY_DETECTION_FIXES_PLAN.md** §7 (flip **P2b** to ? DONE or REJECTED with deltas) and append the result to **docs/QUALITY_ITERATIONS.md** (**[[paper-coauthor-aim]]**); then honestly-update/archive the plan doc per **docs/DOC_STATUS.md** (**[[doc-status-register]]**) ?

P1/P2a are done, P2b is the last open row.

Full validation (for the owner's signoff)

```
- `python run_all_tests.py` (offline, expect ~1490+ passing) + `ruff check .` + `mypy` clean;
keep
coverage at/above the CI floor (`docs` pre-LGTM gate).
- GPU/WSL/data are live ? ALSO run the real-video / golden / native-WASB / GPU paths
previously
owner-only. Enumerate what's runnable; report what passed and what still can't be verified.
- Cross-reference the quality table (docs/RALLY_DETECTION_QUALITY_REPORT.md /
```

```
`docs/NEXT_STEPS.md`  
  ship target) so the launch call is evidence-based.
```

Deliverable = a LAUNCH-READINESS REPORT, then stop for owner signoff

P2b result (A/B numbers + promote/reject + PR) · full test status (offline + the now-unlocked GPU/golden paths: what passed, what couldn't run + why) · open guardrail items (licensing, WASB weight provenance WASB-C3, etc.) · a clear ****GO / NO-GO recommendation**** with reasoning.
****Recommend, do not decide ? owner signs off.****

Ramp-up kit

```
- `docs/RALLY_DETECTION_FIXES_PLAN.md` ($7 tracker, $8 Q2 runbook ? source of truth) ·  
  `docs/RALLY_DETECTION_CODE_REVIEW.md` · `docs/HOW_RALLY_DETECTION_WORKS.md` ·  
  `docs/CODE_MAP.md` ·  
  `docs/NEXT_STEPS.md` (P2 #9b).  
- Code: `backend/pipeline/detectors/tracknet_runner.py::_trim_inactive_tail` (the fix, gated on  
  `temporal_density`) · `backend/config/models.py` (`inactivity_temporal_density` ? the flag to  
  flip) ·  
  `backend/eval/rally_seg_eval.py::_build_preset` (+ the CLI flags) ·  
  `backend/eval/windowing.py`  
  (`WindowingPreset` carries it) · `backend/eval/promotion.py::promote_if_served_safe`.
```

Guardrails

Branch off `origin/master`; ONE PR per item; explicit `git add <paths>` (shared checkout ?
[[gotcha-shared-checkout-branch-collisions]]); never push to master directly; merge per
[[working-agreement-merging]] (this is a behaviour/quality flip ? owner-gated, NOT auto-merge).
Honor `CLAUDE.md` (commercial-clean licensing; paid LLMs = inference only).

330. user (2026-06-29T10:34:10.319Z)

File created successfully at: C:\Users\avidu\OneDrive\Documents\claude-sync\memory\badminton-highlight-indexer\next-session-rally-finding-b-promotion.md (file state is current in your context ? no need to Read it back)

331. assistant (2026-06-29T10:34:13.901Z)

Now port the branch-rename lesson into this (surviving) namespace:

332. assistant (2026-06-29T10:34:19.875Z)

```
---  
name: lesson-branch-rename-closes-open-pr  
description: "Renaming a branch that has an OPEN PR closes that PR ? it does not retarget it.  
Don't rename a branch with an open PR."  
metadata:  
  node_type: memory  
  type: feedback  
---
```

Lesson: renaming a branch closes its open PR

Renaming the head branch of an ****open PR**** ****closes**** the PR ? it does NOT retarget it to the new name. On 2026-06-29 I renamed `docs/preserve-rally-detection-review` ? `fix/rally-detection-edge-cases` via the GitHub rename API (`POST /repos/{o}/{r}/branches/{b}/rename`) expecting open PR ****#393**** to follow; instead GitHub auto-closed #393 (closedAt == rename time). The owner had to roll it forward as a fresh PR ****#394**** from the renamed branch (work was intact ? only the PR object died).

****Why:**** I assumed the rename API preserves open PRs. It does not (at least here) ? the old head ref

is treated as gone and the PR is closed (not merged).

****How to apply:**** Do ****not**** rename a branch that already has an open PR. Either rename BEFORE opening the PR, or warn that the PR will close and plan to re-create it (`gh pr create` from the renamed branch ? reopen may be refused). Prefer just leaving the branch name (it's cosmetic).
Commits/branch survive a rename; the PR object does not. Related: [[gotcha-shared-checkout-branch-collisions]].

333. user (2026-06-29T10:34:20.948Z)

File created successfully at: C:\Users\avidu\OneDrive\Documents\claude-sync\memory\badminton-highlight-indexer\lesson-branch-rename-closes-open-pr.md (file state is current in your context ? no need to Read it back)

334. assistant (2026-06-29T10:34:25.457Z)

Now add the "? Start here" pointer in MEMORY.md (preserving everything) and a Lesson bullet:

335. user (2026-06-29T10:34:32.933Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

336. assistant (2026-06-29T10:34:42.600Z)

Need to Read via the Read tool first (I'd only viewed it via Bash). Let me read both files, then edit.

337. user (2026-06-29T10:34:43.426Z)

```
1      # Memory Index
2
3      ## ? Start here
4      - [? START HERE ? multi-shuttle / prominent-court rally detection](next-session-multishuttle-rally.md) ? **icebreaker for the rally-DETECTION design journey (2026-06-19/20).** WASB's detector sees every shuttle; its single-target tracker discards all but one (sometimes the wrong one). Built a multi-track + court-aware selector, **measured vs GOLDEN truth (GX010128): it does NOT beat the single-track baseline as a general producer ? NO PR.** It's a TARGETED lever (separable-adjacent-court + specific-rally recovery), not an F1 win; Gemini-reference oversold it. The 21-25s recovery is real + visually confirmed. **? NEXT = the one experiment that could flip the verdict: a separable-adjacent-court golden clip (`Boxhill_Doubles`, needs owner GPU run) with recall-preserving (court-membership-only) selection.** Full detail [[next-session-multishuttle-rally]] + [[current-state]] top block. *(Design/scratch only; distinct from packaging + UX/serving tracks.)*
5      - [? START HERE ? batteries-included downloadable app](next-session-packaging-app.md) ? **icebreaker + ramp-up for the packaging track.** **? P3 + P2b + P2c ALL DONE 2026-06-22:** P3 GPU validation ($0, L4?22 rallies?73.7MB reel, proof `gs://?/highlights/p3_`); **P2b-install ? engine [#310] MERGED** (`f8b2c3b` ? seeds truly-local + **motion** default on first `indexer --app`, gate **08** resolved; `scripts/install.py`/`uninstall.py`); **P2c wizard ? khelsutra [#88] MERGED** (`7ae9cee` ? 3-step `SetupWizard` from `/api/capabilities`, 6 langs); **bundle ? engine [#312] MERGED** (`293e5c0` ? re-bundled `backend/ui/` so **indexer --app` SERVES the wizard**, live-boot proven). *(The earlier "RECONCILE" note is RESOLVED.)* **? v1 LIGHT is FEATURE-COMPLETE ? only OWNER steps remain: clean-Windows/macOS-box acceptance + ffmpeg-acquisition UX.** North-star (native-WASB-in-default-download + training + annotation) stays future + counsel(O3)/owner-GPU-gated. Tracked doc `docs/PACKAGING_PLAN.md` (§7); full detail [[packaging-app-plan]]. *(Distinct from the UX/l10n + serving tracks.)*
6      - [? START HERE ? khelsutra UX/l10n/runnable-tool](next-session-ux-localization.md) ? **refreshed 2026-06-22.** The runnable tool is **BUILT + fully localized (6 langs en/kn/hi/es/da/id)**: in-UI upload (T4) ? process ? pick ? **reel watermarked in the user's language** ? download, + PMF locale analytics. DONE this session: A1/A4/A5/A8/A9, T4, MULTILINGUAL-A, B3(#70), T11(#299), **locale?watermark+analytics (LW1 #304 / LW2 #73 / LW3 #306, verified via real watermark video for all 6 locales)**. **? ALPHA IS LIVE + REAL-USE-PROVEN (2026-06-22) ? see [[current-state]] top "SESSION WRAP" for the full ramp-up.** Use **https://api.khelsutra.guru** (SINGLE-ORIGIN: the engine serves the latest SPA + API behind
```

one CF-Access cookie ? the split-origin `app.`+Pages topology broke the cross-origin cookie; `app.` needs a DNS edit the token can't make). Owner uploaded a real match in a browser ? **26 rallies ? reel** (the "build failed" was a Cloudflare 524, reel is fine).

Deploy/Finding-B/cross-origin-fix all done. **? NEXT = the 13-issue triage backlog from real use** (in current-state SESSION WRAP): top = **async `/api/compile` [engine #329]** (kills the 524) + **M7 Cloud-Run L4 GPU** (kills the CPU perf wall, ~\$0.08/match) + **NSFW litmus [engine #332]** (hard gate before dropping the allowlist); plus UX polish (#98/#99/#101/#102/#103) + obs (#326/#328). Launch gate also still needs native-speaker review (es/da/id + watermark) + kn/hi font eyeball. *(Engine/serving track = own [session-handoff](session-handoff.md); don't conflate.)*

7 - [? START HERE ? golden regression fixtures (optional next items)](next-session-golden-fixtures.md) ? **icebreaker for the GOLDEN-FIXTURES track ? implementation COMPLETE 2026-06-22** (8 PRs #186/#189/#191/#192/#193/#194/#314/#317). A clean clone runs rally-seg regression (synthetic + **6 de-attributed REAL fixtures `fx\_r0X`**) with no video/GPU/DB, **cross-OS-enforced** (Win/mac/Linux) + leak-guarded + adversarially audited. Tracker `M1.M2.M3.MX.P1.P2.DOC.P0.P3` all ?. **? NEXT (optional, prioritized):** (1) **M4 quality-floor** on the real fixtures (per-slug F1-floor in `eval\_baselines/regression\_baseline.json` + a windowing fingerprint, CONTROL-only; real F1 is 0.0?0.59 so the floor is per-fixture `current?tol`); (2) the **deferred functional video-name rename** (`eval\_baselines` keys / `MULTICOURT\_CLIPS` ? coupled to the gitignored manifest+data+training-floor ? coordinated, before open-sourcing); (3) 01 protobuf / 02 Tier-1-key-gated (forward-looking). Locked: option (a) per-frame CSV . CONTROL-only . pinned preset. Full detail [[next-session-golden-fixtures]]. *(SEPARATE thread from packaging/UX/serving/quality.)*

8 - [Current state](current-state.md) ? **live micro-checkpoint** (updated every checkpoint): open PRs, what just landed, next steps, open decisions. **Read this first.** Now LIVE-only (handoff + today's checkpoints); kept read-first-short.

9 - [Current-state archive](current-state-archive.md) ? chronological checkpoint history (2026-06-06 ? 06-18, 96 blocks) split out of current-state 2026-06-19. Deep history only; NOT read each session.

10 - [Session handoff](session-handoff.md) ? current state . next steps . open decisions . ramp-up kit. AUTO-refreshed at meaningful checkpoints (PR merged, key decision, phase shift, session wrap) ? see working-agreement-checkpoints staleness tripwire.

11 - [Code map artifact](code-map-artifact.md) ? for CODE/architecture ramp-up, read `docs/CODE\_MAP.md` FIRST (dense anchor-rich map; replaces a file crawl). How to regenerate it.

12 - [File-placement conventions](file-placement-conventions.md) ? **where new files go** (scripts/ vs backend/eval/ vs tools/ vs scratch/ vs training/ vs artifacts/ ?); **read before committing a new file.** Authoritative table = `docs/CODE\_MAP.md` \$0 (PR #138); scratch/gitignored (#136).

13

14 - [Working agreement: attribution](working-agreement-attribution.md) ? never assert fabricated facts; NEVER invent a provenance (a discussion/review/source that didn't happen). The silver-bullet rule.

15 - [Working agreement: checkpoints](working-agreement-checkpoints.md) ? user prefers frequent small PRs + a Claude-consumable current-state file updated at every micro-checkpoint.

338. user (2026-06-29T10:34:56.489Z)

The file C:\Users\avidu\OneDrive\Documents\claude-sync\memory\badminton-highlight-indexer\MEMORY.md has been updated successfully. (file state is current in your context ? no need to Read it back)

339. assistant (2026-06-29T10:34:59.811Z)

Now add the lesson bullet near the other gotcha/lesson entries:

340. user (2026-06-29T10:35:06.103Z)

The file C:\Users\avidu\OneDrive\Documents\claude-sync\memory\badminton-highlight-indexer\MEMORY.md has been updated successfully. (file state is current in your context ? no need to Read it back)

341. assistant (2026-06-29T10:35:24.776Z)

Now a concise top entry in `current-state.md` (the "read first" file) pointing to the

icebreaker. Reading it via the Read tool first:

342. user (2026-06-29T10:35:25.243Z)

```
1      ---
2      name: current-state
3      description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8      ---
9
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12     > **Scope of this file = LIVE only:** the current handoff + *today's* checkpoints + open
PRs/decisions. Keep it read-first-short (?1 page). Older checkpoints are archived in [[current-
state-archive]] ? when today's checkpoints age out (next clean-slate day), move them there,
don't let them pile up here.
13
```

343. user (2026-06-29T10:35:39.987Z)

The file C:\Users\avidu\OneDrive\Documents\claudesync\memory\badminton-highlight-indexer\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

344. assistant (2026-06-29T10:35:45.070Z)

Let me verify the handoff files landed correctly in the namespace.

345. user (2026-06-29T10:35:47.723Z)

```
=== new files present? ===
/c/Users/avidu/OneDrive/Documents/claudesync/memory/badminton-highlight-indexer/lesson-branch-
rename-closes-open-pr.md
/c/Users/avidu/OneDrive/Documents/claudesync/memory/badminton-highlight-indexer/next-session-
rally-finding-b-promotion.md
=== MEMORY.md pointers resolve? ===
2
=== current-state top entry present? ===
1
```

346. assistant (2026-06-29T10:36:01.014Z)

Handoff is written and wired into the next session's auto-loaded memory. Everything verified present.

What I set up (in the main clone's OneDrive-synced memory namespace)

I targeted `?\projects\c--Users-avidu-Projects-badminton-highlight-indexer\memory\` ? the namespace the next session in `C:\Users\avidu\Projects\badminton-highlight-indexer\` will auto-load.

1. **`next-session-rally-finding-b-promotion.md`** ? the full P2b + launch-signoff icebreaker (you-are-here, the exact `fusion\_golden` + `rally\_seg\_eval` A/B commands, the promote-or-reject decision rule, full-validation steps, ramp-up kit, guardrails).
2. **`MEMORY.md`** ? added it as the **first "Start here" entry (auto-loaded each session), plus a Lesson bullet.
3. **`current-state.md`** ? added a **2026-06-29 top block** (the "read first" file) pointing to the icebreaker.
4. **`lesson-branch-rename-closes-open-pr.md`** ? ported the #393 branch-rename lesson here, since the clone that originally held it is being deleted and its namespace would never be loaded again.

Deliberate choices worth knowing

- **I** did not touch ``session-handoff.md`` ? your index scopes it to the serving/maintenance track ("don't conflate"), so I followed the established ``next-session-*.md`` per-track icebreaker pattern instead.
- The handoff makes the next session **git pull --ff-only` first** (main clone is at ``0b4f492``; my merged work is at remote ``0c9c693``), and **flags** that #393 is closed, #394 is the real PR ? so there's no confusion.
- It carries your key facts: **GPU + golden labels + F:\ data are live**, find-B is a **no-op under dense tracking** (so signal only shows on real sparse-tracked footage), and it ends at a **GO/NO-GO recommendation for you to sign off** ? not a launch decision by the agent.

You can either let the next session pick it up automatically (it's under "? Start here") or paste the prompt I gave earlier ? both point to the same place. This clone is safe to delete now; nothing here is unmerged or unsynced.

That closes the loop on my side. ?